Causal Erasure: the Process-Rate Conditional Coordinate

GO-14 statement v1.0 — Theorems R, C and T (the convex program, and the n-transfer lemma), the (H*) *repair* — the cross-block read X, the lemma chain L1–L4, and the split-causal sub-family — T* v2, the plateau: the Collapse identity, periodization, shift-averaging, and a two-sided certificate for the stationary value $\Psi(\Delta)$ that retires (H*), (H**), (H***), (D), (F) and the boundary charge from the plateau at $\Delta = 0, 1, 2$ — and **Theorem R1**: the convexity of the process-rate functional on the cyclostationary class is now a *theorem*, so the plateau chain carries no remaining estimate — and **Lemma W**, the achievability side: the window transfer, the reduction of the missing converse to a single density residual, and *unconditional brackets* on L^∞ at $\Delta = 0, 1, 2$ that narrow the permitted bracket by 51.5× — and **Lemma A′, Theorem D and Lemma M**, the FIR-density side: the density residual is closed, and $L^\infty(\Delta) = \Psi(D; \Delta)$ at $\Delta = 0, 1, 2$ is printed as a *scoped* theorem, in the permitted wording, with five further named scope items — and with the two false n-independence claims of Lemma M *withdrawn on the record*

Geometric Observation program

v1.0 — August 2026 (new at v1.0: §8, the FIR-density side. **Lemma 9** is one line of Hilbert space — σ is the squared distance to a *closed* span, so finite combinations are dense by definition, the infimum is *not attained* by any finite filter, and the hypotheses are $\Phi_R, f_S \in L^1$ and nothing else. **Theorem 14** makes the Wiener-algebra link *unnecessary* rather than proving it, with its hypotheses printed: f_V bounded above and below, continuity of $D \mapsto \Psi(D)$, and the floor Lemma 11; the cap Lemma 10 is *deleted from the proof* as dispensable. **Lemma 12** is restated as an exact three-leg decomposition, and two of its headline claims are *withdrawn*: "every constant n-independent" is false (measured $72.444 \to 12.676$ over $n = 64 \ldots 1024$, the same factor-5.7 decrease W1 refuted) and "no feasibility threshold" is false (at $L = 10$, $n = 64$ the shifted target is $-0.01864 < 0$) — **Lemma M does not discharge W1 or W4; W1 and W2 stand**, and only $C(L, n) = o(n)$ is ever used, so the conclusion survives. Theorem 15 then prints $L^\infty(\Delta) = \Psi(D; \Delta)$ at $\Delta = 0, 1, 2$ **as a scoped theorem, in the permitted wording of §8.8 and only there**: "unconditional" does not attach to it, Ψ remains a two-sided certified bracket, and Corollary 8 continues to govern the quotable numeric statement. The fourteen R-IND-5 restatements F1–F14 are all in force, of which F5–F8 are *not optional*. Registration 085 pending, harness `experiments/go14_density.py`; the two novelty sweeps and the window-side `la_cmi` cross-check are *owed* (Remark 53). v0.9 history: §7, the achievability side. **Lemma 8** transfers any FIR stationary record to a D-feasible finite-window record of $\mathcal{F}_0$ at $O(1/n)$, stated *from the outset* with the twelve R-IND-5 restatements W1–W12 in force: the constant $C(L, n)$ is *monotone decreasing* in n with $\sup_{n \geq n_0(L)} C(L, n) < \infty$ and only $C(L, n) = o(n)$ used (W1); the feasibility threshold $n \geq n_0(L)$ is an explicit hypothesis, with an infeasible witness (W2); the zero-edge, truncated-taps and repaired builds are kept apart and labelled, and steps (1)–(2) are asserted only of the first (W3); and the repair leg is a *separate* estimate (W4). The five steps carry measured certificates; the converse reduces to exactly one residual — FIR-kernel stationary records dense in value — Remark 43. **The headline is Corollary 8**: brackets on $L^\infty(\Delta)$ at $\Delta = 0, 1, 2$ that depend on *no lemma of this document*, narrowing $\Delta = 0$ by 51.5×. The orientation gate of Remark 42 is mandatory for any successor building window records from spectral kernels. The novelty sweep on Lemma W's combination is *owed*, and no novelty language is used for it (Remark 47). v0.8 history: (v0.7 superseded for §6's step (B): what was Lemma (R1) — a pointwise-limit-of-convex-functions *hypothesis* — is now **Theorem 10**, proved on the window-free class $\mathcal{K}_n(\varepsilon, M)$ in four steps, with the matrix Szegő/Wiener–Masani theorem as the only analytic input and the inf-of-affine representation of the leak pivots as the load-bearing new step; the three justification steps recorded at v0.7 are *demoted* — two of them are a genuine gap and an ill-posed statement, and the new route uses neither (Remark 29). Consequently Corollary 6 is *unconditional*: the chain, not the value. Also new: the Scoping Lemma, and the numbered restatements R21–R29, of which R29 supplies the n-uniform spectral floor $\lambda_{\min}(R(\omega)) = 3.1715534 \times 10^{-3}$ that the earlier revisions explicitly declined to claim. The novelty sweep on the (R1) combination was run on 2026-08-07: every ingredient is standard or adjacent-known and is cited as such, the only claim is the joint convexity in spectral moment coordinates, and two page-verification items remain *owed* — see Remark 20, whose caveats travel with any such phrasing. v0.7 history: the plateau — §6 proves the Collapse identity, step (A) periodization and step (B) shift-averaging, states Lemma S and Lemma C-stat, and certifies the stationary value two-sided by a *box-free per-frequency weak-duality* bound that needs no optimality of the fixed point, no convexity of the process-rate functional and no differentiability; the chain is *one-directional*, so what follows is $L^\infty(\Delta) \geq \Psi(\Delta)$ and not an equality, at margins $+0.0147817164/+0.0040574952/+0.0007968190$ over the *sealed* causal-spectral bars. Corollary 7 *supersedes* the v0.6 plateau arithmetic of Corollary 5 entirely; the (H^{**}) and (D)+(F) numbers there are retained as *historical*. Also new: the certified $\Delta = 2$ finite-n anchors, the corrected block infimum, and the numbered errata R14–R20. v0.6 history: the H-repair lemma chain L1–L4 proves $D_1 + D_2 \leq c(\Delta; m) + (2 - \mathbf{1}\{\Delta = 0\})X$ for *every* record of $\mathcal{F}_0$, X the cross-block read, using no optimizer structure at all; consequently (H^*) is a *theorem* on the split-causal sub-family $\mathcal{F}_0^{\mathrm{sc}}(m)$ with this document's own constants, and is demoted to a *corollary* of the strictly weaker, better-margined, monotone-verified hypothesis (H^{**}) on the single scalar X; added: the Π-retraction lemma, the certified ϕ^{sc} brackets, the extended monotone $m \times \Delta$ table that retires the "$m \in \{8, 16\}$ only" caveat, the repaired per-block feasibility step of T^*(ii), the recomputed plateau corollary $0.5514005 > 0.5479448$, and the *named* obstruction that replaces the old repair-path remark. v0.5 history: Theorem T — exact subadditivity, Fekete existence of the limit and the *unconditional* lower transfer side — with (H^*) carrying the boundary-charged reverse direction and the executed within-class certification of the diagonal ladder; six R-IND-5 transfer restatements in force, including the refutation of "$\leq \kappa$ per side" as an $\mathcal{F}_0$ lemma. v0.4 history: Theorems R and C with the certified anchor table and the sandwich-margin interval, seven R-IND-5 restatements. v0.2 history: process-limit conjecture, hard hypothesis, dated erratum to Theorem 1(v); the v0.1 sign error was corrected at v0.2)

Abstract

GO-12 settled the single-record causal-prefix eraser (the Kalman family); the process-rate object — block records erased cell by cell, each against only the causal prefix of an aging reference — was promoted to a first theorem at v0.2 (netted, GO-P-2026-076). v0.3 added the process-limit face: the $n \to \infty$ limit of the coordinate exists at $O(1/n)$, closes at the smoothing-pole rate $\lambda_s^{2\Delta}$ with a constant that is *not* matched by any tested causal-conditioning spectrum, and is driven by an explicit, testable record mechanism (horizon-matched V-cancellation). This revision adds the lower-bound theory: in moment coordinates (H, Γ) the family optimization is a jointly convex program — Theorem R (a schedule-general moment-coordinate representation of nL_a) and Theorem C (joint convexity, so every KKT point is the global family minimum and strong duality yields computable lower bounds). All seven full-space anchors are now *certified two-sided* (widths 3.45×10^{-6}–1.7×10^{-5}), and the sandwich margin is a certified interval: $\min L_a(0) - \text{block}_{16} \in [0.0317872, 0.0317906]$. This revision adds the *n-transfer* theory. Theorem T is unconditional: block concatenation gives exact subadditivity $f(n_1 + n_2) \leq f(n_1) + f(n_2)$, so by Fekete the process limit $L^\infty = \lim_n \phi_n = \inf_n \phi_n$ *exists*, and the lower transfer side $0 \leq \phi_n - L^\infty$ is a theorem — every certified finite-n value bounds the limit from above. The *reverse* direction is not: the boundary charge is carried by a numbered hard hypothesis on dyadic-chain optimizers, because "$\leq \kappa$ per side" is *false* as a family statement (D-feasible U-independent counterexamples reach 26κ; no universal constant exists). The diagonal ladder is now certified *within class*. This revision *repairs* that hypothesis. Introducing the *cross-block read* $X = I(\hat{Y}^{b_1}; W^{b_2} \mid W^{b_1})$, a four-step lemma chain — κ is an *identity* not a bound, $I(E; T^{b_2}) \leq \kappa + X$, and the two per-side charge bounds — gives $D_1 + D_2 \leq c(\Delta; m) + (2 - \mathbf{1}\{\Delta = 0\})X$ for *every* record of $\mathcal{F}_0$, with no optimizer structure anywhere. Three things follow. (a) The refutation is *explained*: the counterexamples are exactly the records with enormous X (18.46 and 29.27 bits), and the bound $I(E; T^{b_2}) \leq \kappa + X$ is *tight* to 5.7×10^{-7} at one of them. (b) (H*) is a *theorem*, with this document's own constants, on the split-causal sub-family $\mathcal{F}_0^{\mathrm{sc}}(m)$ — a convex linear section, so Theorem C certifies it two-sided, at a split-causality value gap that is flat in n (≤ 0.0044 bits, and ≈ 0.0041 at $n = 16$), and onto which an explicit distortion- and block-law-preserving retraction Π maps every record. (c) The transfer results become conditional on (H**): *one scalar*, $X \leq \bar{X}$ at dyadic-chain optimizers, measured 6.3×10^{-3} against an admissible 0.117 ($18\times$ headroom, against (H*)'s $5.5\times$) and monotone decreasing in m over $m \in \{8, 12, 16, 24\}$. The $\Delta = 0$ plateau survives the larger constant at $0.5514005 > 0.5479448$ (margin 3.456×10^{-3}, $203\times$ the quoted bracket width) and the base-24 negative control still fails, now by 3.11×10^{-4}. A separate hole in T*(ii), whose cost would have exceeded the whole repair term, is closed by convexity of the value function $D \mapsto \phi_m(D)$. Three further unconditional results are added. *Theorem K*: the KKT system of the convex program *splits by column block*, because $E[S \mid W] = V$ and Y are both coordinate blocks of W, so the Q-factor and the distortion gradient have disjoint supports; it gives $A_y = \theta N$, $N^{-1} = \theta I + V\Sigma_s^{-1}V^\top$, $A_v = -NV\Sigma_s^{-1}V_S^\top$, hence the first m-uniform regularity of the program ($0 \prec N \preceq I/\theta$, $0 \prec A_y \preceq I$, $\theta \geq 1.119$) and, as a *corollary*, the transpose symmetry that v0.3 could only measure. $\phi_{2m}^{\mathrm{sc}} \leq \phi_m$ holds unconditionally, giving a third and preferable hypothesis (H***) on the split-causality value gap — preferable because that gap, unlike X, is *two-sided certifiable* by Theorem C ($27\times$ headroom). And the repair path itself is now executed rather than promised: an m-uniform conversion theorem reduces (H**) to kernel-decay envelopes plus a noise floor, yielding $\bar{X} = 0.065038$ against an admissible 0.116930; what remains is named exactly — the operator whose inverse produces the optimizer is built *from* the optimizer, so inverse-closedness returns its own hypothesis, and a fixed-point argument in a weighted Banach algebra is the missing step. *This revision adds the plateau* (§6), and it retires most of the machinery above from that particular argument. An exact *Collapse identity* — $2 \ln 2\, n\, L_a = \sum_t \ln \sigma_t - \ln \det N$ for every record, every n and every nondecreasing schedule — turns the two remaining steps into one-liners. *Step (A)*, periodiza-

tion with *independent noise copies* (a load-bearing hypothesis: correlating them breaks the step outright), gives $L^{\text{process}}(D) \le \phi_n$ for every n. *Step (B)*, shift-averaging, produces from any period-n record an *explicit* stationary record of no larger rate and identical per-symbol distortion. Both directions of both steps are inequalities the same way, so the chain is *one-directional*: it yields $L^\infty(\Delta) \ge \Psi(\Delta)$, the stationary spectral value, and it cannot yield the converse. Ψ is then certified *two-sided* by a box-free per-frequency weak-duality bound whose only inputs are admissibility of a pair of frozen filters, positivity of the per-frequency price, and a correct cell infimum — no optimality of the fixed point, no convexity of the process-rate functional, no differentiability, and no moment box: $\Psi(0) \in [0.5627264963, 0.5627264964]$, $\Psi(1) \in [0.5364013784, 0.5364013785]$, $\Psi(2) \in [0.5310500198, 0.5310500199]$. Against the *sealed* causal-spectral allocation this is a plateau of $+0.0147817164/+0.0040574952/+0.0007968190$ at $\Delta = 0/1/2$ — four to eight orders above the certified bracket widths, at two lags the transfer route provably could not reach. *This revision removes the last estimate from that chain.* What v0.7 called Lemma (R1) — convexity of the process-rate functional on the cyclostationary class — is proved here as Theorem 10, on a class carrying no window length anywhere, in four steps: Collapse plus a shifted frame in which Δ is a unimodular phase, blocking by n, the matrix Szegő/Wiener–Masani theorem as the *only* analytic input, and then two convexity arguments — matrix quadratic-over-linear for the block leg and, for the leak leg, the observation that each conditional variance is an *infimum of affine* functions of the moment pair. A Scoping Lemma makes the class restriction automatic, because the shifts of a step-(A) periodization conjugate a constant blocked noise spectrum by a unitary block cyclic shift. Consequently Corollary 6 is *unconditional* — the *chain* is, at three lags, with no (H*), (H**), (H***), (D), (F), no boundary charge, no (R1) as a hypothesis and no (R2); the independent-noise-copy clause survives as a *specification* of the object the argument constructs rather than an assumption about anything unknown. The value Ψ itself remains a two-sided *certified bracket* under this document's floating-point convention, and the chain remains one-directional. Two scoping results are load-bearing and stated as hard terms: (i) the record family — jointly Gaussian with (V, Y), *independent of the reference noise U* — is a numbered hypothesis on the chain rule and on every minimum quoted here (U-coupled records break the identity and invert the sandwich); (ii) all numerical minima remain *class-conditional or finite-n upper bounds*; nothing here relies on them being global. *This revision adds the achievability side* (§7). Lemma 8 converts any depth-L FIR stationary record into a D-feasible finite-window record of $\mathcal{F}_0$ with $\phi_n \le \text{rate}(x) + C(L,n)/n$, where $C(L,n)$ is *monotone decreasing* in n with $\sup_{n \ge n_0(L)} C(L,n) < \infty$, only $C(L,n) = o(n)$ is used, and $n \ge n_0(L)$ is an explicit feasibility hypothesis with an infeasible witness. Its five steps carry measured certificates: edge cells contribute *exactly* zero; interior pivots dominate the stationary one by a *subset*-conditioning argument (the mirror of the bypass's superset step, verified past saturation); the edge charge $\sum_t \delta_t$ is *exactly n-independent* (0.043994832/0.051454447/0.052333440 at $L = 10$, identical to nine decimals over $n = 128 \ldots 1024$ and $L = 4 \ldots 12$); the Szegő noise leg enters with the *helping* sign; and distortion feasibility is discharged constructively by an exactly affine rescale. *The converse reduces to exactly one residual* — FIR-kernel stationary records dense in value — whose remaining gap is a contraction estimate in the Wiener norm. And, depending on *no lemma of this document*,

$$L^\infty(\Delta) \in [0.5627264963, 0.5627656412] \;/\; [0.5364013784, 0.5364458112] \;/\; [0.5310500198, 0.5310994872]$$

at $\Delta = 0/1/2$, the upper ends explicit D-feasible $\mathcal{F}_0$ records on $n = 2048$: a $51.5\times$ narrowing at $\Delta = 0$. *This revision adds the FIR-density side* (§8) and closes the residual. The move is to make the Wiener-algebra link *unnecessary* rather than to prove it: in Collapse coordinates n is a datum and σ is the only derived object, and σ is an *infimum*, so the direction achievability needs is free. Lemma 9 is then one line of Hilbert space — σ is the squared distance to a *closed* span, finite combinations are dense by definition, the infimum is *not attained* by any finite filter, and the hypotheses are $\Phi_R, f_S \in L^1$ and nothing else. Theorem 14 truncates an *arbitrary near-optimal feasible record* — never the fixed point, so the circularity is gone — by Fejér means, whose non-negativity makes the truncated noise a genuine MA(L) by Fejér–Riesz; its hypotheses are printed in full, and the spectral cap is *deleted from the proof* as dispensable, feasibility alone giving $\|g\|_2 \le 2.343168$ and $\|a_y\|_2 \le 1.766965$ because $f_V \in [0.1111, 9.0000]$. Lemma 12 is restated as an *exact* three-leg decomposition with $\eta_n = 2L(1+\varepsilon-D)/(n-2L)$, and **two of its headline claims are withdrawn**: the constants are *not* n-independent (measured $72.444 \to 12.676$ over $n = 64 \dots 1024$ at $L = 6$ — the same factor-5.7 decrease W1 refuted, in the very build Lemma M was written to fix), and there *is* a feasibility threshold (at $L = 10$, $n = 64$ the shifted target is $-0.01864 < 0$, the same threshold W2 recorded). **Lemma M therefore does not discharge W1 or W4; W1 and W2 stand** — and, since only $C(L, n) = o(n)$ is ever used downstream, the conclusion is untouched. Consequently

$$L^\infty(\Delta) \;=\; \Psi(D;\Delta), \qquad \Delta = 0, 1, 2,$$

as a *scoped* theorem (Theorem 15), in the permitted wording of §8.8 and only there, carrying five further named scope items beyond those of Corollary 6. **The word "unconditional" does not attach to this equality**; Ψ remains a two-sided *certified bracket*; the equality identifies two objects and is *not* a licence to quote a value as exact; and Corollary 8 continues to govern the quotable numeric statement.

1 Setting

Sources as in GO-12: V AR(1) with pole a, unit stationary variance; $Y_t = \rho V_t + N_t$ with $\mathrm{Var}(Y) = 1$; reference $S_t = V_t + U_t$, $\mathrm{Var}(U) = \tau^2$. A block record $\hat{Y}^n$ of Y^n (per-symbol distortion $\frac{1}{n}\sum_t E(Y_t - \hat{Y}_t)^2 \le D$) is erased cell by cell; the eraser of cell t holds $S^{t+\Delta}$ and the already-erased cells $\hat{Y}^{t-1}$. The coordinate is

$$L_a(\Delta) \;=\; \frac{1}{n}\sum_{t=1}^{n} I\Big(T^n; \hat{Y}_t \,\Big|\, S^{t+\Delta}, \hat{Y}^{t-1}\Big), \qquad T = (V, Y).$$

Hypothesis 1 (U-independence — hard, load-bearing)**.** *The record family $\mathcal{F}_0$: $\hat{Y}^n$ jointly Gaussian with (V^n, Y^n) and* independent of U^n given (V^n, Y^n)*; concretely $\hat{Y} = A_y Y + A_v V + Z$ with $Z \perp (V, N, U)$ (no U-coupling, $A_u = 0$). Theorem 1 (the chain rule and every item), Conjecture 1, and* every minimum quoted in this document *are stated under this hypothesis; no seal of the process-limit face is valid without it.*

Extension and counter-values (R-IND-5, on record). The chain rule of Theorem 1(i) extends *exactly* to U-coupled records $\hat{Y} = A_y Y + A_v V + A_u U + Z$ if and only if the U-coupling is Δ-lag causal, $A_u[t, s] = 0$ for $s > t + \Delta$. Outside that: the identity is *false* (measured residuals 0.015–0.40 at random U-couplings; 4.59 bits at an optimizer endpoint), and the minima collapse — the trivial Δ-causal U-record $\hat{Y}_t = 0.54\, Y_t + 0.23\, S_t + \mathcal{N}(0, 0.1625)$ is D-feasible and achieves $L_a(0) = 0.539536 < 0.566758$ (the family min), and a general U-coupled certified-feasible record achieves $L_a(0) = 0.092864 < \mathrm{block}_{16} = 0.534968$: the sandwich of item (iii) is *inverted.* The U-coupled variant is a different, far cheaper coordinate and is *open.*

2 The theorem

Write $\lambda_s = a(1 - K)$ for the steady-state Kalman *smoothing* pole (K the steady Kalman gain [37] of S on V); at the netted parameters $(a, \rho^2, \tau^2) = (0.8, 0.49, 0.4)$, $\lambda_s = 0.354554$, $\lambda_s^{-2} = 7.955$.

Theorem 1 (Causal erasure, first netting — GO-P-2026-076)**.** *For every jointly Gaussian record on the window $t = 1, \dots, n$ satisfying Hypothesis 1:*

(i) ***Exact chain rule.*** $n\,L_a(\Delta) = I(T^n; \hat{Y}^n \mid S^n) + C_\Delta$ *with*

$$C_\Delta = \sum_t I\Big(\hat{Y}_t; S_{>t+\Delta} \,\Big|\, S^{t+\Delta}, \hat{Y}^{t-1}\Big) \;\geq\; 0$$

the smoothing-leakage charge*; $C_\Delta = 0$ iff the record is Δ-lag causally simulatable from S. The identity is exact for every record in $\mathcal{F}_0$ and every Δ (numerical residual $\sim 10^{-14}$); for the exact U-coupled extension and its failure outside the family see the counter-values above.*

(ii) ***Collapse is a feasibility face, not an impossibility.*** *Records that collapse ($C_\Delta = 0$) while carrying information exist — e.g. any V-free (N-only) record — so universal non-collapse is* false as first conjectured*. The correct statement: for $\tau^2 > 0$, $\Delta \leq n-2$ and*

$$D \;<\; \rho^2\,\frac{n-\Delta-1}{n},$$

every D-feasible collapsing record must be V-free on all cells $t \leq n-\Delta-1$, which the distortion budget forbids; at $D \geq \rho^2(n-\Delta-1)/n$ a feasible collapsing boundary record exists (exhibited at $\Delta = 6$, $n = 16$, $D = 0.3$). Strictness of $\min L_a >$ block$/n$ in the sub-boundary regime routes through block-optimality plus $C_\Delta > 0$ at the (per-mode unique) block optimizer — not through universal non-collapse.

(iii) ***Class-conditional sandwich, strictly decreasing.*** *With* $\mathrm{block}_n = \min \frac{1}{n} I(T^n; \hat{Y}^n \mid S^n)$ *and* $\mathrm{slice}(\Delta)$ *the netted single-letter slice coordinate,*

$$\mathrm{block}_n \;<\; \min_{\textit{records}} L_a(\Delta) \;\leq\; L_a^{\mathrm{diag}}(\Delta) \;<\; \mathrm{slice}(\Delta),$$

where L_a^{diag} is the diagonal-class value; the upper bounds are strictly decreasing in Δ *(v0.1 said increasing — a sign error: more prefix access can only cheapen causal erasure), and the lower margin closes geometrically at rate $\lambda_s^{2\Delta}$ (overstatement of the diag-class bound over the true minimum $\leq 1.07 \times 10^{-3}$ at $\Delta = 2$, $\leq 2.4 \times 10^{-6}$ at $\Delta = 5$; $n = 16$, $D = 0.3$).*

(iv) ***The smoothing pole — cell-local scope.*** *For* time-local *(cell-diagonal) records, the per-interior-cell leakage ratio* $\mathrm{leak}_t(\Delta)/\mathrm{leak}_t(\Delta+1) \to \lambda_s^{-2}$ *in the window interior; the* aggregate *per-symbol ratio carries the finite-window cell-count prefactor $(n-\Delta-1)/(n-\Delta-2)$. The exponent is* record-dependent*: code-mixing records measurably decay slower (the reverse-water-filling record shows a stable ≈ 0.255/lag, empirically $\approx \lambda_s^2/\rho^2$ — characterization open). No constant measured ratio is quoted — the pole is an asymptotic with a named prefactor and a named scope.*

(v) ***Not bookkeeping.*** *L_a is not the fixed-lag static quadratic: the deficit of the bookkeeping candidate is n-pinned,* $\mathrm{gap}_n = \mathrm{static}(q_{\mathrm{path}}) - \mathrm{block}_n = 0.0213/0.0263/0.0280$ *at $n = 8/16/24$, converging to the spectral gain ≈ 0.0313 (*$\mathrm{block}_\infty = 0.5299499808$ *by per-frequency equal-slope allocation; twice-corrected value — see the dated erratum, Remark 10, and the R20 erratum of Remark 36). Record memory and cross-cell code structure carry real value.*

Remark 1 (Class scoping — load-bearing). The diagonal (stationary-symmetric) class is *provably suboptimal* for L_a: at $\Delta = 0$, $n = 16$, $D = 0.3$ the first-order optimality certificate fails at the diagonal optimum (feasible projected slope 8.6×10^{-3} against 10^{-6} noise) and explicit non-diagonal records achieve $\min L_a(0) \le 0.567353 < 0.572255$ (an improvement of 4.9×10^{-3} bits, two independent parameterizations agreeing to 8×10^{-5}); the full-space search behind Conjecture 1 has since sharpened the beat to $\min L_a(0) \le 0.5667581$ at $n = 16$ (Remark 8). Diagonal optimality *remains proven* for the block coordinate. Every quoted minimum is therefore a class-conditional or finite-n upper bound inside the bracket of (iii); presenting 0.572255 as *the* minimum would be an error.

Remark 2 (Rejected definition). Definition (b) (memoryless eraser) is monotone but has the wrong $\Delta \to \infty$ invariant (floors 0.0136 bits above the netted block coordinate) and is chain-rule-inconsistent; (c) (Kostina–Hassibi causally-conditioned DI [20]) collapses to the full-path endpoint in Gaussian settings. Definition (a) is the thermodynamically operational coordinate pending a reset-protocol construction (open).

3 The convex program: Theorems R and C

Status: lower-bound prover + independent R-IND-5 verification (own evaluator, own polish, own moment box, fresh seeds; representation identity to 2.0×10^{-12} over 279 cells including edges; the composition chain of Theorem 2 re-proved independently; 11,200 verifier Jensen midpoints and 1,566 curvature probes with zero violations, on top of the prover's 32,000). Harness: `experiments/go14_convexity.py`. Prereg 079 pending — nothing in this section is sealed yet.

Moment coordinates. Write $W = (V^n; Y^n)$ (so $T^n = W$), $\Sigma_W = \mathrm{Cov}(W)$, and for a record $\hat{Y} = AW + Z$ of the family $\mathcal{F}_0$ set

$$H = \mathrm{Cov}(\hat{Y}, W) = A\Sigma_W \quad (n \times 2n), \qquad \Gamma = \mathrm{Cov}(\hat{Y}) \quad (n \times n).$$

The family in these coordinates is the convex cone

$$\mathcal{D} = \{(H, \Gamma) : \ \Gamma - H\Sigma_W^{-1}H^\top \succeq 0\}$$

(the slack is $N_{\mathrm{cov}} = \mathrm{Cov}(Z)$), and the distortion is *affine*: $n\,\mathrm{dist} = \mathrm{tr}\,\Sigma_Y - 2\,\mathrm{tr}(H\,\Sigma_W^{-1}\Sigma_{WY}) + \mathrm{tr}\,\Gamma$. Write $P = \Sigma_W^{-1}$ and $Q = \Sigma_W^{-1}\Sigma_{WS}\Sigma_S^{-1}\Sigma_{WS}^\top\Sigma_W^{-1}$.

Theorem 1 (R — moment-coordinate representation, schedule-general). Assume Hypothesis 1. On the window $t = 1, \ldots, n$ let the eraser of cell t hold $S^{se(t)}$ for an *arbitrary nondecreasing access schedule* $se(\cdot)$, and define the schedule-general counts

$$k(j) \ = \ \#\{t : \ se(t) < j\}, \qquad 1 \le j \le n$$

(the number of record cells already erased when S_j enters the interleaved order). Then for every $(H, \Gamma) \in \mathrm{int}\,\mathcal{D}$:

$$\begin{aligned} 2\ln 2 \cdot n\,L_a = {} & \Big[\ln\det\big(\Gamma - HQH^\top\big) - \ln\det\big(\Gamma - HPH^\top\big)\Big] \\ & + \sum_{j=1}^{n}\Big[\ln \mathrm{Var}(S_j \mid S^{j-1}) - \ln \mathrm{Var}\big(S_j \mid S^{j-1}, \hat{Y}^{k(j)}\big)\Big]. \end{aligned}$$

Staircase reduction (explicit). For the coordinate's own schedule $se(t) = \min(t + \Delta, n)$ the counts reduce to $k(j) = \max(0,\, j - \Delta - 1)$. The reduced form is *prefix-staircase-specific*: on non-staircase nondecreasing schedules only the general $k(j) = \#\{t : se(t) < j\}$ is correct — blind use of the staircase formula deviates by up to 1.05 bits (verifier-measured), and off-by-one variants of k are detected at 0.1–0.7 bits.

Proof skeleton. (1) Factor the joint $\ln\det$ of $(S^n, \hat{Y}^n)$ along the interleaved order twice: collecting the $\hat{Y}$-pivots gives $\sum_t \ln \mathrm{Var}(\hat{Y}_t \mid S^{se(t)}, \hat{Y}^{t-1})$; collecting the S-pivots gives $\sum_j \ln \mathrm{Var}(S_j \mid S^{j-1}, \hat{Y}^{k(j)})$ — this regrouping is the whole trick (and the reason single-cell convexity is never needed). (2) U-independence enters *exactly once* as a premise — the residuals (U, Z) are independent given W — but is load-bearing in *two* legs of the computation: $\mathrm{Cov}(S, \hat{Y}) = KH^\top$ with $K = \Sigma_{WS}^\top \Sigma_W^{-1}$ (numerator/leak leg, giving the Q-Schur complement $\Gamma - HQH^\top$ since $Q = K^\top \Sigma_S^{-1} K$), and $\ln\det \mathrm{Cov}(\hat{Y} \mid W, S^\bullet) = \ln\det(\Gamma - HPH^\top)$ (denominator leg: conditioning on S adds nothing given W). (3) The unconditional S-pivots telescope against $\ln\det \Sigma_S$. This re-proves the 076 chain rule of Theorem 1(i) in one line: the bracket is the block term $I(T^n; \hat{Y}^n \mid S^n)$ and the j-sum is the leakage charge C_Δ in its S-side (reversed) form. Verification: residual 1.15×10^{-12} over 144 prover cells and 2.0×10^{-12} over 279 verifier cells including the edges $\Delta \in \{0, n-2, n-1, n\}$ and random non-staircase schedules. □

Remark 3 (U-independence: one premise, two legs — and the non-conflation rule). Correcting only one leg does *not* restore the identity: at fixed dense-U-coupled records ($n = 12$) the leg-by-leg counter-values are 0.32–2.01 bits (as-stated, numerator-only- corrected, and denominator-only-corrected all break; the harness's pinned record measures 2.68/3.51/0.83 bits respectively), while correcting *both* legs restores exactness for arbitrary A_u including anticausal couplings (residual at machine precision). Non-conflation: Δ-lag-causal U-coupling ($A_u[t,s] = 0$ for $s > t+\Delta$) extends the *record-space* chain rule of Theorem 1(i) exactly, but does *not* rescue the moment-form Theorem R as stated — measured moment-form residuals 2.4–3.5 bits at Δ-causal couplings (harness pinned record: chain-rule residual 2×10^{-16}, moment-form residual 3.22 bits at $\Delta = 2$). The two extension results must not be conflated.

Theorem 2 (C — joint convexity; KKT points are global; strong duality). On $\mathcal{D}$: (i) the block bracket $B(H, \Gamma) = \ln\det(\Gamma - HQH^\top) - \ln\det(\Gamma - HPH^\top)$ equals $-\ln\det(I - Z)$ with $Z = R^{1/2} H^\top (\Gamma - HQH^\top)^{-1} H R^{1/2}$ and

$$R \;=\; P - Q \;=\; \Sigma_W^{-1} \Sigma_{W|S} \Sigma_W^{-1} \;\succeq\; 0;$$

under the mapping $C \leftrightarrow H^\top$, $V \leftrightarrow \Gamma$ this is *exactly* the 074 convexity lemma's G-form (GO-13 Theorem 5, block Schur lift) — whose ingredients are the standard matrix-fractional and composition rules [5, Sec. 3.1.7 and 3.2.4] together with Ando's operator concavity [6], the program's contribution there being the *lift into the moment chart*, not the convexity —, so Z is jointly matrix-convex on $\mathcal{D}$ with $Z \prec I$ on $\operatorname{int} \mathcal{D}$ (Sylvester: $\det(I - Z) = \det M_P / \det M_Q$), and B is jointly convex. (ii) Each leak term $-\ln s_j$ is convex: the pivot $s_j = \mathrm{Var}(S_j \mid S^{j-1}, \hat{Y}^{k(j)})$ is a Schur pivot of a joint covariance that is *affine* in (H, Γ), hence an infimum of affine functions of (H, Γ) — concave — and bounded below by $\tau^2 > 0$ (the eraser never sees U_j through a U-independent record). (iii) The distortion is affine and $\mathcal{D}$ is a convex cone. Hence $n\, L_a$ is jointly convex on $\mathcal{D}$; every KKT point of the distortion-constrained program is the *global* family minimum; Slater's condition holds (interior scalar records), so strong duality makes lower bounds computable: any $\mu \geq 0$ and any point x_p give the certificate $p^* \geq \phi_\mu(x_p) - \|\nabla \phi_\mu(x_p)\|\, R_{\text{box}}$, with R_{box} from the per-cell moment bounds $\Gamma_{tt} \le (1+\sqrt{nD})^2$, $|H_{tj}| \le \sqrt{\Gamma_{tt}}$ (verifier: the box is valid and nearly tight — an adversarial one-cell record reaches 10.1748 against the box's 10.1818).

Proof skeleton. (i) is the 074 lift verbatim after the transpose-convention check ($C^\top PC$ with $C = H^\top$ is $HPH^\top$ — mapping exact to 0.0); $R \succeq 0$ by the displayed closed form (verified to 1.7×10^{-14}; $\operatorname{eigmin}(Q) \ge -10^{-15}$, PSD to machine). (ii) is inf-of-affine concavity plus $-\ln$ composition on $[\tau^2, \infty)$. (iii) is inspection. Numerics on record: zero Jensen-midpoint violations (prover 32,000 + verifier 11,200 + 1,566 curvature probes), tangent plane at the $(16, 0)$ winner 400/400 with min slack $+1.13 \times 10^{-2}$, analytic gradient vs FD 1.2×10^{-10}. □

Remark 4 (Per-cell convexity is **FALSE** after the first cell — the regrouping is not a convenience, it is the proof). Theorem 2 proves convexity of the *regrouped* pieces (block bracket, S-side leak sum). The individual per-cell CMI terms of L_a are **not** convex in (H, Γ), and the split is exactly at the first cell.

Cell $t = 0$ is convex, and already proved. With no $\hat{Y}^{t-1}$ to condition on, the $t = 0$ term collapses to the block bracket in scalar form with a *truncated* Q,

$$f_0 = \tfrac{1}{2} \log_2 \frac{\Gamma_{00} - h_0 Q_{k(0)} h_0^\top}{\Gamma_{00} - h_0 P h_0^\top}, \qquad Q_k = K_{[k]}^\top \Sigma_{S,[k]}^{-1} K_{[k]},$$

h_0 the first row of H (verified to 1.3×10^{-15} over $n \in \{3, 4, 5\}$, $\Delta \in \{0, 1, 2\}$; $0 \preceq Q_k \prec P$ with $\operatorname{eigmin}(P - Q_k) \ge 2.7 \times 10^{-2}$). So its convexity is the 074 lift's scalar case and needs no separate argument.

Every cell $t \ge 1$ admits strictly concave directions. Witness ($n = 3$, $\Delta = 0$, $t = 2$): three points of $\operatorname{int} \mathcal{D}$ whose exact midpoint is the base point, with $f_2(x^+) = 0.4209618516$, $f_2(x^-) = 0.1470759393$, $f_2(\text{mid}) = 0.3535350376$, a Jensen gap of $+6.95161422 \times 10^{-2}$ bits the wrong way. All three are strictly feasible ($\operatorname{eigmin} N_{\text{cov}} = 3.77 \times 10^{-1}$, 7.71×10^{-2}, 4.16×10^{-1}), and the gap reproduces on *three* computationally independent routes — the (H, Γ) form, the $4n$-joint form built from the recovered record $A = HP$, $N_{\text{cov}} = \Gamma - HPH^\top$, and a pure $\log\det$-of-submatrices form using no conditional-variance helper at all — agreeing to 5.6×10^{-17}, and at 50 digits the gap is $0.06951614215948869\ldots$, so it is not a floating-point artifact.

The violation survives inside the causal class. Because $\{H : HP \text{ lower-triangular in both blocks}\}$ is a *linear* subspace, midpoints stay causal and the feasible set over it is convex, so a violation there is a violation on $\mathcal{D}$ — and a stronger statement, since it needs no non-causal record. Restricted to that 18-dimensional subspace ($n = 3$) the per-cell Hessian still reaches -3.54, with witness gap $+2.33 \times 10^{-2}$ at $t = 1$ and A lower-triangular to 3.7×10^{-16}.

Negative curvature appears at every $t \ge 1$ for every $n \in \{3, 4, 5\}$ and $\Delta \in \{0, 1, 2\}$, with minimum Hessian eigenvalues between -1.2 and -5.0: it is generic, not marginal. Earlier samplings reported no hits because random pairs are generic and the concave directions are the Hessian's minimum-eigenvector directions; a directed curvature search finds one at the first base point tried.

Theorem 2 is untouched, and the witness shows why: at that same pair the total $n\,L_a$ has Jensen gap -3.27×10^{-1} — convex, as it must be — because the neighbouring cells' convexity absorbs cell 2's concavity ($t = 0, 1, 2$ gaps -6.28×10^{-2}, -3.33×10^{-1}, $+6.95 \times 10^{-2}$). The mass that makes a late cell concave is precisely what the regrouping moves into the S-side leak sum. Nothing in this section ever relied on per-cell convexity, so nothing downstream changes. **Nor could Theorem 10 have implied it**: that theorem convexifies the same two regrouped pieces at the process level, and the regrouping is precisely what makes the argument work (Remark 23).

Remark 5 (The diagonal class is a convex linear section — its ladder is UB-only because class < family). In (H,Γ) the diagonal (stationary-symmetric) class is a *convex (linear) section* of $\mathcal{D}$: midpoints of diagonal records stay in class (verifier: off-diagonal mass in the C_v eigenbasis at (H,Γ)-midpoints is 0 to machine precision). The earlier "non-convex slice" wording is refuted as worded — what is non-convex is the (g,h,z) *parameter chart* (explicit midpoint witness on record), a different statement. Consequently the diagonal ladder of §4 is upper-bound-only *because the class is a strict subset of the family* (Remark on class scoping, §2), not because of any slice non-convexity; and within-class values are certifiable by the same Lagrangian machinery restricted to the section. **That upgrade path has since been executed**: all nine ladder cells and three block values are now certified two-sided *class* values (§5.2). Certifying the class does not certify the family — the ladder remains an upper bound on $\min L_a$ because class < family.

Certificates on record (per-n). $\operatorname{eigmin}(P-Q) = 1.0382\times10^{-2} / 5.5944\times10^{-3} / 4.4309\times10^{-3} / 3.9529\times10^{-3}$ at $n = 8/16/24/32$ — *decreasing in* n; as a finite-window statement this is quoted as "$\geq 3.9\times10^{-3}$ for $n \leq 32$" (the earlier single-number quote is an erratum, corrected here). **The earlier refusal to claim an n-uniform floor is now withdrawn**: at the blocked spectral level $\lambda_{\min}(R(\omega)) = 3.1715534\times10^{-3}$ uniformly in ω and in $n = 1,\ldots,8$, and the finite-window sequence above is convergence to that value *from above* — see Remark 28. The noise-floor parameterization is inactive at every winner and every polished point.

3.1 Theorem K: the KKT system splits by column block

Two *exact* structural facts of this model drive everything in this subsection, and both are checkable in one line:

$$K = \Sigma_{WS}^{\top}\Sigma_W^{-1} = [\,I \;\; 0\,], \qquad \Sigma_W^{-1}\Sigma_{WY} = [\,0;\, I\,]$$

(verified to 2×10^{-15}): the first because $E[S \mid W] = V$ and V *is* a coordinate block of W, so every Q-factor term lives only on the V-columns of $A = H\Sigma_W^{-1} = [A_v, A_y]$; the second because Y is a coordinate block of W, so the distortion gradient lives only on the Y-columns. **The two supports are disjoint**, so the stationarity condition splits by column block.

Theorem 3 (K — KKT structure; *unconditional*, $\mathcal{F}_0$, any Δ, any nondecreasing schedule). Let (H,Γ) be a KKT point of the distortion-constrained program with multiplier μ, and write $N = \Gamma - H\Sigma_W^{-1}H^{\top} = \mathrm{Cov}(Z)$, $\theta = 2\mu\ln 2$, and (v_t, v_t^S, σ_t) for the *record* pivots of the interleaved order (cell t predicted from $S^{se(t)}, \hat{Y}^{t-1}$), collected into $V\Sigma_s^{-1}V^{\top}$ and $V\Sigma_s^{-1}V_S^{\top}$. Then

$$\text{(K1)}\;\; A_y = \theta N, \qquad \text{(K2)}\;\; N^{-1} = \theta I + V\Sigma_s^{-1}V^{\top}, \qquad \text{(K3)}\;\; A_v = -N\,V\Sigma_s^{-1}V_S^{\top}.$$

Verified at every certified optimizer — relative residuals $\leq 3.2\times10^{-8}$ at $(16,0)$, $(24,0)$, $(32,0)$, $(48,0)$, $(16,1)$, $(16,2)$, $(24,1)$ (prover), and $\leq 8.3\times10^{-8}$ over the six of those the harness recomputes from cold starts. The residuals track the polish accuracy of the optimizer, not the identities, which are exact; the harness gates them at 10^{-6} for that reason.

Corollaries, unconditional and m-uniform.

(a) $A_y = \theta N$ is *symmetric positive definite*. **This upgrades the 078-era "mechanism" finding — A_y and the noise kernel transpose-symmetric, all causal asymmetry confined to A_v — from an empirical observation to a corollary of Theorem 3** (measured symmetry defect 1.2×10^{-9}–1.4×10^{-8}, against the 5×10^{-5} at which it was first reported; see Conjecture 1(4)). And (K3) carries the causal asymmetry through V_S, whose column support ends *exactly* at $t+\Delta$ — the structural candidate for the observed sign change at lag Δ. *This is a lead, not a claim*: no proof that the sign change must occur there is offered here.

(b) $0 \prec N \preceq I/\theta$ and $0 \prec A_y \preceq I$, *uniformly in* m — the first m-uniform optimizer-regularity statement in the program.

(c) $\theta \geq 2\ln 2\,\phi_n(D)/(1-D) \geq 1.119$ (convexity of $D \mapsto \phi_n(D)$, Remark 14, together with $\phi_n(1)=0$), hence $N \preceq 0.894\,I$ unconditionally.

Proof skeleton. Stationarity in Γ reads $M_Q^{-1} - N^{-1} + \partial_\Gamma\bigl(-\sum_j \ln s_j\bigr) = -\theta I$. The Cholesky decomposition of J^{-1} along the interleaved order, $J = \mathrm{Cov}(S^n, \hat{Y}^n)$, splits the pivot Gram into a reference half and a record half summing to J^{-1} (verified to 6×10^{-16}); since $(J^{-1})_{\hat{Y}\hat{Y}} = M_Q^{-1}$, the M_Q^{-1} and reference-pivot terms combine into the record-pivot Gram — which is (K2). Stationarity in H then splits by the disjoint supports above: the Y-columns give $N^{-1}A_y = \theta I$, i.e. (K1); the V-columns give (K3). *Caution on reading (K2) and (K3)*: the Gram must be the *record*-pivot one — substituting the reference-pivot Gram makes (K2) false at the 47% level and (K3) false at the 120% level. □

3.2 Certified two-sided anchors

Lagrangian polish + moment-box tangent bound (Theorem 2), prover (P) and independent verifier (V) each end-to-end; per anchor the tighter certified width of the two is cited. Upper ends are the (feasibility-projected) winner values.

(n,Δ)	certified bracket for p^*	width	source
$(16,0)$	$[0.566754682,\ 0.5667581350]$	3.45×10^{-6}	V
$(16,1)$	$[0.5408029,\ 0.5408064658]$	3.6×10^{-6}	P
$(16,2)$	$[0.5358901,\ 0.5358939352]$	3.8×10^{-6}	P
$(24,0)$	$[0.5654096,\ 0.5654138570]$	4.3×10^{-6}	P
$(24,1)$	$[0.5393228,\ 0.5393377781]$	1.5×10^{-5}	P
$(24,2)$	$[0.5342628,\ 0.5342788038]$	1.6×10^{-5}	P
$(32,0)$	$[0.5647249,\ 0.5647418676]$	1.7×10^{-5}	P

The 078 anchors are thereby *certified family values*, no longer upper bounds alone; minimizer *uniqueness* is still not claimed (no flat was found — three polishes agree to 2.8×10^{-14} in value and 2×10^{-6} in coordinates — but gauge flats are not excluded; "an optimizer", never "the"). The harness reproduces four of the brackets end-to-end from cold starts by μ-bisection (convexity makes warm starts a convenience, not a dependence), at widths $4.0/3.0/3.8/9.2\times10^{-6}$ for $(16,0)/(16,1)/(24,0)/(32,0)$ and a block_{16} bracket of width 1.7×10^{-6}.

Corollary 1 (Certified sandwich margin at $n=16$, $D=0.3$)**.**

$$\min L_a(0) - \mathrm{block}_{16} \in [0.0317872,\ 0.0317906]$$

(verifier-tightened interval). The block side inherits the sealed block_{16} *precision through the* proven *diagonal optimality of the block coordinate.*

Remark 6 (Feasible projection of winner upper ends). The stored winner endpoints are distortion-infeasible by $\sim 4 \times 10^{-11}$ (raw optimizer termination). The one-line feasible projection (blend toward the near-copy record) corrects the upper ends by $\sim 10^{-10}$ — immaterial at the 3.5×10^{-6} width scale; all upper ends above are to be read through this remark.

Remark 7 ("Certified" means floating-point certificates). All brackets here are *floating-point* certificates: the weak-duality and tangent-bound inequalities are exact in real arithmetic and are evaluated in IEEE double precision without interval arithmetic. The certified widths exceed the $f64$ evaluation error by roughly seven orders of magnitude; no claim of formally verified arithmetic is made.

4 The process limit

Everything in this section is under Hypothesis 1 and at the netted parameters, $D = 0.3$. Status: probe + independent R-IND-5 verification (two evaluator routes agreeing to 2×10^{-10}; all sealed 076 anchors reproduced); pre-registration (078) pending — no numerical face here is sealed yet.

Conjecture 1 (Process limit of the causal-erasure coordinate — v0.2, R-IND-5-scoped)**.** *Let* $L_n^{\mathrm{fs}}(\Delta) = \min L_a(\Delta)$ *over the full (non-diagonal) family* $\mathcal{F}_0$ *on the window of length* n*, and* $L_n^{\mathrm{diag}}(\Delta)$ *the diagonal-class value.*

(1) ***Existence at*** $O(1/n)$***.*** *Both coordinates converge as* $n \to \infty$ *with* $O(1/n)$ *finite-window drift (model attack: fitted exponent* $\alpha = 0.996$–1.001*). Two-point* $1/n$ *limits, calibrated on the block face (*$\leq 5 \times 10^{-6}$*): full space* $L_\infty^{\mathrm{fs}}(\Delta) = 0.562725/0.536400/0.531049$ *at* $\Delta = 0/1/2$ *(three-n consistency at* $\Delta = 0$*:* $0.5627253/0.5627259/0.5627265$ *from* $n = 16/24/32$ *pairs); diagonal class* $0.568571/0.537401/0.531206/0.529977$ *at* $\Delta = 0/1/2/4$ *(*$\pm \leq 1.2 \times 10^{-5}$*).*

(2) ***Closure rate, constant unclaimed.*** *The excess* $E_\infty(\Delta) = L_\infty^{\mathrm{diag}}(\Delta) - \mathrm{block}_\infty$ *obeys* $E_\infty(\Delta) \sim c\,\lambda_s^{2\Delta}$ as a $\Delta \to \infty$ asymptotic only*: the per-lag ratio rises toward* $\lambda_s^{-2} = 7.955$ from below *(*$5.18 \to 5.93 \to 6.87 \to 7.65 \to$ *within error at* $\Delta = 5 \to 6$*) and is* unresolved beyond $\Delta \approx 6$*; the constant is quoted as the bracket* $c_{\mathrm{diag}} \in [0.105, 0.125]$ only — $c(\Delta) = 0.107/0.112/0.117$ *at* $\Delta = 4/5/6$*, still rising,* not converged through $\Delta = 6$*. The full-family constant is data-supported only as* $c_{\mathrm{fs}} \in (0.07, 0.125]$*. No single-letter identification of* c *is claimed.*

(3) ***Strictly above every tested causal-conditioning spectrum — certified at every computed*** n***; in the limit conditional on (H*) at*** $\Delta = 0$ ***and on the*** $O(1/n)$ ***model at*** $\Delta = 1, 2$***.*** **Both limit conditions are discharged at v0.7** ***— Corollary 7 derives the limit statement at*** $\Delta = 0, 1, 2$ ***from §6, and at v0.8 (R1) is itself a theorem (Theorem 10); what follows in this item is the v0.4-era route and is retained as history.*** *At every computed anchor (*$n \in \{16, 24, 32\}$*,* $\Delta \in \{0, 1, 2\}$*) the* certified lower bracket end *of §3.2 sits strictly above the causal-spectral allocation value (Remark 9) — a certificate-strict gap (smallest exceedance* $\approx 4 \times 10^{-3}$ *bits at* $(24, 2)$*,* $\geq 250\times$ *the bracket width there), not an optimizer report. In the limit, the plateaus* $+0.014781/+0.004057/+0.000795$ *(*$\Delta = 0/1/2$*) above the allocation are* 160–$3000\times$ *the calibrated extrapolation error and carry the wrong shape (drifting per-lag rate, not the allocation's clean* λ*-law). At* $\Delta = 1, 2$ *the limit statement remains* conditional on the $O(1/n)$ finite-window model. *At* $\Delta = 0$ *it is*

no longer *model-conditional: Corollary 5 derives* $L^{\infty}(0) \geq 0.5514005 > 0.5479448$ *from the certified* $n = 32$ *bracket and the repaired transfer constant,* conditional instead on Hypothesis 2 (H**) *— or on (D)+(F), at the smaller margin recorded in Corollary 5. The subadditivity half of the old n-monotonicity face is now closed unconditionally (Theorem 4); what remains open is the cross-block read at optimizers (Question 1). Both closed-form candidates (*$c_{\text{spec}} = 0.0190 \pm 0.0007$*,* $c_{\text{static}} = 0.026668$*) are too small at every* Δ*. No claim is made that no single-letter form exists. The gap is genuine cross-cell code value: the winner cuts leakage* $0.033 \to 0.024$ *by paying block penalty the per-frequency relaxations cannot express.*

(4) ***Mechanism* — the symmetry half is no longer a conjecture.** *At every full-space optimizer computed (*$n = 16/24/32$, $\Delta = 0/1/2$*; seven adversarial starts converge to the same record, KKT certificate* 8.4×10^{-6}*): the* Y*-coupling and noise kernels are time-symmetric, where* time-symmetric means transpose symmetry $A_y = A_y^{\top}$ *(relative residual* $< 5 \times 10^{-5}$ *as first measured; the stronger persymmetry* $JA_y^{\top}J = A_y$ fails *at the* $\approx 6\%$ *level at the window edges — transpose symmetry is the invariant);* all *causal structure lives in the* V*-coupling* A_v*, whose interior-row kernel* changes sign exactly at lag Δ *— horizon-matched* V*-cancellation: the record aligns* V*-content with the span the eraser will hold and anti-correlates* V *beyond the access horizon, suppressing smoothing leakage. This is exactly what the symmetric-by-class diagonal family cannot express (and explains the 076 beat).* ***Status upgrade (v0.6).*** *The first half of this item is* no longer conjectural*:* $A_y = \theta N$ *is symmetric positive definite at every KKT point by Theorem 3(K1), so the transpose symmetry of* A_y *and of the noise kernel, and the confinement of all causal asymmetry to* A_v*, are* corollaries*, not measurements (measured symmetry defect now* 1.2×10^{-9}–1.4×10^{-8}*, i.e. at polish accuracy). What remains conjectural is the* sign change exactly at lag Δ*; Theorem 3(K3) supplies a structural candidate —* A_v *inherits its asymmetry from* V_S*, whose column support ends exactly at* $t + \Delta$ *— but no proof. Falsifiable per-cell at any* n*.*

Remark 8 (Reproducible anchors — raw finite-n values only). The anchors of record are *raw finite-n L_a values*: family min $L_a(0) \leq 0.5667581$ at $n = 16$ (beats the 076-recorded 0.567353 by 5.9×10^{-4}; all start basins converge to it within 10^{-7}); full-space $L_a(0) \leq 0.56474187$ at $n = 32$; the diagonal ladder value $L_a^{\text{diag}}(n{=}48, \Delta{=}6) = 0.5316222730$. The full-space upper ends are now *certified two-sided* (§3.2); the diagonal-ladder values are now certified two-sided *within the class* (§5.2) and remain class-conditional upper bounds on the family minimum (Remark 5). The extrapolated $E_{\infty}(6)$ is quoted *only* with an explicit error bar — $E_{\infty}(6) \approx 4.4 \times 10^{-7} \pm 10\%$ (two-point pair sensitivity); it is *not* a four-digit anchor, the raw $L_a(48, 6)$ above is.

Remark 9 (The causal-spectral allocation is a reference construction). The causal-spectral allocation: equal-slope per-frequency conditional quadratics with conditioning spectrum the Δ-lag causal Wiener error spectrum $S_e^{(\Delta)}$. Values 0.547945/0.532344/0.530253 at $\Delta = 0/1/2$, verified to seven digits by an independent (time-domain) route; it reproduces block$_{\infty}$ at the two-sided endpoint to 1.6×10^{-10}, and its own excess obeys the $\lambda_s^{2\Delta}$ law cleanly. It is a *reference construction only*: it carries no achievability map (not an upper bound on $\min L_a$) and no data-processing direction (not a lower bound). Item (3) of Conjecture 1 is a statement about mismatch with the tested spectra, never a bound.

Remark 10 (Erratum — dated 2026-08-05). v0.2 of this document quoted block$_{\infty} = 0.52991$ in Theorem 1(v), propagated from the pass-1 coarse frequency waterfilling; the converged per-frequency equal-slope value is block$_{\infty} = 0.529950$ (corrected in item (v) above; the spectral gain ≈ 0.0313 is unaffected). That six-digit value is itself superseded at the ninth digit by the anchor-free two-sided certificate of Remark 36(R20): $\text{block}_\infty = 0.5299499808119$. Same date, probe-record erratum: the process-limit probe's diagonal excess at $\Delta = 4$ is 2.66×10^{-5}, not 2.75×10^{-5}.

Dated 2026-08-06 (transfer note). Three further corrections, all carried into §5: (a) the constant was recorded as $c(1) \le 1.1257$; the computed value is 1.1257294, so the recorded bound is false in the fifth decimal and $c(1) \le 1.1258$ is used throughout (Remark 15). (b) The transfer note cites a value 0.564729705 for the certified lower end at $(32, 0)$; no computation on record produces it. The sealed value is $\text{LB}(32, 0) = 0.5647327869$ (`results/GO14-convexity.json`, `s5_32_0`), and it is the only one used in Corollary 5; 0.564729705 must not be cited. (c) The plateau margin is $215\times$ the quoted bracket width, not the $\approx 250\times$ of the transfer note.

5 The n-transfer lemma: Theorem T, and the repair of (H*)

Status: prover + independent R-IND-5 verification (own `slogdet` evaluator, own certified anchors, own re-derivation of the constants and of the corollary arithmetic; six mandatory restatements, all in force in this revision). Harness `experiments/go14_transfer.py`. Registration 080 pending — nothing in this section is sealed. The repair material of §5.1 (the lemma chain, the split-causal sub-family, the Π-retraction, the monotone table, the feasibility repair and the recomputed corollary) and Theorem 3 of §3.1 are newer still: they carry their own harness, `experiments/go14_hrepair.py`, and await registration 081. **Nothing in this revision contradicts a gate of 080**: every 080 gate is either untouched or — in the case of the κ-per-side refutation — *explained* rather than overturned.

Write $\phi_n(\Delta) = \inf_{\mathcal{F}_0} L_a(\Delta)$ for the family value on the window of length n (certified two-sided in §3.2) and $f(n) = n\,\phi_n(\Delta)$. Let

$$\kappa \;=\; \tfrac{1}{2}\log_2 \frac{1}{1-a^2} \;=\; 0.736966 \text{ bits}$$

be the AR(1) boundary information at the netted pole $a = 0.8$. For a split of the n-window into blocks b_1, b_2 define the *per-side boundary charges*

$$D_i \;=\; \sum_{t \in b_i} \Big[I\big(T^{b_i}; \hat{Y}_t \mid \text{own}C_t\big) - I\big(T^{b_i}; \hat{Y}_t \mid \text{big}C_t\big) \Big] \;\ge\; 0,$$

where $\text{own}C_t$ is the conditioning set of cell t inside its own block and $\text{big}C_t = S^{se(t)}, \hat{Y}^{t-1}$ is the big-window one; D_i is exactly what the split loses at the boundary. Write $E = (S^{b_1}, \hat{Y}^{b_1})$ for the block-1 record-and-reference history that the block-2 cells see in $\text{big}C$ but not in $\text{own}C$.

Theorem 4 (T — exact subadditivity, Fekete existence, and the lower transfer side; *unconditional*). Fix Δ and assume Hypothesis 1.

(i) **Exact subadditivity.** $f(n_1 + n_2) \le f(n_1) + f(n_2)$ for all $n_1, n_2 \ge 1$. Take ε-optimal records of the two windows, independent of each other, and concatenate: the result is a feasible record of the $(n_1 + n_2)$-window (the distortion constraint is the cell average), and *at every cell* the big-window per-cell term is at most the own-window term. The numerator $\text{Var}(\hat{Y}_t \mid S^{se(t)}, \hat{Y}^{t-1})$ can only shrink when the conditioning set grows from the own block to the whole window; the denominator $\text{Var}(\hat{Y}_t \mid T^n, S^{se(t)}, \hat{Y}^{t-1})$ is *unchanged*, because a block-diagonal record makes cell t conditionally independent of the other block given its own block's T — verified as an exact equality (relative deviation 2×10^{-14} over all cells, including the unequal split $8 + 4 \to 12$). Letting $\varepsilon \downarrow 0$ gives the inequality for the infima.

(ii) **Fekete [31].** $\phi_n \geq 0$, because each per-cell term is a conditional mutual information; hence $f \geq 0$ is subadditive and

$$L^\infty(\Delta) \;:=\; \lim_{n\to\infty} \phi_n(\Delta) \;=\; \inf_n \phi_n(\Delta)$$

exists in $[0, \infty)$.

Hygiene (both steps were demanded by R-IND-5 and are stated, not assumed). (a) The non-negativity $\phi_n \geq 0$ is what makes Fekete's lemma give a finite limit rather than $-\infty$; it is immediate from the CMI form of the coordinate, but it is a hypothesis of the lemma and is recorded as such. (b) The concatenation argument is run on ε-optimal records and the limit $\varepsilon \downarrow 0$ is taken at the end, so the theorem never needs ϕ_n to be *attained.* (Attainment does hold, by Theorems 1 and 2 plus the moment box, but it is not used.)

Corollary 2 (The lower transfer side is a theorem)**.** $0 \leq \phi_n(\Delta) - L^\infty(\Delta)$ *for every n — unconditionally. Every certified finite-n family value of §3.2 is therefore an upper bound on the limit; in particular $L^\infty(0) \leq \phi_{32}(0) \leq 0.5647419677$.*

5.1 The reverse direction: the cross-block read X

The matching *upper* transfer bound $\phi_n - L^\infty \leq c/n$ needs the boundary charge to be controlled at the optimizers of the doubling chain. At v0.5 that control was a bare hypothesis, because "$\leq \kappa$ per side" is refuted as a family lemma (Remark 12). This revision *repairs* it: a lemma chain, using no optimizer structure whatsoever, reduces the whole reverse direction to a single scalar.

Definition 1 (The cross-block read)**.** For a split of the n-window into $b_1 = \{1, \ldots, m\}$ and $b_2 = \{m + 1, \ldots, n\}$ write $W^{b_i} = T^{b_i} = (V^{b_i}, Y^{b_i})$, $E = (S^{b_1}, \hat{Y}^{b_1})$ and $E' = S^{b_2,1..\Delta}$ (empty at $\Delta = 0$). The *cross-block read* of a record is

$$X \;:=\; I\big(\hat{Y}^{b_1}; W^{b_2} \;\big|\; W^{b_1}\big) \;\geq\; 0,$$

the information the block-1 *rows* draw from block-2 sources beyond what block-1 sources already say. $X = 0$ exactly when the record is *split-causal* at m (Definition 2).

Lemma 1 (L1 — κ is an *identity*, not a bound)**.** $I(T^{b_1}; T^{b_2}) = \kappa$ exactly, *at every split and every n.* Proof. *V is AR(1), hence Markov, so $T^{b_1} \to (V_m, V_{m+1}) \to T^{b_2}$ and both data-processing steps are* equalities*; the surviving term is $I(V_m; V_{m+1}) = \frac{1}{2} \log_2 \frac{1}{1-a^2} = \kappa$. Certificate:* $\max_m |I - \kappa| = 3.8 \times 10^{-14}$ *over $m \in \{3, 4, 6, 8, 12, 16\}$, and the spread over m is 4.1×10^{-14} — the proof predicts a constant, and no drift in m is observed.* □

Lemma 2 (L2 — the cross-read bound, and it is sharp)**.** $I(E; T^{b_2}) \leq \kappa + X$ *and* $I(E'; E) \leq \kappa + X$*.* Proof. *S^{b_1} drops out of the numerator because $U \perp (W, \hat{Y})$: conditionally on W^{b_1} the reference noise carries nothing about T^{b_2}. What is left is $I(W^{b_1}; T^{b_2}) + I(\hat{Y}^{b_1}; W^{b_2} \mid W^{b_1}) = \kappa + X$ by Lemma 1 and Definition 1; the second leg is the same computation with E' in place of T^{b_2}.* □ Sharpness (this is the point). *Over the pinned battery — generic $\mathcal{F}_0$ records, split-causal records, and the two adversarial copy-row classes of Remark 12 — the minimum slack is $+5.67 \times 10^{-7}$, attained* at the two-V-copy counterexample*, i.e. the bound is tight exactly where the universal claim dies. (The second leg's minimum slack is $+0.365$.)*

Lemma 3 (L3 — the block-2 charge). $D_2 \le I(E;T^{b_2}) - I_m$. Proof. *The interleaved chain rule gives the exact identity*

$$D_2 \;=\; I(E;T^{b_2}) - I\big(E;T^{b_2} \mid S',\hat{Y}'\big) - \sum_j I\big(E;S'_j \mid \mathrm{pfx}_j\big),$$

whose first subtracted term is a mutual information (≥ 0)*; the second sum has every term* ≥ 0*, so it is at least its partial sum over* $j \le \Delta+1$*, which telescopes (there* $\mathrm{pfx}_j = S'^{\,1..j-1}$*, no record cell having been erased yet) to* $I(E;S'^{\,1..\Delta+1}) \ge I_m$ *by data processing* ($E \supseteq S^{b_1}$)*. Certificates: identity residual* $\le 9.3\times10^{-11}$*, chain-rule collapse* 6.7×10^{-16}, $\min\big(I(E;S'^{\,1..\Delta+1}) - I_m\big) = +1.47\times10^{-2}$ *over the battery, and the resulting bound holds over the whole battery with minimum slack* $+0.528$*. The zero-claim of Remark 16 enters here and is* $\le 2.5\times10^{-10}$ *on* $\mathcal{F}_0$*.* ☐

Lemma 4 (L4 — the block-1 charge). $D_1 \le I(E';E)$*, and* $D_1 = 0$ exactly *at* $\Delta = 0$ *(the block-1 extension set is empty — measured* 0 *to* 0.0*).* ☐

Theorem 5 (H — the repair inequality; *unconditional on* $\mathcal{F}_0$). For every record of $\mathcal{F}_0$ and every split at m,

$$\boxed{D_1 + D_2 \;\le\; c(\Delta;m) + \big(2 - \mathbf{1}\{\Delta=0\}\big)X} \qquad c(\Delta;m) = \big(2-\mathbf{1}\{\Delta=0\}\big)\kappa - I_m,$$

$I_m = I(S^{b_1};S'^{\,1..\Delta+1})$: add Lemmas 3 and 4, and bound each right-hand side by Lemma 2. *No optimizer structure is used anywhere in the chain* — it is a deterministic, analytic statement about the family. Over the pinned battery (22 cells, *including both refuting counterexamples*) there are zero violations, minimum slack $+0.768$. The index of c is the *block* length; since I is monotone increasing (Remark 15) the base value is the supremum along the dyadic chain, and numerically $c(\Delta;8) = c(\Delta;32)$ to 2×10^{-8} — the constants of v0.5 are unchanged.

Definition 2 (The split-causal sub-family). $\mathcal{F}_0^{\mathrm{sc}}(m) \subset \mathcal{F}_0$ is the set of records whose block-1 cells read no block-2 source; in moment coordinates this is the *linear* condition

$$H[b_1,\mathrm{idx}_2] \;=\; H[b_1,\mathrm{idx}_1]\,\Sigma_{11}^{-1}\Sigma_{12},$$

idx_i the source indices of block i and $\Sigma = \Sigma_W$. Equivalently $X = 0$.

Corollary 3 ((H*) is a *theorem* on $\mathcal{F}_0^{\mathrm{sc}}(m)$, with the same constants). *On* $\mathcal{F}_0^{\mathrm{sc}}(m)$*,* $X = 0$ *and hence* $D_1 + D_2 \le c(\Delta;m)$ *— exactly the inequality (H*) asserts, now proved, with this document's own constants. Numerically:* X *vanishes to* 1.0×10^{-14} *on projected records and the minimum slack of* $D_1 + D_2 \le c(\Delta;m)$ *over them is* $+0.768$*. Moreover* $\mathcal{F}_0^{\mathrm{sc}}(m)$ *is a* convex linear section *of* $\mathcal{D}$ *(moment midpoints of split-causal records stay split-causal: residual* 1.1×10^{-16}*, midpoint* $X \le 5.1\times10^{-15}$*), so the Lagrangian machinery of Theorem 2 restricted to the section certifies* ϕ^{sc} *two-sided exactly as it does for the diagonal section (Remark 5).*

(n,Δ)	certified bracket for ϕ^{sc}_n	split-causality price $n\,\delta_n$ (bits)
$(16,0)$	[0.5669667555, 0.5670144753]	0.004101 (≤ 0.004165)
$(24,0)$	[0.5655717237, 0.5655854048]	0.004117 (≤ 0.004207)
$(32,0)$	[0.5647601440, 0.5648697385]	0.004089 (≤ 0.004382)
$(16,1)$	[0.5414842536, 0.5414911968]	0.010954 (≤ 0.011002)
$(16,2)$	[0.5367829566, 0.5367912904]	0.014358 (≤ 0.014408)

Here $\delta_n = \phi_n^{\mathrm{sc}} - \phi_n$ and the parenthesised figure is the certified (LB-to-UB) upper bound on the price. **The price is flat in** n at ≈ 0.0041 bits at $\Delta = 0$ across $n = 16, 24, 32$: split-causality costs a pure *boundary* constant, not a rate — which is what makes it a plausible object for a transfer argument in the first place.

Lemma 5 (Π — the split-causal retraction)**.** *Let* Π *replace the block-1 rows of a record by their* W^{b_1}*-conditional means plus independent noise matched to restore* $\mathrm{Cov}(\hat{Y}^{b_1})$*. Then* Π *maps* $\mathcal{F}_0$ *onto* $\mathcal{F}_0^{\mathrm{sc}}(m)$ *and*

(i) preserves the per-cell distortion *exactly (measured* 1.1×10^{-16}*);*

(ii) preserves the entire block-1 joint law $(W^{b_1}, S^{b_1}, \hat{Y}^{b_1})$ *exactly (*6.7×10^{-16}*), hence every own-window block-1 term;*

(iii) kills the cross-read, $X(\Pi R) \le 2.1 \times 10^{-14}$*;*

(iv) strictly improves *generic records — over the pinned generic battery (*$n \in \{12, 16, 20\}$*, three styles,* $\Delta \in \{0, 2\}$*) the gain* $n\big(L_a(\Pi R) - L_a(R)\big)$ *lies in* $[-2.74, -0.31]$ *bits (prover's own battery:* $[-3.11, -0.37]$*);*

(v) costs only $\approx 5.0 \times 10^{-3}$ *bits* at optimizers *(*$0.005052/0.005044/0.005039/0.005035$ *at* $m = 8/12/16/24$*,* $\Delta = 0$*).*

Π *is what makes* $\mathcal{F}_0^{\mathrm{sc}}$ a retract *and not merely a subset, and (i)–(iii) are what make the certified* ϕ^{sc} *brackets above meaningful.* ***What*** Π **does not** ***do is bound*** X*: at an optimizer its price is* ≥ 0 *automatically, so "price* ≥ 0*" is consistent with any* X *whatever — see Remark 17(b), where that route is closed. The measured price/X ratio is* 0.7960 *at every* n *and every* Δ*.*

Hypothesis 2 ((H**) — the cross-block read at dyadic-chain optimizers; hard, load-bearing)**.** *Fix* Δ *and a base window length* n*. At every optimizer of the dyadic chain* $n \cdot 2^k$*,* $k = 0, 1, 2, \ldots,$ *the cross-block read of the split into halves satisfies* $X \le \bar{X}$*.*

Corollary 4 ((H*) — what v0.5 assumed — is now a *consequence*)**.** *Under (H****), Theorem 5 gives*

$$D_1 + D_2 \;\le\; c(\Delta; n) + \big(2 - \mathbf{1}\{\Delta = 0\}\big)\bar{X} \;=:\; c^*(\Delta; n),$$

which is (H**) with the constant enlarged by the repair term. (H****) is* strictly weaker and better margined*: it constrains* one scalar *rather than a sum of* $2m$ *conditional mutual informations, the measured value at the* $\Delta = 0$ *optimizers is* $X = 6.35 \times 10^{-3}$ *against the largest admissible* $\bar{X} = 32\big(\mathrm{LB}(32, 0) - 0.5479448\big) - c(0) = 0.116930$ *—* $18\times$ *headroom, against (H***)'s* 0.076971 *inside* 0.4202858 *(*$5.5\times$*) — it is verified* monotone decreasing *in* m *(Remark 13), and it is constructively bounded at any* m *by* Π *(Lemma 5).*

Theorem 6 (The split-causal transfer step; *unconditional*)*.* For every m and every Δ,

$$\phi_{2m}^{\mathrm{sc}}(\Delta) \;\le\; \phi_m(\Delta) :$$

concatenate two independent ε-optimal m-window records. The result is feasible on the $2m$-window (the distortion constraint is the cell average) and is *split-causal at the midpoint* — its block-1 rows read no block-2 source whatever — and its value is at most $\phi_m + \varepsilon$ by the per-cell inequality of

Theorem 4(i). Certified form (upper end of ϕ^{sc}_{2m} against lower end of ϕ_m): the inequality holds with certified slack $3.8/2.4/1.9 \times 10^{-3}$ at $(2m, m) = (16, 8)/(24, 12)/(32, 16)$. Applying Corollary 3 to the $2m$-window $\mathcal{F}^{\mathrm{sc}}_0$-optimizer gives $\phi_m - \phi^{\mathrm{sc}}_{2m} \le c(\Delta; m)/2m$, whence

$$\phi_m(\Delta) - \phi_{2m}(\Delta) \;\le\; \frac{c(\Delta; m) + p_{2m}}{2m}, \qquad p_n \;:=\; n\big(\phi^{\mathrm{sc}}_n - \phi_n\big)$$

the *split-causality value gap* — the same quantity as the price column of the ϕ^{sc} table above.

Hypothesis 3 ((H***) — the split-causality value gap along the chain)**.** *$p_m \le \bar{p}$ for every m in the dyadic chain.*

Remark 11 (Three hypotheses, and why (H***) is the one to prefer)*.* Each of (H*), (H**), (H***) suffices for Theorem 7, and they are ordered by how little they assume and — more importantly — by how *checkable* they are. (H**) implies (H*) with an enlarged constant, by Theorem 5 (Corollary 4); it replaces a bound on a sum of $2m$ conditional mutual informations by a bound on one scalar. (H***) replaces that scalar by another, p_m, which reaches the same conclusion through Theorem 6 rather than through Theorem 5 — and p_m **is a gap between two convex programs, so Theorem 2 certifies it two-sided**, whereas X is only ever *measured* at a computed optimizer. That is the decisive difference: a hypothesis about p is falsifiable by certificate at any m one is willing to pay for. Both bars are the same number, $\bar{p},\ \bar{X} < 32\big(\mathrm{LB}(32, 0) - 0.5479448\big) - c(0) = 0.116930$; the certified instances — each the LB-to-UB upper end, so each an honest *bound* on p_n rather than an optimizer reading — are $p_{16} \le 0.004165$, $p_{24} \le 0.004207$, $p_{32} \le 0.004382$: flat in n, and 27× inside the bar, against X's 18×.

Remark 12 (The "$\le \kappa$ per side" refutation, *refined* — what it refutes is split-causality)*.* The natural lemma — that each side's boundary charge is at most the AR(1) boundary information κ, uniformly over the family — is *false*, and its sharpened form $D_i \le \kappa - I_m$ is false too. Explicit D-feasible records in $\mathcal{F}_0$ (no U-coupling whatsoever; the whole budget respected) spend block-1 rows on copies of block-2 cells:

- two V-copy rows give $D_2 = 0.4563 > 0.4203 = \kappa - I_8$: the *sharpened* constant is exceeded by a mild, feasible perturbation;
- three Y-copy rows give $D_2 = 18.93$ bits $\approx 26\kappa$ at distortion 0.2986: the charge is not bounded by κ at all.

Nothing stops the construction from being repeated: the worst case over the family is $\Theta(m)$ in the block length. **No universal constant exists** — that statement stands unchanged.

What has changed is the diagnosis. These are precisely the records with enormous cross-block read: the harness's own witnesses of the same class measure $X = 18.46$ and $X = 29.27$ bits (against 6.3×10^{-3} at the optimizers), and at the two-V-copy witness the bound of Lemma 2 is *tight to* 5.7×10^{-7}. The refutation is exactly the failure of split-causality, and Theorem 5 absorbs it into the term $(2 - \mathbf{1}\{\Delta = 0\})X$ rather than being defeated by it. Hence the v0.5 blanket rule ("this document never prints $\le \kappa$ per side as an $\mathcal{F}_0$ lemma") is *refined*, not repealed: $D_1 + D_2 \le c(\Delta; m)$ is a *theorem* on $\mathcal{F}^{\mathrm{sc}}_0(m)$ (Corollary 3) and is *false* on $\mathcal{F}_0$; neither form may be printed without its scope. What no longer follows is the v0.5 sentence "no proof of the reverse direction can avoid using optimizer structure": the *inequality* of Theorem 5 avoids it entirely; what still needs optimizer structure is a bound on X (Remark 17).

The two constants above are the verifier's witnesses; they are not in the repository, so the harness pins its own analytic records of the same class and measures $D_2 = 0.7749$ for two V-copy rows (distortion 0.2706, $L_a(0) = 4.219$) and $D_2 = 13.94$ bits $\approx 19\kappa$ for three Y-copy rows (distortion 0.2754, $L_a(0) = 4.740$). Same refutation, independent witnesses; the refutation itself is a gated fact of the harness, not a remark. Note the L_a values: these records are nowhere near optimal — more than seven times $\phi_{16} = 0.5668$ — which is why they never threatened the transfer bound in practice, only its proof.

Remark 13 (The monotone $m \times \Delta$ table (the "$m \in \{8, 16\}$ only" caveat, retired)). v0.5 recorded the optimizer verifications at $m \in \{8, 16\}$ only and flagged the gap explicitly. The table is now $m \in \{8, 12, 16, 24\}$ at $\Delta = 0$ and $m \in \{8, 12, 16\}$ at $\Delta = 1, 2$ (cold-start certified full-space optimizers at $n = 2m$), and *every* quantity of the repair is monotone *decreasing* in m:

Δ	m	$D_1 + D_2$	$I(E; T^{b_2})$	X
0	8/12/16/24	0.076971/0.076934/0.076915/0.076896	0.543069/0.543059/0.543054/0.543049	6.3468/6.336
1	8/12/16	0.090533/0.090487/0.090463	0.569542/0.569504/0.569485	1.8013/1.798
2	8/12/16	0.099029/0.098964/0.098931	0.579319/0.579256/0.579225	2.3894/2.384

Monotonicity is what the caveat was about: *the base value dominates the whole chain*, so a hypothesis verified at the base m is not being extrapolated upward along the doubling chain. *Scope of the netting:* the $\Delta = 0$ row is reproduced end-to-end and gated (strict decrease at every step, each step $\geq 10^{-6}$ against a smallest measured step of 4.96×10^{-6}); the $\Delta = 1, 2$ rows are the prover's, carried on its authority and not re-netted, except that the $m = 8$ optimizers at $\Delta = 1, 2$ are recomputed for Theorem 3. Two-point $1/m$ limits at $\Delta = 0$: $X_\infty \approx 6.31 \times 10^{-3}$, $(D_1 + D_2)_\infty \approx 0.076859$, $I(E; T^{b_2})_\infty \approx 0.54304$; the Π-price of Lemma 5(v) is monotone too, limit $\approx 5.03 \times 10^{-3}$.

Two precision notes on the values v0.5 recorded at $m = 16$. (a) The recorded $D_2 = 0.076926$ and $I(E; T^{b_2}) = 0.543057$ came from a *warm-polish* endpoint sitting at $d - D = +6.4 \times 10^{-5}$, i.e. *distortion-infeasible*; the feasible cold-start certified values are 0.0769149 and 0.543054. (b) The difference, 1.1×10^{-5} resp. 3×10^{-6}, is more than two orders inside the 5×10^{-3} / 2×10^{-2} reproduction bands the 080 harness gates these two quantities at — so this is a *precision note, not a bar change*, and the sealed 080 verdicts are untouched. The table above quotes the feasible cold-start values throughout.

Theorem 7 (T* v2 — boundary-charged reverse direction and the two-sided bound, in *two* forms). Fix Δ and a base n, and write $c^*(\Delta; n) = c(\Delta; n) + \big(2 - \mathbf{1}\{\Delta = 0\}\big)\bar{X}$.

(ii-a) *Family form, conditional on Hypothesis 2 along the chain $n \cdot 2^k$:* $\phi_m(\Delta) - \phi_{2m}(\Delta) \leq \dfrac{c^*(\Delta; n)}{2m}$ for every $m = n \cdot 2^k$, and, summing the geometric telescope, $0 \leq \phi_n(\Delta) - L^\infty(\Delta) \leq \dfrac{c^*(\Delta; n)}{n}$, the lower half being Corollary 2 and *unconditional.*

(ii-b) *Split-causal form,* unconditional*:* the same two inequalities hold for the sub-family value $\phi_n^{\mathrm{sc}}(\Delta) = \inf_{\mathcal{F}_0^{\mathrm{sc}}} L_a$ with the constant $c(\Delta; n)$ itself and *without any of* (H*), (H**), (H***), because on $\mathcal{F}_0^{\mathrm{sc}}$ the charge bound is Corollary 3. (It remains under Hypothesis 1, like everything else in this document, and this scoping is never to be restated as freedom from *every* hypothesis — for the plateau in particular, see Remark 30.)

Scope of (ii-b). It bounds $\phi_n^{\mathrm{sc}} - L^{\mathrm{sc},\infty}$, not $\phi_n - L^\infty$; the two coincide iff the split-causality price $\delta_n = \phi_n^{\mathrm{sc}} - \phi_n$ is $o(1/n)$ in the relevant sense. Measured, $n\delta_n$ is *flat* at ≈ 0.0041 bits over $n = 16, 24, 32$ at $\Delta = 0$ (so $\delta_n = \Theta(1/n)$, of the same order as the transfer term itself, not smaller). That is evidence, not a proof, and (ii-b) is therefore *not* a route to the family statement — it is the unconditional theorem about the sub-family, and the honest statement of what the repair buys without (H**).

Exact superadditivity is refuted, so (ii) cannot be sharpened to $c = 0$: the certified anchors are strictly decreasing, $\phi_8 > \phi_{12} > \phi_{16} > \phi_{24} > \phi_{32}$, with the certified transfer anchors $\phi_8(0) \in [0.570790938, 0.5707933842]$ and $\phi_{12}(0) \in [0.568030932, 0.5681028134]$. The bound is not tight either: the measured drift $n\big(\phi_n - L^\infty\big) \approx 0.0645$ at $n = 32$ sits $6.5\times$ inside $c(0) = 0.4203$.

Remark 14 (The per-block feasibility step of (ii) — a separate hole, and its repair)*.* The restriction step behind (ii) needs each half of the $2m$-optimizer to be a *feasible* m-window record, i.e. $\mathrm{own}_i \geq m\,\phi_m(D)$, and v0.5 tacitly assumed each block distortion is $\leq D$. **It is not.** The $2m$-optimizer's block distortions *straddle* D — $d_1 = 0.299486$, $d_2 = 0.300514$ at $n = 16$; 0.299658/0.300342 at $n = 24$; 0.299744/0.300256 at $n = 32$ — with the *mean* exactly D (to 5×10^{-8}). The correct step is

$$\mathrm{own}_1 + \mathrm{own}_2 \;\geq\; m\big[\phi_m(d_1) + \phi_m(d_2)\big] \;\geq\; 2m\,\phi_m(D),$$

the second inequality by *convexity of the value function $D \mapsto \phi_m(D)$*, which Theorem 2 supplies for free (jointly convex objective, affine distortion constraint $\Rightarrow$ convex optimal value in the right-hand side). Certified on the pinned grid $D \in \{0.26, 0.28, 0.30, 0.32, 0.34\}$ at $m = 8$, $\Delta = 0$: at each interior node the certified *upper* end sits below the chord of the two certified *lower* ends, slack $1.90/1.64/1.47 \times 10^{-3}$. Ignoring the straddle would have cost $m\,\mu\,(d_2 - D) = 8 \times 2.2206 \times 5.14 \times 10^{-4} = 9.13 \times 10^{-3}$ bits at $n = 16$ (μ the distortion multiplier at the optimizer) — *larger than the entire repair term $\bar{X}$*, so the hole was not cosmetic. It is independent of (H*) and of everything else in this section.

Remark 15 (The constants, and their n-uniformity)*.* $I_n = I(S^{b_1}; S'^{\,1..\Delta+1})$ is *monotone increasing in n*, so $c(\Delta; n) = (2 - \mathbf{1}\{\Delta = 0\})\kappa - I_n$ is the supremum of the per-step constants along the whole dyadic chain issuing from base n — which is precisely what the telescope in Theorem 7(iii) requires, and why the constant must be built from I_n and not from I_∞. Numerically I_n is converged to I_{32} by $n = 16$ (to $< 10^{-12}$), and

$$I_{32} = 0.3166798\,/\,0.3482018\,/\,0.3520686 \quad \text{at } \Delta = 0/1/2,$$

each equal to I_∞ to below 10^{-15}. Hence

$$c(0) \leq 0.42029, \qquad \boxed{c(1) \leq 1.1258}, \qquad c(2) \leq 1.1219.$$

Erratum: the transfer note recorded $c(1) \leq 1.1257$; the computed value is 1.1257294, so 1.1257 is false in the fifth decimal. Use 1.1258.

The repaired constants (v0.6). With $c^*(\Delta; n) = c(\Delta; n) + (2 - \mathbf{1}\{\Delta = 0\})\bar{X}$ and $\bar{X}$ at the base-m measured values of Remark 13, $c^*(0) = 0.4266326$, $c^*(1) = 1.161755$, $c^*(2) = 1.169651$ — the $\Delta = 0$ constant grows by 1.5%, the others by 3–4%, because X itself grows with Δ and enters doubled there. Only $c^*(0)$ is used in Corollary 5; $\Delta = 1, 2$ remain $O(1/n)$-model-conditional for the reason given there, and the repair does not change that.

Remark 16 (The zero-claim and the marginalization step are $\mathcal{F}_0$-only). Two steps carry the whole cancellation. (a) The *zero-claim* $I(E; S'_j \mid \mathrm{pfx}_j, T^{b_2}) = 0$ makes the interleaved chain rule cancel the record–reference coupling exactly (residual $\leq 2 \times 10^{-14}$ over pinned $\mathcal{F}_0$ records). (b) The *marginalization* step replaces the block-2 sub-record by an effective 8-window record with the same law (deviation 2×10^{-16}; the effective noise is independent of both W^{b_2} and U^{b_2}). **Both are tagged $\mathcal{F}_0$-only.** They fail outside the family, and the failure is not a technicality: with block-1 rows coupled to U^{b_2} the zero-claim is violated by 0.2223 bits, and with the coupling confined to *block-2 rows only* it is still violated by 0.0950 bits — in the latter case the violation is induced purely through the conditioning, with no block-1 record touching U at all. So U-independence is load-bearing here in the same way it is load-bearing in Theorem 1 (Remark 3), and for a third time.

Corollary 5 (The $\Delta = 0$ plateau over the causal-spectral allocation — recomputed with the repaired constant. **HISTORICAL at v0.7**). ***Superseded in full by Corollary 7.*** *The arithmetic of this corollary — both the (H**) number* 0.5514005 *and the (D)+(F) number* 0.5495664*, and both their margins — is* retained only as history*. It is not the plateau statement of this document; §6 derives the plateau with none of (H*), (H**), (H***), (D), (F) or the boundary charge, at margins larger by a factor of* 4.3 *at* $\Delta = 0$ *and at two further lags this route cannot reach at all. Nothing downstream may cite the numbers below.*

Assume Hypothesis 2 at $\Delta = 0$ *with base* $n = 32$*, with* $\bar{X}$ *taken at the measured value* $X = 6.346782 \times 10^{-3}$*. With the sealed 079 certified lower end* $\mathrm{LB}(32, 0) = 0.5647327869$ *(harness* `go14_convexity.py`*, entry* `s5_32_0`*) and* $c^*(0) = c(0) + X = 0.4202858 + 0.0063468 = 0.4266326$*,*

$$L^\infty(0) \;\geq\; 0.5647327869 - \frac{0.4266326}{32} \;=\; 0.5514005 \;>\; 0.5479448,$$

the right-hand side being the causal-spectral allocation at $\Delta = 0$ *(Remark 9). The margin is* $+3.456 \times 10^{-3}$*:* $203\times$ *the* 1.7×10^{-5} *bracket width quoted for* $(32, 0)$ *in the anchor table of §3.2, and* $376\times$ *the* 9.18×10^{-6} *width the sealed harness actually achieved.* This is a statement about the limit itself, not about the $O(1/n)$ finite-window model. *(At v0.5, under (H*), the same arithmetic gave* 0.5515989 *at* $215\times/398\times$*; the repair costs* 1.98×10^{-4} *of margin and buys a hypothesis with* $18\times$ *headroom instead of* $5.5\times$*.)*

Under the decay route instead. *Replacing (H**) by the two model-scale hypotheses (D) and (F) of Remark 17 gives the proved envelope* $\bar{X} = 0.065038$*, hence* $c^*(0) = 0.485324$ *and*

$$L^\infty(0) \;\geq\; 0.5647327869 - \frac{0.485324}{32} \;=\; 0.5495664 \;>\; 0.5479448,$$

margin $+1.622 \times 10^{-3}$ *(*$95\times$ *the quoted bracket,* $177\times$ *the sealed width). Which hypothesis buys which margin is therefore explicit: (H**) — one measured scalar — buys* 3.456×10^{-3}*; (D)+(F) — two structural, m-uniform statements with a* proved *conversion — buys* 1.622×10^{-3}*. Neither is discharged.*

The margin is not slack in the constant, and the base matters. *Running the same arithmetic from base* $n = 24$ *gives* $0.5654101216 - 0.4266326/24 = 0.5476338 < 0.5479448$*: it* fails *by* 3.11×10^{-4} *(and by* 2.757×10^{-3} *under (D)+(F)) — so the enlarged constant makes the no-slack claim* stronger *than at v0.5, where the base-24 shortfall was* 4.7×10^{-5}*. The hypothesis is genuinely load-bearing, and so is the choice of base window.* $\Delta = 1$ *and* $\Delta = 2$ *remain* $O(1/n)$*-model-conditional: their constants are* $2.7\times$ *larger and their plateaus* $3.6/18\times$ *smaller, so a certified anchor at* $n \approx 260$ *resp.* $n \approx 1320$*, or a sharper constant, would be needed.*

Remark 17 (The reduction, executed — and the obstruction, named). v0.5 recorded a *repair path.* It has since been walked, and it ends at a named theorem rather than at a vague hope. Three things are now on record.

(1) The conversion is proved and is m-uniform by construction. Exactly, $X = \frac{1}{2}\log_2 \det\big(I + N_{11}^{-1} A^{12}\Sigma_{2|1}(A^{12})^\top\big)$ with N the record noise covariance and A^{12} the cross-block kernel. Under

(D) *kernel-decay envelopes*: $|A_v[t,s]| \le C_v \lambda^{|t-s|}$, $|A_y[t,s]| \le C_y \lambda^{|t-s|}$ with $\lambda = \lambda_s = 0.354554$, $C_v = 0.1980$, $C_y = 0.1631$ (fitted, and identical to three digits at $n = 24/32/48$), and

(F) a uniform noise floor $\lambda_{\min}(N) \ge \nu = 0.1495$,

one gets a closed-form bound on X whose sums run over infinite half-lines and which is therefore *independent of* m; the conditional covariance $\Sigma_{2|1}$ is used exactly rather than enveloped (its $-a^{j+k+2}$ boundary term is worth 2.6× and is what makes the bound clear the bar). Audited link by link at real optimizers the total looseness is 10.3× ($\log\det \to \operatorname{tr}$ 1.004×, $N_{11} \succeq \nu I$ 1.22×, the enveloped Schur step 8.4×). **Result:** $\bar{X} = 0.065038$ **against the admissible** 0.116930 **— it clears with** 1.80× **margin, uniformly in** m (free-λ optimisation gives 0.0648; insensitive to the envelope-fit window). So (H**) is no longer a bare hypothesis but a *consequence* of (D) and (F), at the arithmetic cost recorded in Corollary 5.

(2) The spectral premise (D) is not the obstruction — it holds numerically, uniformly in m. At $\Delta = 0$, $m = 8/12/16/24$: θ moves only over 3.0785–3.0835, $\operatorname{cond} N$ over 1.4453–1.4492, $\operatorname{cond} M_Q$ over 2.197–2.217, $\lambda_{\min}(J) = 0.2686$ flat, the record pivots $\sigma_t \in [0.3753, 0.4588]$, and $\|A^{12}\|_F$ is *flat in* m ($5.4823 \to 5.4722 \times 10^{-2}$; $\|A^{12}\|_F/\|A^{11}\|_F = 3.44\%$). The measured kernel is the same Toeplitz kernel plus non-moving boundary corrections at every n, and its per-lag ratios *decrease* with lag ($0.284 \to 0.200$ for A_y, $0.377 \to 0.279$ for A_v), so a single geometric envelope at λ_s is conservative.

(3) The obstruction is ***circularity of the operator****, and it is a genuine research step.* By Theorem 3 the only operator whose inverse produces the optimizer is $N = (\theta I + V\Sigma_s^{-1}V^\top)^{-1}$, and $M := V\Sigma_s^{-1}V^\top = N^{-1} - \theta I$ *exactly*: "M decays exponentially" and "N decays exponentially" are the same statement, so a Demko/Jaffard inverse-closedness theorem [28, 29] applied here *returns its own hypothesis.* The model's own decay (Σ_S, Σ_W, $\Sigma_{W|S}$ all exactly $a^{|t-s|}$) enters the KKT system only through the affine data, never through the operator being inverted. What would inject it is a *fixed-point/bootstrap* argument in a Jaffard [29] (or Gröchenig–Klotz weighted [30]) Banach algebra: show that the KKT map sends a ball $\{\|A\|_{\mathcal{A}_\lambda} \le R\}$ into itself and contracts. A second, independent piece is missing too: (F), the uniform lower bound $\lambda_{\min}(N) \ge \nu$. Theorem 3 gives the *upper* side ($N \preceq I/\theta$, proved) but not the lower; every attempted route closes on itself ($\lambda_{\max}(M) \le 1/\lambda_{\min}(M_Q)$ with $M_Q \succeq N$ gives only $1/\nu \le \theta + 1/\nu$), and coercivity fails because a record that does not use a direction pays nothing there.

Routes closed, so that they are not re-attempted. (a) KKT zeroing cannot work: since $\Sigma_W^{-1}\Sigma_{WY} = [0; I]$, the distortion gradient *vanishes identically* on the cross-read coordinates (Theorem 3), so stationarity there carries no multiplier — the cross-read is a distortion-*free* direction on which the objective is stationary, and the optimizer chooses $X > 0$ because it strictly lowers L_a. KKT yields an equation, not a bound. (b) The Π-retraction cannot bound X either: at an optimizer the Π-price is ≥ 0 automatically, and the measured price/X ratio is 0.7960 at every n and every Δ, so "price ≥ 0" is consistent with *any* X. (c) m-induction restates the same missing estimate: X is monotone in m (Remark 13) but the step compares the m- and $2m$-optimizers. *The next route on record is different in kind*: solve the stationary system (K1)–(K3) in the spectral domain with the GO-13 machinery, where it becomes per-frequency scalar equations — rational symbols with poles off the unit circle would give Toeplitz coefficients decaying at exactly the modulus of the nearest pole, which the numerics place at $\lambda_s = a(1-K) = 0.354554$, GO-14's own netted constant — and then transfer stationary $\to$ finite-window by a stability estimate from the strong convexity of Theorem 2.

What this does not *do.* Corollary 5 cannot drop its hypothesis. It trades (H**) for (D)+(F); the plateau survives either way, at $+3.456\times10^{-3}$ resp. $+1.622\times10^{-3}$, and the base-24 control fails either way.

5.2 The within-class ladder, certified (the Remark 5 upgrade, executed)

The diagonal class is a convex *linear* section of $\mathcal{D}$ (Remark 5), so the Lagrangian machinery of Theorem 2 restricted to the section — with a section moment box $R_{\text{box}} = 73/95/136$ at $n = 24/32/48$, verified attained and hence tight — certifies the ladder two-sided. All nine ladder cells and the three block values are now certified *class* values (per cell the tighter of the two independent end-to-end computations):

(n,Δ)	certified within-class bracket	width
$(24,4)$	$[0.5333160113,\ 0.5333170450]$	1.0×10^{-6}
$(24,5)$	$[0.5332969026,\ 0.5332971560]$	2.5×10^{-7}
$(24,6)$	$[0.5332945088,\ 0.5332947270]$	2.2×10^{-7}
$(32,4)$	$[0.5324808853,\ 0.5324814627]$	5.8×10^{-7}
$(32,5)$	$[0.5324606031,\ 0.5324610143]$	4.1×10^{-7}
$(32,6)$	$[0.5324578912,\ 0.5324584264]$	5.4×10^{-7}
$(48,4)$	$[0.5316458819,\ 0.5316462767]$	3.9×10^{-7}
$(48,5)$	$[0.5316244073,\ 0.5316250208]$	6.1×10^{-7}
$(48,6)$	$[0.5316217661,\ 0.5316233575]$	1.6×10^{-6}
block_{24}	$[0.5332936495,\ 0.5332949450]$	1.3×10^{-6}
block_{32}	$[0.5324577110,\ 0.5324581027]$	3.9×10^{-7}
block_{48}	$[0.5316214213,\ 0.5316220475]$	6.3×10^{-7}

The $(48,6)$ cell is the loosest (its feasible-projection correction is 1.1×10^{-6}, still inside the bracket), which is the same cell the raw-anchor demotion of Remark 8 concerns.

What this does *not* certify. The excess $E_\infty(\Delta)$ is a *two-point* $1/n$ *extrapolation* of these class values against the block values, so the re-assembled E_∞ intervals are *$1/n$-model-conditional, not certificates*, and are tagged as such: $E_\infty(4) \in [2.50, 2.75]\times10^{-5}$, confirming the 2.67×10^{-5} of Conjecture 1 in bar; $E_\infty(5) \in [1.7, 4.8]\times10^{-6}$, sign resolved positive by the tighter re-assembly and *unresolved* by the looser one; and $E_\infty(6) \in [-3.2, 1.5]\times10^{-6}$, whose bracket *contains zero* at every achievable width. The last of these independently re-ratifies the R-IND-5 demotion of $E_\infty(6)$ to the raw anchor $L_a^{\text{diag}}(48,6) = 0.5316222730$ (Remark 8). No claim is made here about how many optimizer endpoints were true class minima; the transfer note's "4/9 endpoints" remark is *withdrawn* as unverifiable from the record.

6 The plateau: Collapse, periodization, shift-averaging, and a two-sided certificate for Ψ

Status: spectral prover + Ψ-bracket prover + (R1) prover + three independent R-IND-5 passes (the bypass pass, ten restatements; the certificate pass, ten further restatements R11–R20 — fresh context, own spectra, own solver, own gradient, own box, own dual, structurally unrelated cell-argmin fixed point rather than the K1–K3 filter recursion; and the (R1) pass, nine further restatements R21–R29, again fresh context, with its own CMI-definition evaluator built from the primitives (V, N, U, Z) and no prover code reused for any rate number). All twenty-nine restatements are in force in this revision. Harnesses `experiments/go14_plateau.py` (registration 082 pending) and `experiments/go14_r1.py` (registration 083 pending, *structurally independent* of 082) — **nothing in this section is sealed**.

The stationary frame, and its order. Write $R_u := \hat{Y}_{u-\Delta-1}$, and read the blocked process in the within-block order $R_1, S_1, \ldots, R_n, S_n$; **the frame and the order are one choice, not two** (Remark 21). In this shifted frame the interleaved order of the coordinate becomes the plain simultaneous order of (R, S), and Δ **enters only as the phase** $z^{\Delta+1}$ **on the record's cross spectra** — that is the entire content of the lag coordinate at stationarity. Per frequency put

$$h(\omega) = \big(\Phi_{RV}, \Phi_{RY}\big), \quad \Gamma(\omega) = \Phi_R, \quad P = \Sigma_W(\omega)^{-1}, \quad Q = \mathrm{diag}(1/f_S, 0),$$

$n(\omega) = \Gamma - hPh^* \geq 0$ the cone, $M_Q = \Gamma - hQh^*$, and $R = P - Q$. The reduction of Q to the rank-one diagonal $\mathrm{diag}(1/f_S, 0)$ is *exact* and is Theorem 3's first structural fact $\Sigma_W^{-1}\Sigma_{WS} = e_1$ (re-verified to 1.5×10^{-15}); $R \succ 0$ in closed form, $\det R = (1/f_V - 1/f_S)/s_N^2 > 0$ (verified to 5.3×10^{-15}; and by Remark 28 $\lambda_{\min}(R(\omega)) = 3.1715534 \times 10^{-3}$ *uniformly* in ω and in the block size $n = 1, \ldots, 8$, not merely on this grid). Write $\Psi(D; \Delta)$ for the value of the stationary program in these coordinates and $\alpha = 1/(2\ln 2)$.

6.1 The Collapse identity

Theorem 8 (Collapse — exact, unconditional). For every n, every nondecreasing access schedule and every record of $\mathcal{F}_0$,

$$\boxed{2\ln 2\; n\, L_a \;=\; \sum_{t=1}^{n} \ln \sigma_t \;-\; \ln\det N} \qquad \sigma_t = \mathrm{Var}\big(\hat{Y}_t \mid S^{se(t)}, \hat{Y}^{t-1}\big),$$

$N = \mathrm{Cov}(Z)$. *Proof.* Conditioning on W reduces $\hat{Y}_t$ to Z_t and $S^{se(t)}$ to $U^{se(t)} \perp Z$, so every denominator of the coordinate is $\mathrm{Var}(Z_t \mid Z^{t-1})$ and the denominator sum telescopes to $\ln\det N$.$\square$ *Certificates:* relative residual $\leq 1.2 \times 10^{-14}$ at all ten certified optimizers ($n = 16/24/32/48 \times \Delta = 0/1/2$); independent R-IND-5 evaluator 5.6×10^{-15} over 84 cells, 1.7×10^{-15} on random non-staircase nondecreasing schedules, 8.2×10^{-16} at the edges; harness 5.65×10^{-15} and 1.73×10^{-15} against a 10^{-12} gate.

The identity re-derives (K1)–(K3) in five lines and makes step (A) below a one-liner.

Remark 18 (R6 — what Collapse actually needs, and the non-conflation rule). Collapse does **not** require U-independence. It requires exactly Δ-*lag-causal* U-coupling, $A_u[t, s] = 0$ for $s > t + \Delta$; Hypothesis 1 ($A_u = 0$) is the special case. Counter-values on record: at $\Delta = 0$ a dense A_u puts the identity 0.2018 bits off and a strictly anticausal A_u 0.3329 bits off, while a Δ-lag-causal A_u is *exactly* 0 (harness: ≥ 0.1485 bits and 6.7×10^{-16} respectively, gated in both directions). **The non-conflation rule of Remark 3 still applies**: this extension is a statement about the *record-space* coordinate and does *not* extend the moment-form Theorem 1, whose measured residuals at Δ-causal couplings are 2.4–3.5 bits. The two extension results must not be conflated.

6.2 Step (A): periodization

Hypothesis 4 ((IC) — independent noise copies; a hypothesis of the *construction*, load-bearing)**.** *In the periodization of Theorem 9 the noise blocks of distinct translates are* independent*: the tiled noise covariance is block diagonal,* $N_{\text{tiled}} = \bigoplus_b N$.

Theorem 9 (A — periodization; Ω_n/n subadditive)*.* Repeat any n-record along $\mathbb{Z}$ under Hypothesis 4. Then (i) the per-symbol distortion is *exactly* preserved; (ii) $\ln\det N$ is *exactly* additive; (iii) *every* σ_t decreases, because the block-b translate's conditioning set is a genuine *subset* of the tiled cell's — $bn + \min(i+\Delta, n) = \min(bn+i+\Delta,\, bn+n) \leq \min(bn+i+\Delta, M)$. By Theorem 8 the tiled per-symbol value is therefore no larger, so

$$L^{\text{process}}(D) \;\leq\; \phi_n(\Delta) \qquad \text{for every } n.$$

This is Theorem 4 in one line instead of a page, and it carries *no* boundary charge because it runs the easy way. *Certificates:* distortion preserved $\leq 4.4 \times 10^{-16}$, $\ln\det N$ additive $\leq 7.1 \times 10^{-15}$, and every σ_t decreases over 1,200 (cell $\times$ tiling) pairs with *zero* violations — the suspected failure mode (a merely *different* rather than *larger* conditioning set) does not occur, for the displayed reason. Measured boundary charge $0.04205/n$, flat to five digits at the certified optimizers.

Remark 19 (R7 — (IC) is load-bearing, not decorative)*.* Correlating the noise copies *breaks* Theorem 9 outright. At cross-block noise correlation 0.3/0.6/0.9 the tiled per-symbol value is 0.7987/0.9218/1.3977 against $\phi_5 = 0.7757$; the harness's own witnesses exceed the untiled value by $+0.0290/+0.1605/+0.6463$ on a random $n = 5$ record and by $+0.0319/+0.1684/+0.6836$ over ϕ_8 at the cold-start $(8, 0)$ optimizer. So (IC) must be stated as a hypothesis of the construction and never dropped in restatement. **What (IC) is, exactly:** a *specification of an object this document constructs*, not an assumption about anything unknown — the tiling is ours to define, and we define it with independent copies. (R7) therefore survives as a *wording rule* — the independent copies must be written down and never left implicit — rather than as a live hypothesis of the chain (Corollary 6, scope item (b)).

6.3 Step (B): shift-averaging

Theorem 10 (R1 — joint convexity of the process-rate functional on the cyclostationary class)*.* Fix a period n, a lag Δ — indeed any n-periodic nondecreasing schedule — and constants $\varepsilon > 0$, $M < \infty$. Let $\mathcal{L}_n$ be the *linear* space of period-n bi-infinite moment kernel pairs (h, Γ) and put

$$\mathcal{K}_n(\varepsilon, M) \;=\; \big\{(h,\Gamma) \in \mathcal{L}_n : \;\Gamma \preceq MI, \quad n(\omega) := \Gamma - hPh^* \succeq \varepsilon I \text{ a.e.}\big\}.$$

Then $\mathcal{K}_n(\varepsilon, M)$ is convex — $\Gamma \preceq MI$ is linear and $n(\cdot)$ is matrix-*concave*, so $\{n \succeq \varepsilon I\}$ is convex — and **it carries no window length anywhere**. On it the process-rate functional is finite and *jointly convex* in (h, Γ).

Proof, in four steps.

(0) Collapse and the shifted frame. By Theorem 8 the coordinate is a sum of log conditional variances of the stationary pair (R, S). In the shifted frame the lag enters only as the unimodular phase $z^{\Delta+1}$ on the record's cross spectra, i.e. as the map $h \mapsto z^{\Delta+1}h$: a *linear invertible change of coordinates* of $\mathcal{L}_n$ which maps $\mathcal{K}_n$ *onto* $\mathcal{K}_n$, because $n = \Gamma - hPh^*$ is unchanged by $h \mapsto z^k h$. Convexity of the objective is therefore Δ-independent. (This is a statement about the *objective*; the shift does move the distortion functional, so it is not a claim that the three lags share a feasible set.)

(1) Blocking. Blocking by n turns the period-n cyclostationary pair into a *stationary* $2n$*-variate* process (R_b, S_b) — a *known evaluation device*, not this program's: the cyclostationary distortion-rate function is obtained this way, by orthogonalising over the polyphase decomposition, in [1], and the same device runs through [2, 3, 4]; no novelty is claimed for it. In the within-block order fixed by Remark 21 the Cholesky factor of the one-step block innovation covariance Λ reproduces the process pivots exactly, so

$$\sum_i \ln \sigma_i \;=\; \ln\det\Lambda \;-\; \sum_i \ln s_i, \qquad s_u = \mathrm{Var}\big(S_u \mid S^{u-1}, R^u\big),$$

the same s as in Lemma 6, read over a full period (Remark 27).

(2) Matrix Szegő (Wiener–Masani) — the only analytic input. This is the classical multivariate prediction theory of [14, 15, 16], cited here *with* its non-degeneracy hypotheses (Remark 22) and claimed by nobody here. Applied to the blocked process it gives the identity the whole proof rests on,

$$2\ln 2\; n\;\mathrm{rate} \;=\; \underbrace{\big[\langle \ln\det M_Q\rangle - \langle \ln\det n\rangle\big]}_{\text{(I)}} \;+\; \underbrace{\langle \ln\det \Phi_S\rangle}_{\text{constant}} \;-\; \underbrace{\sum_i \ln s_i}_{\text{(III)}}\,.$$

The theorem is cited *with* its hypotheses, and they are verified on $\mathcal{K}_n$: $\Phi_Z = n \succeq \varepsilon I$, $M_Q = n + hRh^* \succeq n$ and $\Phi_S \succeq \tau^2 I$ are nonsingular a.e., and $\Gamma \preceq MI$ together with $\Phi_S \preceq \frac{1+a}{1-a} + \tau^2$ puts every $\ln\det$ in L^1. See Remark 22 for what the ε floor is and is *not* doing here.

(3a) (I) is convex. Per frequency, $\langle \ln\det M_Q\rangle - \langle \ln\det n\rangle = -\ln\det(I - Z)$ with $Z = R^{1/2}h^* M_Q^{-1} h R^{1/2}$. Every step here is *textbook convex analysis* and nothing in it is new: the matrix quadratic-over-linear map $(h, T) \mapsto h^* T^{-1} h$ is jointly matrix-convex and Loewner-*nonincreasing* in its denominator [5, Sec. 3.1.7], the composition rule that follows is [5, Sec. 3.2.4], and the operator-concavity ingredient is Ando [6]; $M_Q = \Gamma - hQh^*$ is jointly matrix-*concave* — this, and only this, is where $Q \succeq 0$ is used (Remark 32); so the composition makes Z matrix-convex, and $-\ln\det(I - \cdot)$ is convex and nondecreasing on $0 \preceq Z \prec I$. This is Lemma 7 in its matrix form, i.e. the 074 block Schur lift of Theorem 2(i) — which is itself the block lift of the standard matrix-fractional certificate [5, Sec. 3.1.7 and 3.2.4],[6], and carries that attribution here too (GO-P-2026-074, attribution amendment dated 2026-08-07): *the lift into the moment chart is the program's contribution, not the convexity.*

(3b) (III) is convex — the load-bearing step of this route. **The mechanism is classical and no novelty is claimed for it**: that a prediction-error variance is concave in the spectral density *precisely because* it is an infimum, over frozen predictors, of spectrum-linear functionals is the standard device of minimax-robust prediction [7, 8]. What is this document's is only the *packaging* — the causally conditioned cross-record instance $\mathrm{Var}(S_u \mid S^{u-1}, R^u)$ read as an infimum of functionals affine *jointly* in (h, Γ), with the $\mathcal{F}_0$ restriction making the minorant linear. Each frozen *admissible* predictor (Definition 4) gives a minorant $\hat{s}$ that is exactly *affine* in (H, Γ) —

linear because $A_u = 0$, which is where $\mathcal{F}_0$ enters and why Remark 24 matters — and satisfies $\hat{s} \geq s$ always, with equality at that record's own optimum. Hence s_i is an *infimum of affine functions*, therefore concave; it is bounded in $[\tau^2, \mathrm{Var}(S)]$; and $-\ln$ of a positive concave function is convex. **This is exactly Definition 4 together with Remark 33, which this document already contained but used only inside the Ψ certificate.** The two halves of the argument must be kept apart (Remark 23).

Conclusion. (I) is convex, the middle term is a constant, and $-$(III) is convex; the distortion is affine and $\mathcal{K}_n(\varepsilon, M)$ is convex. Hence the process-rate functional is jointly convex there. ☐

Theorem 11 (Scoping Lemma — the class restriction is automatic). Step (B) is only ever applied to step-(A) periodizations with independent noise copies (Hypothesis 4). The blocked noise spectrum of such a record is *constant* in ω; its k-shift conjugates that spectrum by a block cyclic shift $U_k(\theta)$ that is *unitary* on $|z| = 1$ (Remark 25 fixes the side and the sign), so the eigenvalues are the same at every frequency and for every k. One pair (ε, M) therefore serves the whole shift orbit *and its convex hull* ($\lambda_{\min}$ is concave, $\lambda_{\max}$ convex). Finally $n \succ 0$ is w.l.o.g.: the program is convex and Slater's condition holds, so the infimum over the closed convex set equals the infimum over its relative interior — and the rate in fact *blows up* at the cone boundary, so the boundary is never active on a minimisation. *Certificates:* blocked noise spectrum constant in ω to 0.0×10^0; $U_k U_k^* = I$ to 2.3×10^{-16}; eigenvalue invariance 5.6×10^{-16} at all ω and all k; hull floor slack $+0.091$ to $+0.69$ with the Γ cap respected; the rate rises $0.383 \to 11.224$ as $\lambda_{\min}(n)$ falls $10^{-1} \to 10^{-8}$.

Certificates for Theorem 10. Step (1): the sequence pivots reproduce (σ_t, s_j) in both frames to 1.7×10^{-15}, $\sum$ block log-pivots $= \ln\det\Lambda$ to 1.8×10^{-15}, and the identity itself to 1.3×10^{-15} over 24 $(n, \Delta, \text{frame})$ cells (prover: 1.8×10^{-15}; verifier: 1.33×10^{-15} over 24 cells). Step (2): via an exact aliasing route with exact finite-support blocked transfer functions — *zero* block-lag truncation — the full decomposition holds to 3.6×10^{-15} at $N_f = 4096$ over $n \in \{1, 2, 3, 4, 6\} \times \Delta \in \{0, 1, 2\}$, *including* $n = 2$; the prover's coarser 1.8×10^{-10} was its own DMAX, and the chord sweep decays geometrically ($1.8 \times 10^{-3} \to 1.6 \times 10^{-13}$ at DMAX $2 \to 28$). Step (3a): 4,200 prover chords at blocks $n = 1..8$ with Γ pushed to 1.9×10^{-14} of the cone boundary, zero violations. Step (3b): 900 verifier and 1,080 prover *per-instance* certificates with all three links separate (affineness 4.4×10^{-16}; $\hat{s}(\text{mid}) = s(\text{mid})$ 4.4×10^{-16}; $\hat{s} \geq s$ at both endpoints with minimum slack $+9.9 \times 10^{-6}$), concavity margin $+1.909 \times 10^{-5}$, and $\min s = 0.421289$ against the $\tau^2 = 0.4$ floor. *Control:* a predictor peeking two slots ahead is inadmissible and breaks the minorant 240/240 (verifier; prover 286/288), worst -0.293 — the test has power. Conclusion: zero Jensen violations over 1,944 verifier chords (six adversarial families: $\lambda_{\min}(n) \sim 10^{-6}$, a 10^{-3} notch, sign-alternating kernels, near-deterministic and large-gain records), zero over 2,160 second differences, zero over 2,352 prover chords, 576 shift-averaging orbits, 126 cross-period chords and 60 non-staircase schedules; window-length independence flat to 10^{-15} from $B = 16$.

Remark 20 (Attribution and novelty for Theorem 10). Theorem 10 is **four standard ingredients, combined**, and novelty is disclaimed on *each* of them in the sentence that introduces it above: (a) blocking a cyclostationary pair to a stationary multivariate one and evaluating spectrally is a known evaluation device [1, 2, 3, 4]; (b) the matrix Szegő / Wiener–Masani theorem is classical [14, 15, 16], and is cited with its non-degeneracy hypotheses (Remark 22); (c) matrix quadratic-over-linear composed with $-\ln\det$ is textbook convex analysis [5, Sec. 3.1.7 and 3.2.4],[6] — and so, therefore, is the 074 lemma (GO-13 Theorem 5) that step (3a) invokes; (d) inf-of-affine concavity of a prediction error is the classical mechanism of minimax-robust prediction [7, 8]. **No novelty attaches to (c) or to the mechanism in (d)**; what is claimed for (d) is the packaging and the coordinates only.

The only claim made. To our knowledge the resulting *joint convexity in spectral moment coordinates* — of a causally conditioned, lagged-reference process rate, on a window-free convex set — has not been stated. **The sweep's caveats travel with that sentence and must be printed with it**: the (R1) sweep of 2026-08-07 was *metadata-only* (title and abstract; the arXiv API returned HTTP 429 throughout, forcing the HTML search endpoint), it read *no full text*, and *no Semantic Scholar and no general-web/Scholar pass was run* (the search budget was already exhausted when it started). The nulls in channel (d) are additionally an artefact of the date range — that literature is 1983–85, before arXiv — and are *not* quoted here as a null set.

The genuinely defensible point. **The moment chart is load-bearing.** The same functional is *non-convex in record coordinates* — 1206/3960 negative second differences, with a directed search reaching -23.4 — so **the choice of chart, not the convexity machinery, is the contribution**.

Adjacent work, and how this differs. The existing convexity results for rate-distortion-type functionals convexify in the *estimation-error covariance* [9, 10, 11] or in the *causal filter coefficients* in the time domain [12]; the Charalambous/Stavrou NRDF line convexifies in the causal *reproduction kernel.* Theorem 10 convexifies in (h, Γ) — cross-spectrum *and* record spectrum. The closest single item is [13], which *evaluates* the causally conditioned directed information rate of Gaussian sequences, i.e. the same species of object; its abstract states no convexity result.

Two items are ***owed****, and no "first" phrasing is used until they are discharged*: (i) page-verification of [13] for a convexity statement; and (ii) a re-run of the sweep's channels 1, 3 and 4 through Semantic Scholar and a full-text engine — the 1983–85 robust-prediction layer and any journal-only convexity result are invisible to an arXiv-abstract sweep. The attributions [7] and [8] are metadata-confirmed but not page-verified.

Remark 21 (R21 — the frame and the within-block order must be printed *as a pair*). **This document adopts the frame $R_u := \hat{Y}_{u-\Delta-1}$ together with the within-block order $R_1, S_1, \ldots, R_n, S_n$**, and these two choices travel together in every statement above: it is this pair that produces the phase $z^{\Delta+1}$ and Lemma 6's $s = \mathrm{Var}(S_u \mid S^{u-1}, R^u)$. The other pairing — $R_u := \hat{Y}_{u-\Delta}$ with the order $S_1, R_1, \ldots, S_n, R_n$ — is *equally valid* and gives the same value. **Mixing them silently returns the lag-$(\Delta+1)$ rate**: reading this document's frame in the other pairing's order is wrong by up to 7.64×10^{-2} bits (measured; it reproduces the lag-$(\Delta+1)$ value to 3.3×10^{-16}, and the smallest cell error is 1.5×10^{-3}, so the mistake is not uniformly loud). This is the one place in the plateau where a silent wrong theorem is available. Accordingly the leak pivot is written with R^u — including lag 0 — everywhere, matching Lemma 6; the finite-window $s_j = \mathrm{Var}(S_j \mid S^{j-1}, \hat{Y}^{k(j)})$ of Theorem 1 is the same object under the schedule-general count $k(j)$, and off-by-one variants of k are detected at 0.1–0.7 bits (Theorem 1).

Remark 22 (R22 — the ε floor is *not* a hypothesis of the Szegő step). Wiener–Masani [14, 15, 16] needs only $\Phi \succeq 0$ with $\ln\det\Phi \in L^1$; it does *not* need a spectral floor. Verified on Szegő's own extremal family $f = |1 - ce^{-i\omega}|^2$ up to and *including* $c = 1$, where the spectrum *vanishes* at a frequency: residual $\leq 2.7 \times 10^{-14}$. **Do not write that the analytic input requires the floor.** What $\varepsilon > 0$ actually does here is three bookkeeping jobs: it keeps $\mathcal{K}_n$ convex and closed, it keeps the functional finite, and it gives the Scoping Lemma 11 a single class for the whole shift orbit. (Separately, the *finite-lag evaluator* for σ and s converges slowly on deep narrow notches — that is the evaluator caveat of Lemma 6, not a property of the lemma.)

Remark 23 (R23 — the s-lemma is two statements, and only the second one has a topology in it). (i) *An infimum of affine functions over any index set whatsoever is concave.* No density, no continuity, no topology, no completeness. (ii) *The closure argument is needed only to identify that infimum with s*: the finite-support causal predictors are dense in the closed span of the past, and $w \mapsto \hat{s}_w$ is a continuous quadratic in the L^2(past) norm, so the infimum over them is s. Truncating the family gives a *different* function which is still concave (0/180 violations at truncation depths 4 and 12), and the truncations decrease to s monotonically and geometrically (machine zero by depth 32). Stating (i) and (ii) in one sentence invites the reader to think concavity is at risk; it is not.

Remark 24 (R25 — outside $\mathcal{F}_0$ the moment coordinates are not even well-posed). A_u is invisible to $H = \mathrm{Cov}(\hat{Y}, W)$, and $\tau^2 A_u A_u^\top$ can be absorbed into $\mathrm{Cov}(Z)$; so outside $\mathcal{F}_0$ two records with *identical* (H, Γ) — held identical to 2.2×10^{-16} — have rates differing by **at least** 0.136 **bits** (the R-IND-5 verifier's random battery reproduces 0.1383; this document's pinned deterministic A_u-ladder reaches 1.635). *Print this as a lower bound, never as a maximum.* Consequently "convex in the moment coordinates" is not a well-posed phrase outside $\mathcal{F}_0$: Hypothesis 1 is here a *well-posedness condition on the coordinates*, a sharper form of the non-conflation rule of Remark 3. The exact place it enters is the *leak leg* $\mathrm{Cov}(R, S) = H\Sigma_W^{-1}\Sigma_{WS}$, which is what makes the joint J affine in (H, Γ) in step (3b).

Remark 25 (R26 — the conjugation has a side and a sign). In Lemma 11 the correct statement is

$$\Phi^{(k)}(\theta) \;=\; U_k(\theta)^*\, N\, U_k(\theta),$$

with the *wrapped* entries of U_k carrying $e^{+i\theta}$ (residual 1.2×10^{-16}). **The other three sign/side combinations are false by $O(1)$:** $U_k N U_k^*$ with $e^{+i\theta}$ and with $e^{-i\theta}$ both miss by 4.12×10^{-1}, and $U_k^* N U_k$ with $e^{-i\theta}$ by 2.59×10^{-1}. The statement is only useful if the convention is printed with it.

Remark 26 (R27 — the `np.roll` defect, scoped). Earlier tooling shifted a periodized record by `np.roll` of the window matrices. That is *not* a shift: Σ_V is Toeplitz, not circulant, so a rolled record's derived noise is not the rolled noise. **Scope: it is a *moment-coordinate* defect** — 27/27 rolled (H, Γ) pairs leave the cone — whereas in *record* coordinates the same roll is wrong by only $\leq 4.3 \times 10^{-9}$ (this document's re-run: 2.8×10^{-9}), so pre-fix record-space numerics are neither over- nor under-condemned. With genuine shifts of the bi-infinite period-n record, rate shift-invariance holds to 3.33×10^{-16} — better than the 10^{-12} first claimed, and it did not hold at all before the fix.

Remark 27 (R28 — the legs are individually well defined only at full period). In the decomposition of step (2) the two legs are individually well defined only when read over a *full* period: $\sum_j \ln s_j$ is offset-invariant to 8.9×10^{-16} over a full period and *not* over sub-periods. Sub-period reads of either leg are not quantities of this document.

Remark 28 (R29 — the n-uniform spectral floor, which this document previously declined to claim). At the *blocked spectral* level

$$\lambda_{\min}\big(R(\omega)\big) \;=\; 3.1715534 \times 10^{-3} \qquad \text{uniformly in } \omega \text{ and in } n = 1, \ldots, 8$$

(spread over n: 3.6×10^{-15}), and $Q(\omega) = \mathrm{blockdiag}\big(\Phi_S(\omega)^{-1}, 0\big)$ *exactly* at every block size ($\leq 1.4 \times 10^{-14}$), with $\lambda_{\max}(Z) < 1$ strictly on the class interior (measured 0.9154). **This supplies the n-uniform floor that the finite-window certificate block of §3 explicitly declined to claim**, and it explains the recorded decrease of $\mathrm{eigmin}(P - Q) = 1.038 \times 10^{-2}/5.594 \times 10^{-3}/4.431 \times 10^{-3}/3.953 \times 10^{-3}$ at $n = 8/16/24/32$ as *convergence to it from above* — monotone, and still

7.8×10^{-4} above the floor at $n = 32$. The finite-window sequence is therefore an approximation of the blocked quantity, not evidence against a floor.

Remark 29 (The three v0.7 justification steps, demoted — and why the new route needs none of them). v0.7 justified (R1) by three steps. They are recorded here, with verdicts, and *the proof above uses none of them. (i) "F_T is convex on the linear section cut out by the cyclostationary class"* — **true but useless**: the right object carries no window length at all, which is why Theorem 10 is stated on $\mathcal{K}_n(\varepsilon, M)$. *(ii) "$F_T/T \to$ rate pointwise"* — **a genuine gap**. F_T is a *different functional*: it carries a truncated schedule, no pre-window past, and a truncated W; its convergence to the process rate needs its own boundary-charge argument, which this campaign does not have. It is *measured* true ($T(F_T/T-\text{rate})$ flat to five or six digits from $T = 2n$) — but measured is not proved, and **the route above does not use it**. *(iii) "a pointwise limit of convex functions is convex"* — **ill-posed as worded**, and it breaks exactly where one would expect: the domain D_T *moves with* T, so the functions are not functions on a common space. It is repairable by pulling back along the linear restrictions, but only *given* (ii). The new route eliminates both.

Theorem 12 (B — shift-averaging produces an explicit stationary record). Let x be a period-n record on $\mathbb{Z}$ and let $\bar{x}$ be the shift-average of its moment functions over the n shifts ($\bar{A}$ the diagonal average of A, $\bar{\Gamma}$ the period average, the noise spectrum explicit). Then $\bar{x}$ is

- (i) *in the cone*: the residual noise spectrum $\bar{n}(\omega)$ is nonnegative, with a named reason — the kernel-dispersion term is a variance, hence PSD, and it is exactly what makes the inequality strict;
- (ii) *stationary*, and *realisable*: the record is constructed, not merely asserted;
- (iii) of *identical per-symbol distortion*; and
- (iv) of *no larger rate*, by Theorem 10: $\text{value}(\bar{x}) \le \overline{\text{value}(x)} = \text{value}(x)$.

Step (B) is now unconditional, and the class restriction it needs is free: the n shifts have equal rate, the average lies in $\mathcal{K}_n(\varepsilon, M)$ by Lemma 11, is stationary, has identical per-symbol distortion, and has rate at most the average, which is the original. **Wording.** Step (B) *delivers an explicit stationary record of no larger rate and identical per-symbol distortion.* It does *not* say that the process infimum is attained: attainment is never invoked, the infimum comparison follows by definition, and the lower-semicontinuity "soft spot" recorded in the probe is a red herring. *Certificates:* executed at the certified optimizers, the constructed stationary records evaluate to 0.5631757405/0.5629264128/0.5628390620 at $n = 8/12/16$, $\Delta = 0$ (harness, cold-start: 0.5631757430/0.5629263942/0.5628390182), distortion preserved to machine precision, converging from *above* like C/n^2; 8/8 adversarial edge cases (near-deterministic noise, sign-alternating kernels, rank-2 noise, strictly anticausal A_v) hold with cone margin ≥ 0.2. There is no schedule leak: on $\mathbb{Z}$ the schedule *is* shift-invariant, and the finite-n truncation appears only on the $\le$ side of Theorem 9. At $\Delta = 1, 2$ the same construction gives 0.5364993215 and 0.5311645087, reproduced by the harness to ten digits.

Definition 3 (The class $\mathcal{P}(D; \Delta)$ — written down once, and used in that same form in three places (F10))**.** $\Psi(D; \Delta)$ is the infimum of the stationary rate over $\mathcal{P}(D; \Delta)$, the class of *feasible stationary records of* $\mathcal{F}_0$: records x on $\mathbb{Z}$ that are stationary, lie in $\mathcal{F}_0$ (Hypothesis 1), have residual noise spectrum $n(\omega) > 0$ almost everywhere and $\Phi_R \in L^1$, and satisfy the per-symbol distortion constraint $\mathrm{dist}(x) \le D$. *No further restriction is imposed* — no spectral floor, no moment box, no rationality, no kernel-summability and no window length. **This is the class in Corollary 6, in Theorem 13 and in Theorem 14, in that same form each time.** (R-IND-5 restatement F10: earlier revisions left this class to an unqualified clause that was doing load-bearing work in three separate places. The identification was re-derived rather than taken: Theorem 13's dual is an infimum over $\{h,\, n > 0\}$ per frequency with no further restriction, so the 082 bracket brackets exactly this Ψ; Corollary 6 runs through shift-averages, which lie inside the class by Lemma 11; and Theorem 14 covers exactly it.)

Corollary 6 (The chain, and its direction — *the chain is unconditional*). *$\phi_n \ \ge\ $* rate(*periodize*) $\ge$ rate(*shift-average*) $\ge\ \Psi(D;\Delta)$*, with $\Psi(D;\Delta)$ the infimum over the class $\mathcal{P}(D;\Delta)$ of Definition 3, whence*

$$\boxed{L^{\infty}(\Delta)\ \ge\ \Psi(D;\Delta)}$$

with Theorem 10 supplying step (B)'s convexity as a theorem *rather than a hypothesis.* ***Every step is an inequality the same way and each produces an explicit feasible object, so the chain is one-directional****: it cannot yield the reverse inequality. The converse would need an unwritten truncation/achievability lemma — and the stationary optimum is not rational at any finite order, so the FIR-density step is not free; it is a boundary-charge argument again, in the other direction.*

The permitted wording, verbatim, and the scope that travels with it. "$L^{\infty}(D) \ge \Psi(D;\Delta) \ge$ LB(Δ) for $\Delta = 0, 1, 2$, with no (H*), (H**), (H***), (D), (F), no boundary charge, no (R1) as a hypothesis — it is now a theorem — and no (R2); margins $+0.0147817164/+0.0040574952/+0.0007968190$." *The scope that travels with that sentence, and must be printed with it:*

(a) the family $\mathcal{F}_0$ (Hypothesis 1) — a definition*, and by Remark 24 a* well-posedness condition *on the moment coordinates;*

(b) the independent-noise-copy clause (Hypothesis 4) — a specification of an object this document constructs*, not an assumption about anything unknown; (R7) survives as a* wording rule *— the tiling must be specified with independent copies and never left implicit — rather than as a live hypothesis;*

(c) the classical Wiener–Masani/Szegő input [14, 15, 16], cited with *its non-degeneracy hypotheses, which are verified to hold on $\mathcal{K}_n(\varepsilon, M)$ (Theorem 10, step (2), and Remark 22);*

(d) the floating-point house convention on the Ψ bracket (Remark 7).

"Unconditional" attaches to the chain and never to the value: *Ψ remains a two-sided* certified bracket *under (d), the chain remains one-directional (Remark 30), and the permitted headline is unchanged.*

Remark 30 (What the chain does *not* give). Corollary 6 is a *lower* bound; the reverse inequality is not available from it, and is not available from anything in §§6–7. **Amended at v1.0 (dated 2026-08-08).** The reverse inequality is now available, but *only* from Theorem 15, and *only* inside that scoped statement with its five further scope items (i)–(v). Outside Theorem 15 this remark continues to govern in full: **no free-standing reverse inequality, and no statement that the process limit coincides with Ψ, may be printed anywhere else in this document**;

"unconditional" still never attaches to the value Ψ nor to that equality; and the *quotable numeric* headline is unchanged — it remains Corollary 8, never a single number. The permitted headline at $\Delta = 0$ is the bracket

$$L^{\infty}(0) \;\in\; [\,0.5627265,\; 0.5647420\,],$$

of width 2.0×10^{-3}: the lower end from Corollary 6, the upper end the certified $n = 32$ anchor of §3.2 through Corollary 2. *This bracket is superseded numerically — and only numerically — by Corollary 8, which replaces the upper end by an explicit D-feasible record on* $n = 2048$ *and narrows the width by* $51.5\times$*; the prohibition in this remark is unaffected by that narrowing and continues to govern.* Equally, the plateau is not *scope*-free even though the chain is unconditional: it removes (H*), (H**), (H***), (D), (F) and the boundary charge, and Theorem 10 now removes (R1) as well — but the four scope items (a)–(d) of Corollary 6 travel with every statement of it, and it remains under Hypothesis 1 throughout. In particular **the word "unconditional" may be attached to the chain and never to the Ψ value**.

Remark 31 (Where the $O(1/n)$ drift lives)*.* At $\Delta = 0$ the drift splits exactly along the two steps: $n(\phi_n - \Psi) = 0.0645 = 0.0420$ (periodization / boundary) $+\,0.0225$ (cyclostationarity) + residual. The entire (H*)/(H**)/(H***) machinery was fighting only the first 65%.

6.4 The two ingredients of the certificate

Lemma 6 (S — evaluation of the leak term)**.** *Under Theorem 8 and bivariate Wiener–Masani/Szegő [14, 15, 16], the stationary rate is*

$$\text{rate} = \alpha\big[\langle \ln M_Q\rangle - \langle \ln n\rangle\big] \;+\; \alpha\big[\langle \ln f_S\rangle - \ln s\big], \qquad s = \operatorname{Var}\big(S_u \mid S^{u-1}, R^u\big),$$

with the closed form $\langle \ln f_S\rangle = \ln(\tau^2 a/\lambda_s) = -0.1025391956$*.* ***Scope (non-degeneracy).*** *The Szegő step needs the S-side spectrum bounded away from* 0 *and* ∞*; here* $f_S \in [0.5111,\, 9.400]$*, so it cannot degenerate — but the hypothesis is part of the lemma and must travel with it.* Certificates: *identity residuals* $-2.2\times10^{-16}/-4.4\times10^{-16}/-1.1\times10^{-15}$ *at the three certified points; the closed form agrees with the grid to* 2.9×10^{-16} *across* $N_f = 256$–16384*.* ***Evaluator caveat.*** *On records with a deep narrow notch in the noise spectrum the* finite-lag *evaluator for* σ *and* s *converges very slowly (residual* 2.4×10^{-4} *at* $P = 50$*, still* 1.5×10^{-4} *at* $P = 1600$*). This is a property of the* evaluator*, not a failure of the lemma (which is* P*-independent to* 4.4×10^{-16} *at the anchors), and it does not touch the certificate, whose leak term is an exact integral of* frozen *filters. By-product:* $\text{rate}_{\text{block+leak}} \le \text{rate}_{\text{true}} \le \text{rate}_{\text{collapse}}$ *there, so the empirical gates used the aggressive side.*

Lemma 7 (C-stat — per-frequency joint convexity)**.** *Fix a frequency. If* $Q \succeq 0$ *and* $R = P - Q \succeq 0$*, then*

$$(h,\Gamma) \;\longmapsto\; \ln\!\big(\Gamma - hQh^*\big) - \ln\!\big(\Gamma - hPh^*\big) \;=\; -\ln(1 - Z), \qquad Z = \frac{hRh^*}{M_Q},$$

is jointly convex on the cone. Proof. ***Every step is standard convex analysis and no novelty is claimed for any of it.*** $\varphi(h,t) = hRh^*/t$ *is jointly convex on* $t > 0$ *(quadratic-over-linear,* $R \succeq 0$ *— [5, Sec. 3.1.7]) and* nonincreasing *in* t*;* $M_Q(h,\Gamma) = \Gamma - hQh^*$ *is jointly* concave *(*$Q \succeq 0$*);* h *enters* φ*'s first slot affinely. The composition rule for a convex, coordinatewise-monotone outer function [5, Sec. 3.2.4] gives* Z *convex; and* $u \mapsto -\ln(1-u)$ *is convex increasing on* $u < 1$*. In its matrix form the same argument runs on the operator-concavity ingredient of Ando [6].* □

Hypotheses used are exactly $Q \succeq 0$ **and** $R = P - Q \succeq 0$*, and they are used for different things*

— *see Remark 32, which supersedes the earlier (R12) wording. No matrix lift is needed and the 074 lemma (GO-13 Theorem 5, the block Schur lift of Theorem 2(i), and itself a lift of the standard certificate [5, Sec. 3.1.7 and 3.2.4],[6]) is cited here only as the* antecedent *of which this is the scalar instance — the proof above does not use it.* Certificates: 60,000 *random chords with* Γ *pushed to* 10^{-9} *of the cone boundary, zero violations (harness; the verifier's own sweep likewise).* Control: *with* $Q \succ P$ *the statement* fails — 12,532/20,000 *violations, worst* +13.2*, in the verifier's sweep;* 60,000/60,000 *violations, worst* −21.4*, in the harness's* $Q' = 1.3P$ *control. The psd hypothesis is not decoration.*

Remark 32 (R24 — which psd hypothesis is *demonstrated* necessary; this supersedes (R12)). The earlier text called $Q \succeq 0$ and $R = P - Q \succeq 0$ "both load-bearing". That is too strong, and the evidence separates them. $R = P - Q \succeq 0$ **is demonstrated necessary**: the R-breaking control $Q' = 1.3P$ violates the statement, 165/645 and 185/505 on a second seed (this document's re-run: 198/673, worst +2.14). $Q \succeq 0$ **is a hypothesis of this proof route, not of the statement**: it is what makes M_Q Loewner-concave, and that is the only place it is used — but breaking it while keeping $R \succeq 0$ produces *no* violation. Over 22,000 chords across four non-psd-Q families with $R \succeq 0$ ($Q'' = -0.2I$, $Q''' = -2P$, an indefinite Q built by scaling P's eigenvalues by $+0.9/-1.5$, and $Q'''' = -(P+I)$) there are *zero* violations, every chord on the convex side by at least 5.0×10^{-8}. So the honest statement is: one hypothesis is necessary as far as the evidence goes, the other is a convenience of the route.

Definition 4 (Admissible frozen filters — a hypothesis with content)**.** A pair (C_0, B_0) is *admissible* if C_0 is *monic causal in* S and B_0 is *causal including lag* 0 *in* R. For such a pair the *minorant*

$$\hat{s}(x; C_0, B_0) \;=\; \big\langle |C_0|^2 f_S + |B_0|^2 \Gamma + 2\,\mathrm{Re}\big(\overline{C_0} B_0 h_1\big) \big\rangle$$

is *affine* in x and satisfies $\hat{s} \geq s$ *always.* **This definition is now used twice**: inside the certificate of Theorem 13, and — as the inf-of-affine representation of the leak pivots — in step (3b) of Theorem 10, where it is the load-bearing step.

Remark 33 (R11 — admissibility is a hypothesis, not an adjective). $\hat{s} \geq s$ was checked over 2,880 (record × frozen-filter) pairs — 120 adversarial records × 24 filter choices including foreign optima, truncations to 1/2/5/20 lags, oscillating and near-unit-root filters — with $\min(\hat{s} - s) = -2.2 \times 10^{-16}$ (machine zero, attained only where the filter *is* that record's own optimum). **A non-monic** C **(**$C = 0.3$**) gives** $\hat{s} - s = -0.5447$ **and *inverts* the bound**, and fed into the dual it produces "lower bounds" *above* the true rate. This is the one place where a sign slip inverts the whole certificate.

6.5 The certificate

Theorem 13 (The box-free per-frequency certificate). Let $\mu \geq 0$, let (C_0, B_0) be admissible (Definition 4) with $\hat{s}_0 > 0$ any linearisation point, and set

$$\beta(\omega) = \mu - \alpha\,|B_0|^2/\hat{s}_0, \qquad p_1 = -\alpha\, C_0 \overline{B_0}/\hat{s}_0, \qquad p_2 = -\mu\, z^{\Delta+1}.$$

If $\beta > 0$ everywhere then, with $\Psi(D;\Delta)$ the infimum over the class $\mathcal{P}(D;\Delta)$ of Definition 3,

$$\Psi(D;\Delta) \;\geq\; \mathrm{const}(\mu, C_0, B_0, \hat{s}_0) \;+\; \Big\langle \inf_{h,\, n>0} \Big[\alpha \ln\Big(1 + \tfrac{hRh^*}{n}\Big) + \beta\,(n + hPh^*) + 2\,\mathrm{Re}(p^* h) \Big] \Big\rangle.$$

The three inputs are: (i) admissibility of the frozen filters, (ii) $\beta > 0$, and (iii) a correct cell infimum. In particular the bound requires *no optimality of the fixed point, no convexity of the process-rate functional, no differentiability of the rate, and no moment box*; the separability across frequencies replaces the tangent plane entirely, and Lemma 7 is then used only to *compute* each cell infimum reliably, never to establish validity. *Proof.* Feasibility gives rate $\geq$ rate $+ \mu(\text{dist} - D)$; Definition 4 gives $-\ln \hat{s} \leq -\ln s$; concavity of ln gives $-\ln u \geq -u/u_0 - \ln u_0 + 1$; and the result is separable across ω, so the infimum passes inside the average. □

Remark 34 (R17 — state the certificate this way, and why). The dual bound survives brutal mistreatment: 25 random *non-optimal* anchors per Δ give valid bounds; mistuning μ by $\times 0.95$ or $\times 1.10$ degrades but never invalidates; truncating the frozen filters to 12/4/1 lags loses $0/3.6 \times 10^{-6}/1.08 \times 10^{-2}$. The KKT solve only makes the bound *tight*, never *valid*. **This is a genuine strengthening over the moment-box route**, which does *not* survive: a 10^{-4} perturbation of the anchor collapses the box bound to 0.1377. Both agree at the anchor to 2.2×10^{-12}, and **the moment-box computation is therefore recorded as a corroboration, never as the certificate**. Its constants, for the record: $\langle \Gamma \rangle \leq (1 + \sqrt{D})^2 = 2.395445115$ (Minkowski) and $\langle |\Phi_{RV}| \rangle, \langle |\Phi_{RY}| \rangle \leq 1 + \sqrt{D} = 1.547722558$ (pointwise + integral Cauchy–Schwarz, using $\langle f_V \rangle = \langle f_Y \rangle = 1$) — far tighter than the finite-n box, which carries $\sqrt{nD}$ where this carries $\sqrt{D}$ per symbol. A free structural check found en route: $\langle \Gamma \rangle = 1 - D = 0.7$ *exactly* at all three Δ, so the stationary optimum is a proper test channel ($E[\hat{Y}^2] = E[\hat{Y}Y]$). **Consequence: (R2), the global optimality of the stationary program, is *discharged* — it is no longer required anywhere in the plateau chain.**

Remark 35 (The L^2 box does not exist — constructively). The Hölder pairing above is L^1/L^∞, and it must be: the distortion constraint controls $\langle \Gamma \rangle$ but *not* $\langle \Gamma^2 \rangle$. Constructively, feasible records at fixed distortion and fixed $\langle \Gamma \rangle$ have $\langle \Gamma^2 \rangle = 1.667/1.965/2.929/6.770$ as the spike narrows over 64/16/4/1 bins — $\langle \Gamma^2 \rangle$ *diverges.* **No L^2 moment box exists here**, and any certificate written against one is invalid.

Remark 36 (Errata and scope notes carried by this section). *(R14) — a proof-step correction.* The cell guard must not be justified by "Hessian $\succeq 2\beta\lambda_{\min}(P)I$": that argument is false, because $\varphi(hRh^*)$ is radially *concave.* The correct strong-convexity modulus is

$$m \;=\; \beta\,\lambda_{\min}(P+Q), \qquad \text{guard} \;=\; \frac{2|u|^2}{\beta\,\lambda_{\min}(P+Q)},$$

u the Wirtinger residual at the cell's stationary point. The corrected modulus is 3.5% larger here (0.14231 against 0.14725) and numerically inert (guards 1.0×10^{-26} against 9.7×10^{-27}; the bound moves by 0.0×10^0), but the sentence is replaced, not patched.

(R15) — outward rounding. The quotable brackets of §6.6 are rounded *outward* at 10^{-10}. The 12-digit line that circulated earlier was rounded *inward* (the lower ends up by $\approx 4 \times 10^{-13}$, the $\Delta = 1$ upper end down by 4×10^{-16}) and is *withdrawn*; only the 10-digit form is quotable. The raw floating-point widths are $\leq 4.4 \times 10^{-12}$.

(R18) — $\Delta = 2$ has no headroom in the tuning. At $\Delta = 2$, $\mu \times 0.95$ costs -2.4×10^{-5} of margin, $\mu \times 1.10$ costs -2.2×10^{-3}, and a $\pm 5\%$ error in the $\hat{s}$ linearisation costs $-4.9 \times 10^{-5}/-1.7 \times 10^{-4}$. Validity is never lost, only margin. $\Delta = 2$ **carries the campaign's smallest margin** and must not be promoted ahead of the other two.

(R19) — the $O(1/n)$ constant is Δ-dependent. $n(\phi_{16} - \Psi) = 0.064507/0.070483/0.077503$ at $\Delta = 0/1/2$ (harness: $0.064506/0.070483/0.077503$). Consequently $c(0) \in [0.06447, 0.06449]$ —

itself a correction of the earlier "0.064495 $\pm$ 3 $\times$ 10^{-5} relative", which was merely the $n = 32$ point; the sequence is monotone decreasing over n = 8–48 with spread 3.0×10^{-4} relative and Richardson$(32, 48) = 0.064471$ — **is a $\Delta = 0$ statement only. No Δ-uniform $O(1/n)$ constant is claimed anywhere in this document.**

(R20) — the block infimum. The anchor-free two-sided certificate of §6.7 pins $\text{block}_\infty \in [0.5299499808, 0.5299499809]$; the value to use is $\text{block}_\infty = 0.5299499808119$. The committed `results/GO14-process-limit.json` entry `s6_block_inf` = 0.529949985183839 is **high by** 4.4×10^{-9} and is corrected here. Nothing downstream moves: the 0.0313 spectral gain of Theorem 1(v) is unaffected.

Truncation — and its sign. Lag truncation of the frozen filters is on the *safe* side for the lower bound: a truncated filter is still admissible, so $\hat{s} \geq s$ and the bound can only loosen. It has the *wrong sign* for the upper bound — the rate evaluator's own s is P-truncated, so the computed rate *under*-states the true rate. The upper ends here are safe because the evaluation is flat to 4.4×10^{-16} at the anchors over P = 50–1600; the sign is stated rather than assumed.

6.6 The certified brackets and the plateau

Minimum lower end and maximum upper end over the grid family $N_f \in \{1024, 2048, 4096, 8192\} \times P \in \{60, 100, 140, 200\}$, rounded *outward* at 10^{-10} (R15):

$$\Psi(0) \in [0.5627264963,\ 0.5627264964], \quad \Psi(1) \in [0.5364013784,\ 0.5364013785], \quad \Psi(2) \in [0.5310500198,\ 0.53105001$$

The multipliers are $\theta = 3.086038362097/3.130079068857/3.151357871769$, identical to 12 digits across the whole grid family; the lower-end spread over the family is $4.0 \times 10^{-13}/1.4 \times 10^{-12}/7.7 \times 10^{-13}$ and the upper-end spread $\leq 6.7 \times 10^{-16}$. The integrands are analytic and periodic, so the rectangle rule is spectrally accurate (confirmed, not assumed). These are *floating-point* certificates in the sense of Remark 7: no interval arithmetic.

Corollary 7 (The plateau at $\Delta = 0, 1, 2$ — *this supersedes Corollary 5*)**.** *Under Hypothesis 1 and Hypothesis 4 — and, since Theorem 10 discharges (R1),* conditional on none *of the hypotheses this campaign has been fighting, with the four scope items (a)–(d) of Corollary 6 travelling with the statement —*

$$L^\infty(\Delta) \ \geq\ \Psi(D; \Delta) \ \geq\ \text{LB}(\Delta),$$

and against the sealed *causal-spectral allocation values of* ***`results/GO14-process-limit.json`*** *(**`s6_cand`**),*

Δ	*sealed bar*	*quotable* LB	*margin*
0	0.5479447799144537	0.5627264963	+**0.0147817164**
1	0.5323438832146611	0.5364013784	+**0.0040574952**
2	0.5302532008457406	0.5310500198	+**0.0007968190**

Every margin clears the certified bracket width by four to eight orders of magnitude ($\Delta = 2$, the tightest, by 1.8×10^8 bracket widths). ***This retires (H*), (H**), (H***), (D), (F), the boundary charge*** **and** ***(R1) from the plateau argument, at three lags — two of which the transfer route provably could not reach*** *(it would have needed certified anchors at $n \approx 260$ and $n \approx 1320$). The permitted wording for the whole chain, with its scope, is printed verbatim in Corollary 6; "unconditional" is a statement about that chain and never about the value Ψ, which remains a two-sided certified bracket under Remark 7.*

Remark 37 (The bars, and a superseded pair of numbers). (R16.) The margins above are computed against the *sealed* causal-spectral values 0.5479447799144537/0.5323438832146611/0.5302532008457406 of `results/G014-process-limit.json` (`s6_cand`), and against nothing else. The Ψ-bracket prover used a rounded bar triple that is correct at $\Delta = 0$ but *wrong* at the other two lags — at $\Delta = 1$ in a *non-conservative* direction, its margin coming out 8.8×10^{-7} too large, and at $\Delta = 2$ conservatively by 8.2×10^{-7}. **That triple is withdrawn: no margin anywhere in this document is computed against it, and none may be.** No verdict moves — all three margins above still clear by four to eight orders — but the $\Delta = 1$ margin recorded against that triple is withdrawn and replaced by +0.0040574952. The harness reads the sealed bars from the committed JSON at run time and cross-checks them against its pinned literals.

Cross-checks, all consistent. An independent window-Cholesky evaluator re-evaluates the certified point to $\leq 1.6\times10^{-15}$ at all three Δ and at three window sizes. Every sealed certified $\mathrm{LB}(\phi_n)$ sits *above* Ψ^{UB} at $n = 8$–48, with $n(\phi_n^{\mathrm{UB}} - \Psi^{\mathrm{UB}}) = 0.064535/0.064516/0.064507/0.064497/0.064495/0.064487$ — monotone decreasing, exactly as (R19) requires. All independent constructive upper bounds sit above Ψ^{UB}; a D-ladder over $D = 0.26$–0.34 sits strictly below the sealed 081 $\phi_8(D)$ lower ends, smooth and monotone; and 400 random feasible stationary records per Δ give zero violations (minimum slack $+0.090/+0.110/+0.116$).

6.7 "Does it prove too much?"

The identical machinery run on the *block* program ($se \equiv n$; no leak, so the cell dual needs *no anchor at all*) returns UB = LB = 0.5299499808119 against the independently known block infimum — agreement 1.2×10^{-11}, and it does *not* produce a value above the known answer. The full Δ-ladder $0, \ldots, 9$ matches the recorded ladder at every Δ,

$$\Psi(\Delta) = 0.562726496/0.536401378/0.531050020/0.530117530/0.529973408/0.529953058/0.529950369/0.529950029$$

with $\mathrm{LB}(9) - \mathrm{block}_\infty = +7.3 \times 10^{-10} \geq 0$: the machinery approaches the known value *from above* and never crosses it. The per-lag excess ratios rise monotonically *from below* toward $\lambda_s^{-2} = 7.954917$ and cross at $\Delta \approx 6$; the full-family closure constant is $c_{\mathrm{fs}} \approx 0.098\pm0.004$, strictly below the diagonal-class $c_{\mathrm{diag}} = 0.111 \pm 0.006$ as it must be. **Keep the two appearances of λ_s apart**: it *is* the Δ-closure exponent, and it is *not* the kernel pole — the true dominant singularity is a complex pair of modulus $\rho^* = 0.27 \pm 0.02$, uniform in Δ (measured at three Δ), strictly inside $\lambda_s = 0.3545538$, which therefore survives only as a *conservative envelope* for hypothesis (D) (making $\bar{X} = 0.065038$ loose).

6.8 The $\Delta = 2$ finite-window cross-check, certified

The Ψ prover's scope note — that no *committed* sealed ϕ_n existed at $\Delta = 2$ — is discharged. The same certify machinery that produced the sealed anchors (cold start, deterministic) gives

(n, Δ)	certified bracket for $\phi_n(2)$	width
$(16, 2)$	[0.535891743313, 0.535893963469]	2.22×10^{-6}
$(24, 2)$	[0.534274270522, 0.534279127269]	4.86×10^{-6}
$(32, 2)$	[0.533465073456, 0.533471490973]	6.42×10^{-6}

— at or better than the sealed-cell width band (3.0–9.2×10^{-6}); nothing had to be disclosed as short. Diagnostics: gradient norms $1.19/1.28/1.11 \times 10^{-8}$; the noise-floor parameterization is *inactive* ($\lambda_{\min}(N) \approx 0.1404$ at all three n, and $\lambda_{\max}(N)$ under Theorem 3's $1/\theta$ cap, so the optimizer is interior); dist $- D$ strictly negative, so each upper end is a *raw feasible* evaluation with zero projection correction; an independent evaluator agrees to 5.6×10^{-16}; and a second, unrelated cold start lands within 5.3×10^{-7}, inside the certified widths, at all three n. Controls $(16, 0)$, $(16, 1)$, $(24, 0)$ reproduce the sealed brackets first, every upper end within 1.2×10^{-7}.

Cross-check against Corollary 7: no violations. Every certified $\mathrm{LB}(\phi_n(2))$ sits strictly *above* $\Psi(2)$'s upper end, by $4.84/3.22/2.42 \times 10^{-3}$ — 500–2000× the bracket widths — and $n(\phi_n^{\mathrm{UB}} - \Psi^{\mathrm{UB}}) = 0.0775031/0.0774986/0.0774871$, monotone decreasing as (R19) requires. The committed `s4_fs16_D2` $= 0.5358939351814885$ lies strictly inside the new $(16, 2)$ bracket.

Remark 38 (Bookkeeping: two valid $(16, 0)$ bracket conventions — this document uses one of them). Two different $(16, 0)$ brackets are in circulation and *both are valid records of different things*: the R-IND-5 *width-of-record* bracket $[v^* - 3.45 \times 10^{-6}, v^*] = [0.566754682, 0.5667581350]$, and the `results/G014-convexity.json` bracket produced by the harness's own cold start. **This document uses the *width-of-record* convention throughout** — it is the convention already printed in the anchor table of §3.2 — and every $(16, 0)$ bracket quoted anywhere above is that one. Mixing the two in a single comparison is an error.

Hygiene notes carried forward. All brackets in this section are floating-point certificates (Remark 7); the finite-n lower endpoints are BLAS-sensitive at $\sim 10^{-7}$ through $r_n \cdot R_{\mathrm{box}}$, which is exactly why the width gates live in the artifact-self-consistency tier of CI and why a governed runner must *re-derive* these numbers and never copy them. Two artifact-hygiene items are on record and are not cited as evidence anywhere above: the committed `spectral/transfer.log` kernel-transfer defect is an A_v lag-orientation mismatch (a flat $6.76 \times 10^{-2} = |a_v(+1) - a_v(-1)|$ swamping a genuine $O(1/n)$ boundary layer) and must be regenerated before use; and the internal check "block + leak = rate" is an *algebraic identity*, not an independent validation.

7 Achievability: Lemma W, the FIR-density residual, and the unconditional brackets

Status: achievability prover + independent R-IND-5 pass on Lemma W (fresh context, own evaluator from the model primitives by *two* routes — the definitional CMI and the Collapse route — agreeing to 1.55×10^{-15}; the prover's evaluators supplied no rate or distortion number). The pass is **PASS on substance with one refutation of wording**, and its twelve numbered restatements W1–W12 are *all in force in this revision*: W1 and W3 rewrite the statement of Lemma W itself, W2 adds a hypothesis to it, W4 removes a leg from its proof, and W5–W12 fix the surrounding wording. Registration 084 pending, harness `experiments/go14_achiev.py` — *nothing in this section is sealed.*

This section runs the chain of §6 in the *other* direction, and it does not close it. What it delivers is (i) a window-transfer lemma with measured certificates, (ii) a reduction of the missing converse to exactly one functional-analytic residual, and (iii) — depending on no lemma of this document at all — a 51.5× narrowing of the permitted bracket of Remark 30.

7.1 Three builds, kept apart (W3)

Fix Δ, a depth-L two-sided FIR pair (a_v, a_y) and a causal MA factor q of the record noise, i.e. a *stationary* record $x = (a_v, a_y, q)$ of $\mathcal{F}_0$ on $\mathbb{Z}$. Three different finite-window objects are built from x, and **they must not be conflated**:

(Z) the **zero-edge build** $x_{\mathrm{Z}}^{(n)}$: the interior rows $t \in [L, n-L)$ carry the full depth-L taps, the $2L$ edge cells are *pure independent noise* of variance ε, and N_{cov} is block diagonal (edge block εI, interior block the stationary MA Toeplitz matrix);

(T) the **truncated-taps build** $x_{\mathrm{T}}^{(n)}$: every cell keeps whatever taps fall inside the window and the MA Toeplitz matrix covers the whole window;

(R) the **repaired records**: either build with its noise covariance multiplied by the unique scalar c that makes the *window* distortion exactly D.

Steps (1) and (2) of Lemma W prove statements about the zero-edge build (Z) and do *not* apply to the truncated-taps build (T): the edge cells of (T) are not independent noise and its N_{cov} is not block diagonal. **Every constant quoted in this section, and every bracket endpoint in §7.4, comes from the repaired truncated-taps build (T)+(R)**, which is the build that is feasible at the window sizes used. Feasibility is what forces (R): see W2 below.

Lemma 8 (W — window transfer)**.** *Let x be a depth-L FIR stationary record of $\mathcal{F}_0$ with residual noise spectrum bounded below (spectral floor), and let* $\mathrm{rate}(x)$ *be its stationary rate at lag Δ.* ***Hypothesis (feasibility threshold, W2):*** *there is an $n_0(L)$ such that the repairing rescale c is strictly positive for $n \geq n_0(L)$; the lemma is asserted only for such n. Then for every $n \geq n_0(L)$ the repaired window record $x^{(n)}$ is a D-feasible record of $\mathcal{F}_0$ on the window of length n, and*

$$\phi_n(\Delta) \;\leq\; L_a\big(x^{(n)}\big) \;\leq\; \mathrm{rate}(x) + \frac{C(L,n)}{n},$$

*where **(W1)** $C(L,n)$ is* monotone decreasing in n *with*

$$\sup_{n \geq n_0(L)} C(L,n) \;<\; \infty,$$

*and where **only** $C(L,n) = o(n)$ **is used** in every consequence drawn from the lemma. **No claim that C does not depend on** n **is made anywhere**; the object that is n-independent is the* edge charge $\sum_t \delta_t$ *of step (3), not the constant.*

Remark 39 (W1 — what was refuted, and why the conclusion survives). The prover's statement asserted a bound $C(L) < \infty$ carrying no dependence on the window length. **That assertion is false in the verifier's own measurements, for the very build the proof covers**: for the zero-edge build (Z) the constant $n\,(L_a(x^{(n)}) - \mathrm{rate}(x))$ runs $71.76 \to 20.38 \to 15.16 \to 13.39 \to 12.65$ over $n = 64, 128, 256, 512, 1024$ at $L = 6$, $\Delta = 0$ — a factor 5.7, and still falling. The cause is *isolated*: it is the **repair leg**, whose n-scaled cost converges *from above* to $\mu \cdot n(\mathrm{dist} - D)$. The conclusion is intact, because the constant is monotone decreasing in n in every swept row, so $\sup_n C(L,n) < \infty$; and $C(L,n) = o(n)$ is all any consequence uses. The truncated-taps build (T) is far better behaved — $0.08033 \to 0.08022$ over the same n at $L = 4$, $\Delta = 0$ — but it too decreases in n, and it too is quoted with the honest wording.

Remark 40 (W2 — the feasibility threshold is a hypothesis, not a formality). The zero-edge build is *infeasible* at small n: at $n = 64$, $L = 10$ the repairing rescale is $c = -0.832/-0.922/-0.948$ at $\Delta = 0/1/2$, i.e. $c \leq 0$ and no rescaled record exists; at $n = 96$ it is still negative $(-0.061/-0.113/-0.128)$ and at $n = 128$ it is positive $(+0.253/+0.217/+0.206)$. At $L = 6$ the same build is already feasible at $n = 64$ $(c = +0.070/+0.024/+0.011)$. So $n_0(L)$ is real, is L-dependent, and the lemma as the prover stated it carried no such hypothesis. The harness gates the $n = 64$, $L = 10$ infeasibility as a must-fail control.

Remark 41 (W4 — the repair leg is a separate estimate). Step (5) below discharges *feasibility* — exactly, constructively and in closed form. It does *not* bound the *rate cost* of the repair. That cost is a separate estimate, and it is the one that behaves differently in the two builds: n-independent in the truncated-taps build (measured $0.054031 \to 0.054005$ over $n = 256 \ldots 2048$, converging from above to $\mu \cdot n(\text{dist} - D) = 0.054002$), n-dependent in the zero-edge build, where it is exactly the source of the W1 decrease. **Steps (1)–(5) do not cover it.**

7.2 The five steps, with their measured certificates

Throughout, $\delta_t := \ln \sigma_t - \ln \sigma_{\text{stat}}$ is the per-cell log-excess of the window record's interleaved pivot over the stationary one, and Collapse (Theorem 8) is what makes the whole argument one line per step.

(0) Collapse. $2 \ln 2\, n\, L_a = \sum_t \ln \sigma_t - \ln \det N$, a theorem, every n, every schedule, every record of $\mathcal{F}_0$.

(1) The edge cells contribute exactly zero. For the zero-edge build the $2L$ edge cells have $\sigma_t = \varepsilon$ exactly and their numerator and denominator coincide, so their per-cell rate contribution is 0; and since N_{cov} is block diagonal, the $2L \ln \varepsilon$ they put into $\ln \det N$ cancels identically. Measured: per-cell CMI 0.000×10^0 at all three Δ, and $\ln \det N - [\,2L \ln \varepsilon + \ln \det T_{\text{int}}\,]$ at machine zero.

(2) Interior pivots dominate the stationary one. The conditioning set available to the window pivot at an interior cell is a genuine *subset* of the stationary one, so $\sigma_t \geq \sigma_{\text{stat}}$, i.e. $\delta_t \geq 0$. **This is the mirror of step (A) of the bypass, run the other way**: there a *superset* argument gave the inequality in the lower-bound direction. Verified past saturation — $\Delta \in \{0, 1, 2, 3, 5, 9, 20\}$, including $\Delta \geq n$ where the schedule saturates and the coordinate becomes the block program: $\min_t \delta_t \geq -7.8 \times 10^{-16}$ and *zero* negative cells at every Δ.

(3) The edge charge is finite — and is exactly n-independent. $\sum_t \delta_t$ is not merely bounded uniformly in n: it is *identical to nine decimals* across $n = 128, 256, 512, 1024$ and $L = 4, \ldots, 12$. At $L = 10$,

$$\textstyle\sum_t \delta_t \;=\; 0.043994832 \;/\; 0.051454447 \;/\; 0.052333440 \qquad (\Delta = 0/1/2),$$

and **the recorded triple is the $L = 10$ one** (the probe does not say which L; at $L = 4$ the same quantity is 0.044031439, equally n-independent). The per-cell profile is geometric at ratio ≈ 0.037 and is at the $f64$ floor by the sixth interior cell. The analytic input is the standard rational-spectral-factorization/Riccati fact [17, 18] that the finite-window one-step prediction errors of a process with rational spectrum and a spectral floor converge to the stationary value geometrically — *the same citation class* as the Wiener–Masani/Szegő input of Theorem 10, not a new hypothesis. **Non-circularity, checked**: that citation is applied to the depth-L FIR record itself, whose spectrum is rational by construction, so it does *not* assume the density of §7.3 that the argument is meant to isolate.

(4) The noise leg helps. $\ln\det T_m \geq m\langle \ln n(\omega)\rangle$ for the MA Toeplitz block T_m (Szegő, with the min-phase factor), so this leg enters the bound *with the helping sign.* Measured, and itself n-independent (flat in m over $m = 16, \ldots, 512$ to $\leq 1.1 \times 10^{-11}$):

$$\ln\det T_m - m\langle \ln n(\omega)\rangle \;=\; +8.622 \times 10^{-3} \;/\; +1.079 \times 10^{-2} \;/\; +1.076 \times 10^{-2} \qquad (\Delta = 0/1/2),$$

with the min-phase identity $\langle \ln n(\omega)\rangle = 2\ln q_0$ confirmed to 0.00×10^0.

(5) Feasibility, discharged constructively. The window distortion is *exactly affine* in the scalar noise rescale c (residual $\leq 1.7 \times 10^{-16}$ over a three-point secant test), so the c landing at dist $= D$ is available in closed form and lands there to $\leq 1.7 \times 10^{-16}$ at every n and every Δ. **Theorem C's convexity repair was available and was not needed.** By W4 this discharges feasibility only.

The structural finding worth keeping. In Collapse coordinates the converse boundary charge is *cheap*, because the two legs have opposite and *both favourable* signs: the numerator leg $\sum_t \delta_t \geq 0$ is geometrically summable, and the noise/ln det leg is ≤ 0 outright. That is why the (H*)/(H**)/(H***) machinery of §5 has no counterpart on this side.

Remark 42 (W5 — the orientation gate is the σ_t gate only). The first window build the prover wrote had a *lag-orientation defect* in a_v — the same class as the `np.roll` defect of Remark 26 — producing a spurious constant offset in σ_t that *masqueraded as a non-vanishing edge charge.* It is now a hard gate with a reversed-kernel control, and the control has power: the reversed build sits $+1.256\times10^{-2}$ above the stationary pivot *at every interior cell* against $\leq 4.4\times10^{-16}$ for the canonical build, and it converts the n-independent edge charge into an $O(n)$ one — $\sum_t \delta_t = 1.76 \to 3.87 \to 8.08 \to 16.50$ at $n = 64/128/256/512$, a slope of $+3.289 \times 10^{-2}$ per cell, against a canonical slope at the $f64$ floor (-1.2×10^{-16} per cell, i.e. the canonical edge charge does not move with the window length at all). **The gate is the σ_t gate, and only that.** The per-cell *distortion* has *zero* power against this defect: it agrees to $\pm1.1 \times 10^{-16}$ in the canonical and both reversed builds, and analytically so, since reversing a real two-sided kernel conjugates its symbol on $|z| = 1$ and the distortion is invariant under conjugation. A distortion check is therefore *decorative* here and is gated as a negative control. Also on record: a_y **is real zero-phase by (K1)** (Theorem 3), so its kernel is symmetric and carries *no* orientation — reversing a_y alone leaves the rate unchanged to machine precision; only a_v carries orientation. **Any successor building window records from spectral kernels must run this gate.**

7.3 The residual: one thing, and what it is

Remark 43 (The residual, stated as one thing). Exactly one thing stands between Lemma W and a converse theorem: **FIR-kernel stationary records are dense in value**, i.e. $U(L)$, the repaired depth-L stationary value, decreases to Ψ. The evidence, and its scope:

(a) **(W6) The measured list is at** $L \in \{2, 3, 4, 5, 6, 8, 10\}$, not "$L = 2 \ldots 10$": $U(L) - \Psi = 4.8 \times 10^{-4}$, 4.6×10^{-5}, 4.3×10^{-6}, 3.7×10^{-7}, 2.9×10^{-8}, 1.2×10^{-10}, 7.9×10^{-14} at $\Delta = 0$.

(b) **(W8) "R measured $\leq 1.5 \times 10^{-12}$" names its reference**: the reference is *the grid fixed point* at $N_f = 4096$, $P = 180$. Against the *082 certified lower endpoints* the gap is $+4.0 \times 10^{-11}/+7.2 \times 10^{-11}/+5.3 \times 10^{-11}$ at $\Delta = 0/1/2$ — positive, as weak duality forces. And **the entries at** $L \geq 11$ **are** $f64$ **noise**: at $\Delta = 0$ they go *negative* (-1.2×10^{-13} at $L = 11$, -1.3×10^{-13} at $L = 12, 13, 14$). **Those entries are not cited as convergence evidence anywhere.**

(c) **(W7) The squaring, in its sharp form.** With $T(L)$ the ℓ^2 norm of the discarded taps, $(U(L) - \Psi)/T(L)^2$ is 4.3–4.8, *constant over* $L = 1, \ldots, 6$ at $\Delta = 0$ (and 4.0–4.5 at $\Delta = 1$, 2.9–4.6 at $\Delta = 2$). *That* is the quotable form: the value loss is the square of the kernel loss, which is what a stationary Lagrangian requires — the first-order term vanishes. The cross-link to the recorded dominant singularity $\rho^* = 0.27 \pm 0.02$ of §6.7 is a **consistency check, not an independent reproduction**: a value-ratio band of 0.048–0.100 admits ρ^* anywhere in $\approx [0.22, 0.32]$, so ρ^* landing inside is weak evidence; the $\Delta = 1, 2$ *kernel* per-lag ratios (0.27–0.30) are the better witness.

(d) **The proof route, and where it stops.** (i) Continuity of the rate under uniform kernel convergence with a spectral floor is sketchable and believed complete — the frozen filters lie in a uniformly bounded H^2 ball, so two-sided continuity follows by evaluating each record's $\hat{s}$ at the *other*'s optimal filters; the floor is real and comfortable ($n(\omega) \in [0.1495, 0.2167]/[0.1378, 0.2095]/[0.1404, 0.2082]$). (ii) That the stationary fixed point's kernels lie in the Wiener algebra: the fixed-point map *preserves* $\mathcal{W}$, by Wiener's $1/f$ theorem and matrix Wiener–Lévy — **but that is invariance, not convergence**. Making (ii) a proof needs a *contraction estimate in the Wiener norm*, which is not available here. It is a small, well-posed functional-analysis problem, not an open modelling question — but it is not done.

Consequently the only statement permitted about the pair (L^∞, Ψ) from this side is the one printed verbatim in Corollary 8, with its $+R(\Delta)$ term never dropped. **Amended at v1.0 (dated 2026-08-08).** The residual named in (d) is *closed* — by §8, and by making route (ii) unnecessary rather than by proving it: Theorem 14 nowhere uses the optimum's regularity, kernel summability or $\mathcal{W}$-membership, so the contraction estimate in the Wiener norm is no longer needed for density. The citation route to (ii) remains *blocked* and is recorded as such: the classical Wiener–Hopf factorisation theorems have hypothesis $\Phi \in \mathcal{W}$, and $\Phi \in \mathcal{W}$ *iff* the kernels are, which is the claim — the citation relocates the bootstrap, it does not discharge it. What replaces this remark's prohibition is Theorem 15 and nothing wider: outside that scoped theorem, no statement that the process limit coincides with Ψ may be printed, here or anywhere, and Remark 30 governs as amended.

7.4 The unconditional brackets — the headline of this registration

Corollary 8 (Unconditional brackets on L^∞ at $\Delta = 0, 1, 2$)**.** ***The permitted wording, verbatim:*** "$L^\infty(\Delta) \in [\Psi^{\mathrm{LB}}(\Delta),\, V(10, 2048; \Delta)]$ for $\Delta = 0, 1, 2$ — the lower end by Corollary 6, the upper end an explicit D-feasible $\mathcal{F}_0$ record on $n = 2048$ through Corollary 2. Under Lemma W the upper end may be replaced by $U(L)$, giving $L^\infty \leq \Psi + R(\Delta)$ with R measured $\leq 1.5 \times 10^{-12}$ and proved zero only conditional on the FIR-density link." *Numerically,*

$$\begin{aligned}
\Delta = 0: &\quad [0.5627264963,\ 0.5627656412] \quad \textit{width } 3.914 \times 10^{-5}\\
\Delta = 1: &\quad [0.5364013784,\ 0.5364458112] \quad \textit{width } 4.443 \times 10^{-5}\\
\Delta = 2: &\quad [0.5310500198,\ 0.5310994872] \quad \textit{width } 4.947 \times 10^{-5}
\end{aligned}$$

with the upper ends outward-rounded at 10^{-10} *on the safe side (raw:* 0.562765641106*,* 0.536445811112*,* 0.531099487172*). At* $\Delta = 0$ *this is a* **51.5×** ***narrowing*** *of the* 2.0155×10^{-3} *bracket of Remark 30, which it supersedes numerically.* ***These brackets depend on*** **no lemma of this document** *— not Lemma W, not the residual, not (H*), (H**), (H***), (D), (F). The three inputs are: membership of the upper record in* $\mathcal{F}_0$*, its exact D-feasibility, and correct evaluation. All three are checked,* *not assumed: exact feasibility to* $\leq 1.7\times10^{-16}$ *at every* n*, and the CMI route against the Collapse route to* 1.1×10^{-15} on the certificate records themselves*. The scope items (a)–(d) of Corollary 6 travel with the lower end, as always; and, as always, "unconditional" is said of a chain and of these two* inequalities*, never of the value* Ψ *(Remark 7, Remark 30).* ***Note added at v1.0 (dated 2026-08-08).*** *The permitted wording above is* unchanged *and continues to govern:* ***this corollary, and not Theorem 15, is the quotable numeric statement about*** L^∞*, and the* $+R(\Delta)$ *term is never dropped from it. The "FIR-density link" its last clause names is now supplied by Theorem 14, but that changes nothing here: it licenses only the* scoped *Theorem 15, whose reverse direction may not be quoted outside that theorem, and it does not convert* Ψ *from a certified bracket into an exact value.*

Remark 44 (W10 — the anchor cross-check is 8/8)*.* Every $V(10,n)$ sits strictly *above* the corresponding certified ϕ_n upper end, as a suboptimal D-feasible record must: $\Delta=0$ at $n=16,24,32$; $\Delta=1$ at $n=16,24$; $\Delta=2$ at $n=16,24,32$ — **eight comparisons,** 8/8 (the probe's "7/7" undercounts its own table). The upper bound beats the standing $\mathrm{UB}(32,0)=0.5647420$ from $n=48$ on ($V(10,48;0)=0.5643999$).

Remark 45 (W9 — the constructive constants, and the comparison that is actually forced)*.* The constructive $O(1/n)$ constants $n\,(V(10,n)-U(10))$ at $n=2048$ are

$$0.08017\ /\ 0.09100\ /\ 0.10131 \qquad (\Delta=0/1/2),$$

Δ-dependent and quoted per-Δ only (R19); no Δ-uniform constant is quoted here or anywhere. **Their "correct sign" is forced only at equal n.** What is forced is $\phi_n\leq V(10,n)$ — and that *is* the 8/8 cross-check of Remark 44. At equal n the constructive constant duly exceeds the optimal one: at $\Delta=0$, $n(V(10,n)-U(10))=0.080645/0.080484/0.080404$ at $n=16/24/32$ against the recorded optimal-side 0.064507/0.064497/0.064495; at $\Delta=2$, 0.102064/0.101809/0.101682 against 0.0775031/0.0774986/0.0774871. The probe instead compared the $n=2048$ constructive constants against the recorded optimal-side triple 0.064507/0.070483/0.077503 and reported +24%/+29%/+31%; **that comparison is not forced and is withdrawn in that form**, because those are finite-n values, the constructive ones are quoted at $n=2048$, and both sequences decrease in n. The legitimate limit comparison is against the recorded limiting constant $c_\phi(0)\approx0.06447$. **(W12) The decomposition** of the $\Delta=0$ constant is

$$0.026163\ (\text{edge})\ +\ \mu\cdot n(\text{dist}-D)\ =\ 0.026163+2.2261\times\mathbf{0.024258}\ =\ 0.080165$$

against the measured 0.080168 at $n=2048$ — closing to 3.7×10^{-6}, and the residual is itself $O(1/n)$ ($2.98\times10^{-5}\to3.73\times10^{-6}$ over $n=256\ldots2048$). The distortion constant is $n(\text{dist}-D)=0.02426$, *not* 0.02430; with 0.02430 the identity closes only to $\sim10^{-4}$.

Remark 46 (W11 — the safe side, stated the way it is true)*.* $U(L)$'s σ comes from a P-lag innovation solve, which *overstates* σ and hence the rate — the conservative direction for an upper bound. The correct wording is: **non-increasing in P by construction; verified flat to 3.3×10^{-16} over $P=20,\ldots,400$ and $N_f=1024,\ldots,8192$.** The earlier "monotone decreasing over $P=40\ldots260$, flat to 12 digits" is *testing floating-point ties* — the verifier's own run reports "decreasing: False" for exactly that reason — and is withdrawn. The safe side has power where power exists: $P=1$ gives 0.5867, $P=8$ gives 0.56272650.

7.5 Does the achievability machinery prove too much? No

Three checks, all on the side of not proving too much. *(i)* Run on the *block* program ($se \equiv n$), the window records built here give upper bounds that stay strictly *above* the independently known $\text{block}_\infty = 0.5299499808119$ and decrease monotonically in n; the R-IND-5 record adds that the block-schedule window records converge to it like $\approx 0.105/n$ *from above*. *(ii)* The Δ-ladder of a fixed record approaches its saturation value from above and *never crosses* block_∞: at $n = 256$ the depth-10 record evaluates to 0.563040/0.542314/0.539317/0.538904/0.538849 at $\Delta = 0/1/2/3/4$ and is flat at 0.538840 from $\Delta = 12$ on, i.e. $+8.89 \times 10^{-3}$ above block_∞ throughout. On the *certified* Ψ-ladder of §6.7 the same approach-from-above is recorded down to $+1.70 \times 10^{-12}$ at $\Delta = 12$ (R-IND-5 record; not re-derived here). *(iii)* The machinery never produces a bound below a known value: this is the same test that §6.7 ran on the lower-bound side, and it passes on this side too.

Remark 47 (The novelty sweep on Lemma W's combination is *owed*). Lemma W's *combination* — Collapse, plus subset-conditioning monotonicity of the interleaved pivots, plus Toeplitz innovation monotonicity, assembled as a **signed two-leg boundary argument** in which the two legs carry opposite and both-favourable signs — has **not** been swept. **No novelty language of any kind is used for it**, here or anywhere in this document, and none may be used until the sweep is run and its verdict, permitted language and residuals are recorded in the manner of Remark 20. The ingredients are individually classical; it is the combination that is unswept.

8 FIR density: Lemma A′, Theorem D, Lemma M — and the ***scoped*** equality $L^\infty(\Delta) = \Psi(D; \Delta)$ at $\Delta = 0, 1, 2$

Status: FIR-density prover + independent R-IND-5 pass on Theorem D, Lemma A′ and Lemma M (fresh context; own evaluator built *from the model primitives only* — exact analytic autocovariances, σ as a Cholesky pivot of the interleaved (R, S) covariance, $\langle \ln n \rangle$ by Jensen-on-the-roots; *no FFT grid, no Toeplitz normal equations, and no prover code in any rate number*, the prover's stored kernels used only as *data*). The pass refutes **nothing** in Theorem D or Lemma A′ and refutes **two headline claims of Lemma M**. Its fourteen numbered restatements F1–F14 are *all in force in this revision*, and **F5–F8 are not optional**: printing Lemma M unrestated would put a *second* false n-independence claim on the record one registration after the first was caught (Remark 39). Registration 085 pending, harness `experiments/go14_density.py` — *nothing in this section is sealed.*

This section closes the residual of Remark 43, and it closes it by making the Wiener-algebra link *unnecessary* rather than by proving it. The move is one line: in Collapse coordinates the stationary rate is $[\ln \sigma - \langle \ln n \rangle]/(2 \ln 2)$, so n is a *datum* of the record and σ is the only derived object; and since σ is an *infimum* over admissible filters, the direction achievability needs is the free one. That kills the uniform-convergence requirement which had forced the earlier route into the Wiener algebra $\mathcal{W}$, and it is why the circularity recorded in Remark 43(d)(ii) is gone: the proof approximates an *arbitrary near-optimal feasible record*, never the fixed point.

8.1 Two admissibility notions, defined side by side and never conflated (F2)

Definition 4 fixes a pair (C_0, B_0) with C_0 *monic causal in* S and B_0 *causal including lag* 0 *in* R, and its minorant $\hat{s}$ bounds the S-*side* pivot $s = \mathrm{Var}(S_u \mid S^{u-1}, R^u)$ from above. Lemma A′ needs the *mirror* of that, and it attaches to a *different pivot*.

Definition 5 (A′-admissible filters — the mirror of Definition 4)**.** A pair (V, V_S) of symbols is *A′-admissible* if V is *monic causal in* R (taps at lags $0, 1, 2, \ldots$ with the lag-0 tap equal to 1) and V_S is *strictly causal in* S (taps at lags $1, 2, \ldots$ only). For such a pair,

$$\hat{\sigma}(x; V, V_S) \;:=\; \mathrm{Var}\Big(\textstyle\sum_i V_i R_{u-i} + \sum_j (V_S)_j S_{u-j}\Big) \;=\; \big\langle |V|^2 \Phi_R + |V_S|^2 f_S + 2\,\mathrm{Re}(\overline{V} V_S \Phi_{RS})\big\rangle$$

is *affine* in the moment coordinates of x and satisfies $\hat{\sigma} \geq \sigma := \mathrm{Var}(R_u \mid R^{u-1}, S^{u-1})$ *always.* **Definitions 4 and 5 are different objects and must never be conflated**: the first is monic in S and bounds s; the second is monic in R and bounds σ. They are used in different arguments — Definition 4 inside Theorem 13 and step (3b) of Theorem 10, Definition 5 only here.

8.2 Lemma A′ (F1, F3, F14)

Lemma 9 (A′ — pivot upper semicontinuity)**.** *Let x be a stationary record with $\Phi_R, f_S \in L^1$. Then:*

(1) $\sigma(x)$ equals the infimum of $\hat{\sigma}(x; V, V_S)$ over finitely supported *A′-admissible pairs.*

(2) For any finitely supported A′-admissible pair with $\hat{\sigma}(x; V, V_S) \leq \sigma(x) + \varepsilon$ — chosen first, before the approximant — *and any second record x' with $\Phi'_R, f_S \in L^1$,*

$$\sigma(x') \;\leq\; \sigma(x) + \varepsilon + C_V \,\big\|(\Phi'_R, \Phi'_{RS}) - (\Phi_R, \Phi_{RS})\big\|_{L^1}, \qquad C_V = \sup_\omega \big(|V|^2 + |V_S|^2\big) < \infty.$$

***The hypotheses are** $\Phi_R, f_S \in L^1$ **and nothing else**: no spectral floor, no H^2 ball, no uniform convergence, no membership in $\mathcal{W}$.*

Proof of *(1). $\sigma(x) = \mathrm{dist}^2\big(R_u, \mathcal{M}\big)$ in $L^2(\Omega)$, where $\mathcal{M}$ is the* closed *linear span of $\{R_{u-i}\}_{i\geq 1} \cup \{S_{u-j}\}_{j\geq 1}$. Finite linear combinations are dense in $\mathcal{M}$* by the definition of a closed span*, and the distance to a set and to its closure coincide; so the infimum over finitely supported A′-admissible pairs equals the infimum over $\mathcal{M}$, which is $\sigma(x)$. This is the Hilbert-space projection fact [35, Ch. 4] and nothing more.* □

Proof of *(2). $\hat{\sigma}(\cdot; V, V_S)$ is an* affine *functional of the pair (Φ_R, Φ_{RS}) with coefficient symbols $|V|^2$ and $2\overline{V} V_S$, both bounded because the filter is finitely supported and fixed* before *x' is named; the f_S term is common to the two records. Hence $|\hat{\sigma}(x') - \hat{\sigma}(x)| \leq C_V \|\cdot\|_{L^1}$, and $\sigma(x') \leq \hat{\sigma}(x')$ by Definition 5.* □

***The infimum in (1) is not attained by any finite filter**; only ε-optimality is ever used, here and downstream. That is the whole content of the lemma, and it is why nothing about the regularity of any optimum enters.*

Certificates (harness §s1–s2). *(F14) The K-lag ladder below is instantiated on the $L = 10$ Dirichlet truncation at $\Delta = 0$,* and that record is named because the same lemma instantiated on the *optimum* gives different numbers — the prover's $+2.81 \times 10^{-2}/+2.50 \times 10^{-3}/+1.78 \times$

$10^{-5}/+2.64\times10^{-10}$ at $K=1/2/4/8$ against this record's $+1.2718\times10^{-2}/+1.0900\times10^{-3}/+7.9614\times10^{-6}/+1.3020\times10^{-10}$. **Both are valid**; they are two instantiations of one lemma, and neither is a correction of the other. The ladder decreases to σ *from above* and reaches it exactly: $|\hat\sigma-\sigma|=1.67\times10^{-16}$ at $K=60$, and the minimum excess over the whole ladder is -1.67×10^{-16}, i.e. machine zero. A$'$(2) is verified on 22 adversarial record pairs including 12 random FIR records far from the optimum ($\|\Delta\Phi\|_1$ up to 14.19): *zero* violations of either inequality, minimum slacks $+2.83\times10^{-11}$ and $+1.83\times10^{-5}$.

Remark 48 (F3 — the admissibility controls, and a defect that destroyed their power twice). Admissibility is a hypothesis with content, and the controls that show it are easy to get wrong. **The prover's first pass at the peeking controls had no power at all**: it recorded S-peek $+\mathbf{0.252}$ and R-peek $+\mathbf{0.031}$ — *positive*, i.e. the "controls" made the pivot *larger* — because of a lag-sign defect in its helper. **The R-IND-5 verifier independently hit and fixed the same defect in its own helper**, which is what confirms the prover's disclosure is accurate rather than merely candid. The corrected controls, each *fully optimised* over its enlarged index set so that it has power, all *break* the minorant:

S-peek (uses S_u), fully optimised	-0.010659
R-peek (uses R_{u+1}), fully optimised	-0.021585
R-peek two-step (R_{u+1},R_{u+2})	-0.023042
S-peek two-step (S_u,S_{u+1})	-0.012182
non-monic $V(0)=0.3$, rest optimal	-0.182975
non-monic $V(0)=0.3$, fully re-optimised	-0.341884

The run of record is `t5b.py`. The earlier `t5_ctl2.py` still carries the defect, is marked defective, and must not be cited — by this document or by any successor. The harness rebuilds all six from the primitives and gates them on the side of failure.

8.3 The cap, the floor, and which of them Theorem D needs (F4c, F4d)

Lemma 10 (B$'$ — the spectral cap; *recorded, and not used*)**.** *Feasibility alone bounds $\langle\Gamma\rangle\le 3.449116$, so capping Γ at a level M costs arbitrarily little as $M\to\infty$ and puts the kernels in L^∞; the three inequalities of the cap step hold with slack even on the $\langle\Gamma^2\rangle$-divergent feasible family that Remark 35 asserts exists (built explicitly:* $\max\Gamma=146.48$, $\langle\Gamma^2\rangle\to3.80$ *as the spike narrows).* ***(F4d) This lemma is deleted from the proof of Theorem 14 and is recorded here as an a-priori fact only.*** *It is* dispensable*: by Theorem 14(a), feasibility* alone *already gives*

$$\|g\|_2\;\le\;\rho+\sqrt{D/\min f_V}\;=\;2.343168,\qquad \|a_y\|_2\;\le\;1+\sqrt{D/\sigma_N^2}\;=\;1.766965,$$

so the Fejér means of the two signal kernels converge in L^2 with no cap at all*. No statement below depends on Lemma 10.*

Lemma 11 (C$'$ — the white-noise floor; *load-bearing*)**.** *Adding δ white noise to a feasible record is* affine *in distortion,* helps $\langle\ln n\rangle$*, and raises σ by at most $\alpha\,\delta\,\|V\|_2^2/\sigma$ for any fixed A$'$-admissible V. Hence $\Psi(D;\Delta)$ is unchanged if the class of Definition 3 is further restricted to records with $n\ge\nu>0$, and* **no convexity is used** *(the earlier convex-mixing regularisation is unnecessary).* ***(F4c) This lemma is what is load-bearing****, and the control has power: on a feasible family whose noise is* not *bounded below, the Fejér error in $\langle\ln n\rangle$ does* not *vanish — at $\nu=3.100\times10^{-6}$*

it is 0.371*, against* 9.9×10^{-4} *at* $\nu = 0.151$*, while the* L^1 *error is flat at* $\approx 1.3 \times 10^{-2}$ *throughout. The Lipschitz constant of* $n \mapsto \langle \ln n \rangle$ *on* L^1 *is* $1/\nu$*, and it blows up exactly as the floor is removed.*

8.4 Theorem D, with its hypotheses printed (F4)

Theorem 14 (D — FIR density). Let $\Psi(D; \Delta)$ be the infimum over the class $\mathcal{P}(D; \Delta)$ of Definition 3, and let $U_{\mathrm{tr}}(L)$ denote a depth-L FIR stationary record's value (see Remark 51 for the labelling). Then

$$\Psi(D; \Delta) \;=\; \inf_{L} \big\{\text{value of a depth-}L\text{ FIR stationary record of } \mathcal{P}(D; \Delta)\big\},$$

i.e. **FIR-kernel stationary records are dense in value**. **The hypotheses are, in full:**

(a) f_V **bounded above and below**: here $f_V \in [0.1111, 9.0000]$ (exactly $[(1-a^2)/(1+a)^2, (1-a^2)/(1-a)^2]$ at $a = 0.8$). This is what turns *feasibility* into L^2 bounds on the kernels — $\langle |g-\rho|^2 f_V \rangle \le D$ gives $\langle |g-\rho|^2 \rangle \le D/\min f_V$ — and hence into L^2 convergence of the Fejér means. It is a *model* hypothesis and it travels with the theorem.

(b) **Continuity of** $D \mapsto \Psi(D)$ **at** $D = 0.3$, which follows from convexity of the value function (Remark 14); the proof picks a near-optimal record at a slightly *smaller* distortion and needs the loss to vanish as the shift does.

(c) **Lemma 11** (the floor), which is load-bearing.

(d) **Lemma 10 (the cap) is *not* a hypothesis: it is deleted from this proof** — feasibility alone supplies the L^2 bounds quoted there.

Proof. Fix $\varepsilon > 0$ and pick $x \in \mathcal{P}(D-\eta; \Delta)$ within ε of $\Psi(D-\eta; \Delta)$, with η small enough that (b) makes $\Psi(D-\eta) - \Psi(D) \le \varepsilon$. Apply Lemma 11 to put its noise above a floor $\nu > 0$ at a cost $\le \varepsilon$. Truncate with **Fejér means** [32] of order L: the taps are supported in $[-L, L]$, the sup-norm is non-increasing, and by (a) the signal kernels converge in L^2 and the noise in L^1. Crucially $n^{(L)} = K_L * n$ is a *non-negative* trigonometric polynomial, because the Fejér kernel $K_L \ge 0$ and $n \ge \nu > 0$; hence by **Fejér–Riesz** [33, 34] $n^{(L)} = |q_L|^2$ for a causal polynomial q_L of degree L, i.e. a legitimate MA(L) record noise. Lemma 9 then closes the σ leg (choose the finite filter *first*, at x; the L^1 distance of the spectra vanishes), and L^1 convergence with the floor closes the $\langle \ln n \rangle$ leg. Repairing the distortion back to D is a scalar rescale of the noise, affine and vanishing as $L \to \infty$. Letting $\varepsilon, \eta \downarrow 0$ gives the claim. □

Nothing in this proof touches the optimum's regularity: it approximates an arbitrary near-optimal feasible record, not a fixed point. This was checked, not asserted — the construction was run on three deliberately non-optimal records (the $\langle \Gamma^2 \rangle$-divergent spiky-noise record, a detuned-kernel record and an oscillating-noise record) and cap+floor+Fejér works on all three with the rates converging to *each record's own* value.

Remark 49 (F12 — the Fejér build, measured to $L = 200$, and a numerical warning that must travel with it). The prover ran $L \le 14$, where the Fejér gap is still 3.9×10^{-4}. Pushed to $L = 200$ at all three lags:

- **convergence is** $\Theta(L^{-2})$: fitted log-log slopes $-1.94007/-1.94006/-1.94009$ over $L = 8 \ldots 200$ with local slopes running to $-1.9891/-1.9890/-1.9890$; the gap above the 082 certified lower endpoint at $L = 200$ is $2.177 \times 10^{-6}/2.591 \times 10^{-6}/3.279 \times 10^{-6}$.

- **positivity holds at every rung**: $\min_\omega n^{(L)} \geq 0.1331$ over 42 rungs (the minimum is attained at $\Delta = 2$, $L = 1$).
- **realisability is certified by a root-free route.** A cepstral (Kolmogorov) factorisation [36, Ch. 13] gives $\left\| |q_L|^2 - n^{(L)} \right\|_\infty \leq 4.16 \times 10^{-16}$ and an MA tail beyond lag L of $\leq 4.85 \times 10^{-17}$ — a genuine *finite* MA. **Polynomial root-finding is not fit for purpose here beyond** $L \approx 96$, and this warning must travel with any successor: on the same rungs the root-finding route degrades 3.86×10^{-16} ($L = 32$) $\to 1.13 \times 10^{-13}$ (64) $\to 4.64 \times 10^{-7}$ (96) $\to \mathbf{1.88 \times 10^{-1}}$ (128), against the root-free 3.33×10^{-16} at that same rung. **This is a numerical artefact of degree-$2L$ root-finding, not a mathematical failure of Fejér–Riesz**, and it is gated as a must-fail control precisely so that no successor mistakes one for the other.
- **the Fejér mean preserves $\langle n \rangle$ exactly** (the lag-0 coefficient is untouched by the window; measured 0.00×10^0), so the distortion repair of the theorem's last step is asymptotically free.

8.5 Lemma M, restated — what it delivers, and the two headline claims that are withdrawn (F5–F9)

Remark 50 (F5–F8 — withdrawals, *not optional*). Lemma M was written to repair W1 and W4 by removing the rescale: run the depth-L stationary program at the *shifted* distortion $D - \eta_n$, so that the zero-edge window record is D-feasible with no repair. **The exact distortion identity it rests on passes** — rebuilt cell by cell from the primitives over 18 rows, $\max|\text{built} - \text{formula}| = 2.22 \times 10^{-16}$, with the **edge cell equal to $1 + \varepsilon$ exactly** (0.00×10^0) and the interior cell equal to D_{stat} to 1.67×10^{-16}; and the shifted build is D-feasible *with no rescale*, $|\text{dist} - D| \leq 2.78 \times 10^{-16}$ over $n = 64 \ldots 1024$. **Two headline claims around it are false and are withdrawn here.**

*(F5) "every constant n-independent" is **withdrawn**.* Measured in the very build Lemma M was written to fix,

$$C(6, n) \;=\; 72.444 \;/\; 20.434 \;/\; 15.194 \;/\; 13.426 \;/\; 12.676 \qquad (n = 64, 128, 256, 512, 1024),$$

a factor 5.715 and still falling — **the same factor-5.7 decrease that Remark 39 refuted**.

*(F6) "≈ 18.7 at $L = 6$, $\Delta = 0$" is **withdrawn as an upper bound***: it is exceeded at $n = 64$ (72.444) and at $n = 128$ (20.434). The asymptote is ≈ 12.0.

(F7) the cause, and it is a mathematical error and not numerical looseness. $D \mapsto U(L; D)$ is **convex** (measured: every second difference of the D-ladder over $D = 0.20 \ldots 0.30$ is positive, minimum $+2.140 \times 10^{-3}$), so the tangent estimate $\mu \cdot \eta$ is a **lower** bound on the repair cost, not an upper one. The multiplier must be taken *at the shifted point.* At $L = 6$, $n = 64$ the shift is $\eta = 0.161769$ and the secant slope is 9.40818 against the tangent $\mu = 2.2261061$ — a factor 4.226.

*(F8) "no feasibility threshold" is **withdrawn**.* At $L = 10$, $n = 64$ the shifted target is

$$D - \eta_n \;=\; -0.018636 \;<\; 0,$$

with the depth-L noise multiplier $\lambda = -0.83506$ — **the same threshold, at the same (n, L), that Remark 40 recorded** ($c = -0.832$ there). Lemma M *relocates* W2's threshold; it does not remove it.

Consequently: Lemma M does *not* discharge W1 or W4. W1 and W2 stand, exactly as stated in Lemma 8 and Remarks 39 and 40, and every consequence drawn from Lemma 8 continues to use only $C(L, n) = o(n)$.

Lemma 12 (M — the distortion margin, as an exact three-leg decomposition (F9))**.** *Fix L, Δ and $\varepsilon > 0$, and run the depth-L stationary program at the shifted target $D - \eta_n$ with*

$$\boxed{\eta_n \;=\; \frac{2L\,(1+\varepsilon-D)}{n-2L}}$$

*— **exactly, with denominator** $n-2L$ **and not** n. Then for $n \geq n_0(L)$, i.e. for those n at which the shifted target is attainable (Remark 50(F8) exhibits an (n,L) at which it is not), the zero-edge window record is D-feasible with no rescale, and $C(L,n) = n\,[L_a(x^{(n)}) - U(L;D)]$ admits the* exact *decomposition*

$$C(L,n) \;=\; -2L\,\mathrm{rate}(x_n) \;+\; \frac{\sum_t \delta_t - \mathrm{Szegö}(L,n)}{2\ln 2} \;+\; n\big[\,U(L;D-\eta_n) - U(L;D)\,\big],$$

with $\mathrm{Szegö}(L,n) = \ln\det T_{n-2L} - (n-2L)\langle \ln n(\omega)\rangle$*. Each leg is separately bounded, so* $\sup_{n\geq n_0(L)} C(L,n) < \infty$*. **Only** $C(L,n) = o(n)$ **is used downstream**, so the conclusion of Lemma 8 — and everything drawn from it below — is untouched by the withdrawals of Remark 50.* Certificates: *the decomposition residual is* $\leq 1.85 \times 10^{-13}$ *at every one of* 14 *rows over* $L \in \{4, 6, 10\}$*,* $n \in \{64, \ldots, 1024\}$*; and* η_n *agrees with the closed form above to* $\leq 2.78 \times 10^{-17}$ *at every row.*

8.6 The ladder is $U_{\mathrm{tr}}(L)$ (F11), and the steps of Lemma W, correctly counted (F13)

Remark 51 (F11 — label the ladder for what it is)*.* The tabulated ladder of Remark 43(a) — and every $U(L)$ quoted in §7 — is $U_{\mathrm{tr}}(L)$**, the *truncation-and-repair* value**: the Dirichlet truncation of the stored kernels with the exact scalar noise rescale. **It is not the depth-L optimum.** Direct minimisation over depth-L FIR records gives strictly smaller values at every rung — 0.5647679 against $U_{\mathrm{tr}}(1) = 0.5674834$ at $\Delta = 0$, 0.5629403 against 0.5632032 at $L = 2$, 0.5627482 against 0.5627729 at $L = 3$, and similarly at $\Delta = 1, 2$ — with $U_{\mathrm{tr}}(L)$ above the direct optimum by at least 9.16×10^{-6} at every one of the nine rungs. **Both sequences decrease toward Ψ from above and neither crosses it**: all nine direct optima sit strictly *above* the 082 certified lower endpoints, minimum margin $+1.12 \times 10^{-5}$. This is a *sharper* does-not-prove-too-much test than truncation alone, and it is the form the harness gates. (The R-IND-5 addendum's full-strength run — 8000 evaluations $\times 4$ restarts, against the harness's pinned 900×2 — reports 0.562747377388/0.536414037072/0.531061238971 at $L = 3$ with margins $+2.0881 \times 10^{-5}/+1.2659 \times 10^{-5}/+1.1219 \times 10^{-5}$; the two runs differ only in the sixth-to-seventh decimal and neither is used as a bar.)

Remark 52 (F13 — Lemma W's steps, and a floor that does not reproduce)*.* Two bookkeeping corrections, both on record. *(i)* The correct reference to Lemma 8 in any chain below is **"steps (1)–(4), with (3) carrying the classical citation, and (5) superseded by Lemma 12"**. Writing "Lemma W(1,2,4)" *undercounts*: **step (3) is load-bearing** and carries the only analytic input that is cited rather than re-proved, namely the standard rational-spectral-factorization/Riccati fact [17, 18]. Its hypotheses were re-verified independently and hold: the joint (S, R) spectrum is $\succeq 0.2394\, I$ uniformly in ω, and the depth-L FIR record's spectrum is rational *by construction*, so the citation is not applied to the object whose density is being proved. *(ii)* The probe's "floored at $n \geq 0.1288$" is **unreproduced and is withdrawn**. The record noise floors are 0.1495/0.1378/0.1404 at $\Delta = 0/1/2$ (as already printed in Remark 43(d)), and the minimum over the entire Fejér ladder to $L = 200$ is 0.1331. Nothing downstream moves — every floor in play is comfortably positive — but the number is replaced, not patched.

8.7 Does the density machinery prove too much? No

Four checks, all on the side of not proving too much. *(i)* Every rung of the U_{tr} ladder, at every $\Delta \in \{0, 1, 2\}$ and every $L = 1, \ldots, 10$, sits strictly *above* the 082 certified lower endpoints, minimum margins $+3.997 \times 10^{-11} / + 7.213 \times 10^{-11} / + 5.267 \times 10^{-11}$ — positive, as weak duality forces, and independently reproducing the values recorded in Remark 43(b). *(ii)* The Δ-ladder of a fixed depth-L record is monotone and stays above block$_\infty$, margin $+2.822 \times 10^{-4}$ at $\Delta = 14$ for $L \in \{4, 6, 10\}$. *(iii)* The direct minimisation of Remark 51 likewise never falls below the certified endpoints, 9/9. *(iv)* **Sharpness, stated honestly.** Theorem 14's modulus — the L^1 bound of Lemma 9(2) plus the floor — is *valid* at every one of 28 rungs (zero violations, minimum slack $+1.898 \times 10^{-8}$) and *loose*, by 3.3×10^{-2} relative at $L = 1$ down to the $f64$ floor at the deepest Dirichlet rungs. The reason is structural and is not a defect: the modulus is *first order* in the kernel error while the truth is *second order* — W7's squaring law (Remark 43(c)), the first-order term vanishing at the optimum. Density needs only that the modulus $\to 0$; the $T(L)^2$ law is sharper behaviour that this bound does not explain, and no claim is made that it does.

8.8 The scoped equality

Theorem 15 ($L^\infty = \Psi$ at $\Delta = 0, 1, 2$ — *a scoped theorem*). Under Hypothesis 1 and Hypothesis 4, with $\Psi(D; \Delta)$ the infimum over the class $\mathcal{P}(D; \Delta)$ of Definition 3,

$$\boxed{L^\infty(\Delta) \;=\; \Psi(D; \Delta) \qquad \text{for } \Delta = 0, 1, 2.}$$

Proof. "$\geq$" is Corollary 6. "$\leq$": by Theorem 14 there are depth-L FIR stationary records of $\mathcal{P}(D; \Delta)$ of value $\to \Psi(D; \Delta)$; by Lemma 8 steps (1)–(4) with Lemma 12 in place of step (5), each such record transfers, for $n \geq n_0(L)$, to a D-feasible window record of $\mathcal{F}_0$ with $\phi_n \leq \mathrm{rate}(x) + C(L, n)/n$ and $C(L, n) = o(n)$; and $L^\infty = \inf_n \phi_n$ by Theorem 4. Let $n \to \infty$ at fixed L, then $L \to \infty$. □

The scope this statement carries, and which must be printed with it. Beyond the four scope items (a)–(d) of Corollary 6, which travel with the "$\geq$" half as always:

(i) Lemma 8 step (3)'s classical rational-spectral-factorization / Riccati citation [17, 18], *with* its hypotheses — verified to hold here (Remark 52(i)), *not* re-proved;

(ii) Fejér–Riesz [33, 34] and Fejér convergence [32];

(iii) the model hypothesis $f_V \in [0.1111,\ 9.0000]$ (Theorem 14(a));

(iv) continuity of $D \mapsto \Psi(D)$ at $D = 0.3$, from convexity (Theorem 14(b));

(v) a window threshold $n \geq n_0(L)$ (Remark 40, Lemma 12).

The word "unconditional" does not attach to this equality. It attaches only to Corollary 6's chain and to Corollary 8's two inequalities, and it never attaches to the value. Ψ remains a two-sided *certified bracket* (Remark 7); **the equality identifies two objects, it is not a licence to quote a value as exact, and Corollary 8 continues to govern the quotable numeric statement.**

The permitted wording, verbatim, as returned by the R-IND-5 pass of 2026-08-08. The following is the ruling as written, and it is the only form in which this equality may be stated:

> *THE RULING: YES, $L^{\inf} = \Psi$ MAY BE PRINTED at $\Delta = 0, 1, 2$ — as a SCOPED THEOREM, in the permitted wording, and ONLY there. Lemma M's refutations do NOT break it (only $o(n)$ is used). The permitted wording carries, beyond cor:onedir's scope: (i) Lemma W step (3)'s classical rational-spectral-factorization/Riccati citation WITH its verified hypotheses; (ii) Fejer–Riesz and Fejer convergence; (iii) the model hypothesis f_V in $[0.1111, 9.0000]$; (iv) continuity of $D \to \Psi(D)$ at $D = 0.3$ from convexity; (v) a window threshold $n \geq n_0(L)$. "UNCONDITIONAL" DOES NOT ATTACH TO THIS EQUALITY — it attaches only to cor:onedir. Ψ remains a two-sided CERTIFIED BRACKET; the equality IDENTIFIES TWO OBJECTS, it is NOT a licence to quote a value as exact, and cor:brackets CONTINUES TO GOVERN the quotable numeric statement.*
>
> *STILL FORBIDDEN: "hypothesis-free"; "unconditional" on the Ψ value or on this equality; any FREE-STANDING reverse inequality outside the scoped theorem; any novelty language for Lemma A''s packaging, Lemma W's combination, or Theorem D until the sweeps run.*

Those prohibitions are house rules of this document from this revision on. In particular the *reverse direction* of Theorem 15 is stated and invoked *only* inside that theorem — it is never written free-standing anywhere else, in this document or in any seal, registry entry or successor — and the quotable numeric statement remains the bracket of Corollary 8.

Remark 53 (What is *owed*, and what may not be said until it is delivered). Three items are outstanding and are recorded here so that no successor treats them as done. *(1)* A **novelty sweep on Lemma 9's packaging** — an infimum of affine functionals over *finitely supported* admissible filters, used as an upper-semicontinuity device for FIR density. Every ingredient is classical; the combination is unswept. *(2)* The **novelty sweep on Lemma 8's combination**, already owed at v0.9 (Remark 47). *(3)* The **window-side `la_cmi` cross-check**: the definitional-CMI validation of §7 was run stationary-side only. **No novelty language of any kind is used for Lemma 9, Lemma 8 or Theorem 14**, here or anywhere in this document, and none may be used until (1) and (2) are run and their verdicts, permitted language and residuals are recorded in the manner of Remark 20.

Question 1 (Open). *(0)* ***(R1) is closed*** *— it is Theorem 10, and Corollary 7 now stands on Hypothesis 1 and the explicit construction of Hypothesis 4 alone. What remains open here is the* reverse *inequality in Corollary 6 — a truncation / achievability lemma at stationarity, which the non-rationality of the stationary optimum makes non-trivial.* ***§7 takes that face apart****: Lemma 8 supplies the truncation half with measured certificates, and what was left was the single residual of Remark 43 — FIR-kernel stationary records dense in value.* ***At v1.0 that residual is closed****: it is Theorem 14, proved by a route that makes the Wiener-norm contraction estimate* unnecessary *rather than supplying it, and the consequence is Theorem 15 —* a scoped theorem, in the permitted wording of §8.8 and only there*. Outside that statement, nothing may be said about L^∞ beyond the brackets of Corollary 8 (which supersede the numbers, though not the prohibition, of Remark 30 as amended), and the quotable numeric statement is unchanged. What remains genuinely open on this face is not the density but the* sharpness *of its modulus (§8: valid at every rung, loose because it is first order* *against a second-order truth) and the two* owed *novelty sweeps of Remark 53. The* ***novelty sweep on the (R1) combination*** *— blocked cyclostationary Szegő, plus the inf-of-affine leak, plus matrix quadratic-over-linear, giving convexity of a causally-conditioned process-rate functional in spectral moment coordinates — against Wiener–Masani/Helson–Lowdenslager [14, 15, 16] and the 070/073 conditional-RDF attributions [19] was* run on 2026-08-07*; its verdict, its permitted language and its caveats are Remark 20, and the* two page-verification items it leaves owed *are recorded there.* ***No language stronger than that remark's is used anywhere in this document****; (1)* A uniform-in-m decay bound on the optimizer's cross-block kernel — no longer needed for the plateau *(Corollary 7 retires it there), but still the residue of the old n-monotonicity face after Theorem 4 closed its subadditivity half unconditionally and Theorem 5 reduced the rest to the single scalar X. The open problem is now* named*, and it is not "bound $I(E; T^{b_2})$": (i) no* uniform family *bound can exist (Remark 12); (ii) no KKT/exchange argument can force $X = 0$, because the distortion gradient vanishes identically on the cross-read coordinates (Theorem 3) — the cross-read is a distortion*-free *direction on which the objective is stationary and the optimizer chooses $X > 0$ because it strictly lowers L_a, so stationarity yields an equation, not a bound; (iii) what is missing is an* optimizer decay estimate*: measured, $\|A^{12}\|_F/\|A^{11}\|_F = 3.44\%$ with the cross-block kernel decaying geometrically at per-lag ratio ≈ 0.28–0.35, bracketing the smoothing pole $\lambda_s = 0.354554$, and $\|A^{12}\|_F$ is* flat in m*. A Demko/Jaffard-type [28, 29] exponential off-diagonal decay bound for inverses of banded positive-definite operators, applied to the KKT system of Theorem 3, would give $\|A^{12}\| \le C\lambda^{\bullet}$ uniformly in m and close the gap — a standard* type *of theorem, much softer than (H**). The conversion from such a bound to $\bar{X}$ is already proved (Remark 17); the two pieces still missing are the fixed-point/bootstrap step that breaks the circularity $M = N^{-1} - \theta I$, and the uniform lower bound $\lambda_{\min}(N) \ge \nu$. Alternatively (H***) (Remark 11) may be attacked directly, since the split-causality value gap is two-sided certifiable. Separately, certified anchors at* $n \approx 260/1320$ *would carry the $\Delta = 1, 2$ plateaus without any of the three hypotheses; and the n-uniform dual certificate of the GO-13 spectral machinery would bypass the transfer route altogether; (2) per-cell convexity (Remark 4) — the within-class certification of the diagonal ladder (Remark 5) is no longer open, it is executed in §5.2; (3) the U-coupled coordinate (Hypothesis 1 dropped): a different, far cheaper object, definitionally open; (4) the reset protocol realizing definition (a); (5) identification of the closure constant beyond $\Delta \approx 6$, and the innovations characterization of the limit (vector NRDF closed forms were refuted in general — Stavrou–Tanaka–Tatikonda [11] — so only scalar forms are expected cheaply).*

House rules: the novelty gate is *cleared* for the four objects swept (arXiv + DBLP + Crossref; S2 residual is belt-and-suspenders only), conditional on the five must-cite adjacents (Berta et al. [21, 22]; Boyd–Mandal–Crutchfield [23, 24]; Venkat–Weissman–Carmon–Shamai [25]; Sandberg et al. [26]; Anderson 2018 [27]) — headline novelty language may be promoted only with the five citations in force, and nothing above claims it yet. R-IND-5 passes 1 (+closure), the Conjecture-v0.2 pass, the Theorems-R+C pass (seven mandatory restatements) and the n-transfer pass (six mandatory restatements — the (H*) demotion, the κ-per-side refutation, the constants, the $\mathcal{F}_0$-only tags, the corollary arithmetic and the ladder tagging), the Ψ-certificate pass (ten mandatory restatements R11–R20) and the **(R1) pass** (nine mandatory restatements R21–R29, all in force in this revision, and four named targets — the block-innovation Cholesky, the matrix Szegő step at $n > 1$, the inf-of-affine representation with its closure point, and the ε-floor Scoping Lemma — all passed) are on record (`go14-causal-erasure-PROBE.md`); seal GO-P-2026-076 governs the theorem's numerical faces; the process-limit faces of §4 await prereg 078 (harness `experiments/go14_process_limit.py`); the convex-program faces of §3 await prereg 079 (harness `experiments/go14_convexity.py`: representation identity incl. non-staircase schedules, PSD/closed-form/Sylvester certificates, 4,200-pair Jensen sweep, tangent plane, cold-start certified brackets at $(16,0)/(16,1)/(24,0)/(32,0)$ + block$_{16}$, sandwich-margin window, and the two-leg / non-conflation U-gates); the transfer faces of §5 await prereg 080 (harness `experiments/go14_transfer.py`: per-cell subadditivity with block-local denominators, the interleaved-prefix set identity, the $\mathcal{F}_0$-only zero-claim with its U-coupled counter-values, the (H*) refutation netted as a gated fact, the optimizer charges at $m \in \{8, 16\}$, the constants and the corollary arithmetic including the failing $n = 24$ base, and reproduction of the transfer anchors and of the within-class ladder). The scope proposed for 080 is: Theorem 4 and Corollary 2 (unconditional), the transfer anchors, the within-class ladder, and the conditional Theorem 7 / Corollary 5.

The v0.6 material — Theorem 3 and its corollaries (§3.1), the H-repair lemma chain L1–L4 and Theorem 5, the split-causal sub-family with its certified ϕ^{sc} brackets and Theorem 6, the Π-retraction identities (Lemma 5), the extended monotone $m \times \Delta$ table, the convexity-of-$\phi_m(D)$ hygiene of Remark 14, the recomputed corollary arithmetic under (H**) and under (D)+(F), and the (H***) value gaps — awaits **registration 081**, harness `experiments/go14_hrepair.py` (s1 the L1 identity and its m-independence; s2 the L2 bound with its tightness witness; s3 L3/L4 and the H-repair inequality over a 22-cell pinned battery including both refuting counterexamples; s4 the split-causal sub-family, its constants and the linearity of the section; s5 the Π-retraction identities; sK the three Theorem-K identities, the two structural facts, the Cholesky split, A_y's transpose symmetry and the m-uniform bounds; s6 the block-distortion straddle and the value-function convexity; s7 the constants and both corollary arithmetics with the base-24 negative control; s8 the monotone m-table; s9 $\phi^{\mathrm{sc}}_{2m} \le \phi_m$ and the (H***) gaps). Nothing in §3.1 or §5.1 is sealed.

The v0.7 material — §6 in full: Theorem 8 with Remark 18, Theorem 9 with Hypothesis 4 and Remark 19, Theorem 10, Theorem 12, Corollary 6 with Remark 30, Lemma 6, Lemma 7, Definition 4 with Remark 33, Theorem 13 with Remarks 34 and 35, the errata of Remark 36 (R14, R15, R18, R19, R20 and the truncation sign), the certified brackets of §6.6, Corollary 7 with Remark 37, §6.7, the $\Delta = 2$ anchors of §6.8 and the bookkeeping convention of Remark 38 — awaits **registration 082**, harness `experiments/go14_plateau.py` (s1 the Collapse identity on staircase, edge and non-staircase schedules with the R6 Δ-lag-causal U-extension gated in *both* directions; s2 the three sub-claims of step (A) with the R7 correlated-noise-copy control; s3 step (B) at cold-start optimizers and at random period-n records — cone membership, distortion preservation, and the no-larger-rate inequality; s4 Lemma 7's Jensen sweep at 10^{-9} from the cone boundary with the R12 $Q \succ P$ control; s5 the minorant over 2,880 record×filter pairs with the R11 non-monic control; s6 R17 — validity of the dual at deliberately non-optimal anchors, mistuned μ, truncated filters and mistuned linearisation; s7 reproduction of the quotable brackets, of the margins against the *sealed* bars, of block$_\infty$ (R20) and of the Δ-dependent $O(1/n)$ constants (R19); s8 the "does it prove too much" block program and the Δ-ladder). The scope proposed for 082 is: Theorem 8, Theorem 9 and Theorem 13 (unconditional given Hypothesis 1 and Hypothesis 4); Lemmas 6 and 7; the certified Ψ brackets; the $\Delta = 2$ finite-n anchors; and Theorem 12 / Corollary 6 / Corollary 7. Nothing in §6 is sealed, and **no seal of 082 may print an equality between L^∞ and Ψ —** that statement belongs to Theorem 15 alone, is scoped, and is proposed for registration 085, never for this one (Remark 30 as amended).

The v0.8 material — Theorem 10 with its four-step proof and its certificate block, the Scoping

Lemma 11, the restatements R21–R29 (Remarks 21, 22, 23, 32, 24, 25, 26, 27, 28), the demotion of the three v0.7 justification steps (Remark 29), the unconditional form of Corollary 6 with its verbatim permitted wording and scope items (a)–(d), and the withdrawal of the "no n-uniform floor" refusal in §3 — awaits **registration 083**, harness `experiments/go14_r1.py`, and is registered *separately from* 082 because the two objects are structurally independent: 082 certifies the value Ψ two-sided, 083 discharges the convexity step of the chain, and neither uses the other. The harness sections are: s1 Collapse against the definitional CMI plus the R21 frame/order pair, with the mixed pairing gated as a must-fail control; s2 the block-innovation Cholesky identity in both frames; s3 the matrix Szegő decomposition by an exact aliasing route, with the R22 extremal family including the vanishing-spectrum case and the DMAX sweep that settles $n = 2$ as truncation; s4 the three inf-of-affine links separately, the concavity margin, the R23 truncated-family check and the peeking-predictor control; s5 the Jensen and curvature sweeps over six adversarial families, with two controls — the record-parameter chart being non-convex, and both legs being concave while the rate is not; s6 the Scoping Lemma including R26's side and sign with a wrong combination gated as failing; s7 the R24 psd battery (22,000 chords) and the R25 well-posedness witness; s8 the R29 floor, the exact Q reduction and $\lambda_{\max}(Z) < 1$; s9 shift-invariance with the R27 `np.roll` control. The scope proposed for 083 is: Theorem 10, Lemma 11 and the consequent unconditionality of the *chain* in Corollary 6 — never of the value Ψ — together with R21–R29. The harness contains no optimizer, no fixed point and no root find, so no gate races a stopping point, and it reads no bracket, width or certificate endpoint. **The novelty sweep on the (R1) combination was run on 2026-08-07**; its verdict, the language it permits and the two page-verification items it leaves *owed* are Remark 20, and no language stronger than that remark's appears anywhere above.

The v0.9 material — §7 in full: the three-build taxonomy of §7.1, Lemma 8 with the restatements W1–W4 built into its statement and hypotheses (Remarks 39, 40, 41), the five steps of §7.2 with their certificates, the orientation gate of Remark 42 (W5), the residual of Remark 43 (W6, W7, W8), Corollary 8 with its verbatim permitted wording and Remarks 44 (W10), 45 (W9 and W12) and 46 (W11), and §7.5 — awaits **registration 084**, harness `experiments/go14_achiev.py`, and is registered *separately from* 082 and 083 because it is the first material on the *achievability* side: 082 certifies the value Ψ two-sided, 083 discharges the convexity step of the lower-bound chain, and 084 nets an explicit family of D-feasible records and the window transfer that turns them into a statement about L^{∞}. The harness sections are: s1 the *mandatory* orientation gate — the canonical build's interior σ_t offset, the reversed-kernel must-fail control at every interior cell, and W5's negative control that the per-cell *distortion* has zero power against the defect; s2 the edge cells contributing exactly zero, with N_{cov} block-diagonality; s3 the subset/monotonicity step over $\Delta \in \{0, 1, 2, 3, 5, 9, 20\}$ including past saturation; s4 the exact n-independence of $\sum_t \delta_t$ and reproduction of the recorded $L = 10$ triple; s5 the Szegő leg's sign, its flatness in m and the min-phase identity; s6 the exactly affine repair and the W2 feasibility threshold, with the $n = 64$, $L = 10$ infeasibility gated as a must-fail control; s7 the *monotone decrease* of $C(L, n)$ in n over a pinned (L, n) grid — the W1 gate, which gates monotonicity and never n-independence; s8 the unconditional brackets, exact feasibility and the 8/8 anchor cross-check; s9 the "does it prove too much" checks against block_{∞}. The scope proposed for 084 is: Lemma 8 with W1–W4 as stated, the five steps, the residual *as a residual*, and Corollary 8. **The harness contains no optimizer, no fixed point and no root find** — its stationary kernels are pinned data, and a record's provenance is irrelevant to whether it is a valid feasible upper-bound certificate — so no gate races a stopping point, and no gate reads or gates a bracket width or certificate endpoint. **The novelty sweep**

on Lemma W's combination is *owed* (Remark 47), and no novelty language for it appears anywhere in this document.

The v1.0 material — §8 in full: the two admissibility notions of §8.1 (Definition 5 beside Definition 4, F2), Lemma 9 with its Hilbert-space proof and the six admissibility controls of Remark 48 (F1, F3, F14), Lemma 10 as a *recorded* a-priori fact and Lemma 11 as the load-bearing one (F4c, F4d), Theorem 14 with its printed hypothesis list (F4) and Remark 49 (F12), Remark 50 (F5–F8) and Lemma 12 (F9), Remarks 51 (F11) and 52 (F13), Definition 3 (F10), and Theorem 15 with the verbatim permitted wording of §8.8 — awaits **registration 085**, harness `experiments/go14_density.py`, and is registered *separately from* 082, 083 and 084 because it is the terminal object on this face: 082 certifies the value Ψ two-sided, 083 discharges the convexity step of the lower-bound chain, 084 nets the window transfer, and 085 nets the *density* that closes the converse. The harness sections are: s1 Lemma 9(1) — the K-lag admissible ladder decreasing to σ from above and reaching it exactly, with all *six* rebuilt, fully optimised admissibility controls gated as must-fail; s2 Lemma 9(2) on ≥ 20 adversarial record pairs including records far from the optimum; s3 the Fejér build — positivity at every rung to $L = 200$, *root-free* Fejér–Riesz realisability, the $\Theta(L^{-2})$ rate, the exact preservation of $\langle n \rangle$, and the root-finding route gated as a must-fail control at $L = 128$ (F12); s4 the cap dispensable (the L^2 bounds from feasibility alone) and the floor load-bearing (the $\langle \ln n \rangle$ error that does *not* vanish at $\nu \approx 3 \times 10^{-6}$); s5 Lemma 12 — the exact distortion identity rebuilt from primitives with the edge cell at $1 + \varepsilon$ exactly, D-feasibility with no rescale, and the *three must-fail controls for F5–F8*: the factor-5.7 *decrease* of $C(6, n)$ (so n-independence is false), the 18.7 figure exceeded at $n = 64$, and the negative shifted target at $(L, n) = (10, 64)$; s6 the three-leg decomposition residual at every row; s7 does-it-prove-too-much — every U_{tr} rung above the 082 certified lower endpoints, the Δ-ladder above block_∞, and the *direct* minimisation records above Ψ^{LB} (F11); s8 the modulus valid at all 28 rungs. The scope proposed for 085 is: Lemma 9, Lemma 11, Theorem 14, Lemma 12 *as restated*, and Theorem 15 *with its five scope items and in the permitted wording only.* **The only search in that harness is s7's direct minimisation, and its gate is on the *set* of feasible records a pinned fixed-budget search evaluates, never on a stopping point**; no fixed point is solved and no bracket, width or certificate endpoint is gated anywhere. **The novelty sweeps on Lemma 9's packaging and on Lemma 8's combination are *owed*, as is the window-side `la_cmi` cross-check** (Remark 53), and no novelty language for Lemma 9, Lemma 8 or Theorem 14 appears anywhere in this document.

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